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Essays on the role of information in shaping short-term price dynamics: evidence from a high-frequency setting

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Declaration of Originality

I, Tesfaye Salarin, hereby declare that the work presented in this thesis is my own original research, conducted during my doctoral studies at the Department of Economics and Management of the University of Trento between autumn 2023 and spring 2026. Where information, ideas, or material derived from other sources have been used, these have been fully acknowledged and cited in accordance with academic conventions.

I further declare that this thesis has not been submitted, in whole or in part, for any other degree or professional qualification at this or any other institution.

This thesis was written with the assistance of AI language tools for language editing, structural suggestions, and LaTeX formatting.

All intellectual content, research design, empirical analysis, and conclusions are entirely my own. No AI tool was used to generate or fabricate data, results, or citations.

Dedication

I'll be the moon that shines on your back

To my beloved *Inat*.
Look what I've become.
I know you'd be proud of me.
I say it, I do it. I'm a man of my word.
Never have I ever doubted about your presence.
May this reach your heart. For eternity, your son.

*From a lost baby abandoned on the streets of Addis Ababa 29 years ago,
now holding a PhD.*

Preface

This dissertation is a collection of three empirical research papers contributing to the ongoing debate on whether public information continues to play a meaningful role in the mechanisms shaping short-term price dynamics in a financial environment increasingly governed by granular, high-frequency trading data. The overarching aim is to investigate whether and how new information (i.e., public news events) translates into price movements, order flow dynamics, and adverse selection around the release. The papers were written between autumn 2023 and spring 2026 during my doctoral studies at the Department of Economics and Management of the University of Trento. Parts of the second paper were initiated and developed under the supervision of a host professor at University College London during autumn and winter 2024–2025. The third paper was co-supervised by a practitioner from the IMI Corporate & Investment Banking Division of Intesa Sanpaolo. Throughout, the thesis has benefited from the suggestions of researchers and practitioners met at conferences, workshops and, above all, during visiting research periods.

It all started in early 2021. At the time, I was approaching the end of my Master’s degree in Economics and Finance at Ca’ Foscari University of Venice. When I first came across the GameStop short squeeze I decided almost immediately to dig into it, because it seemed to be the exact opposite of what the theory I was being taught predicted. For readers who do not recall the episode: in January 2021, shares in the struggling American video game retailer GameStop became the object of an extraordinary short squeeze. Short interest stood at roughly 140% of the free float, so as the price rose, short sellers were forced to buy back shares to cover their exposure, hence pumping the price even further up. The buying pressure did not originate from revisions of the company’s fundamentals, rather from an emotional attachment to the company by retail investors, who collectively coordinated on social media. Surprisingly, the massive wave of buying pressure behind GameStop was completely disconnected from traditional financial metrics (i.e., revisions of the company’s fundamentals, corporate earnings, or intrinsic valuation). Instead, the reason for all of this was as simple as fascinating: for millions of individual retail investors, GameStop was far more than a ticker symbol on a balance sheet. It was a cherished memory of their youth, before the closing of physical stores and the transition of gaming toward digital platforms. This deep emotional attachment to GameStop quickly translated into a broader cultural movement, a ‘*David vs. Goliath*’ crusade aimed at penalizing institutional short-sellers who had heavily bet on the company’s bankruptcy.

At its peak on 28 January the price exceeded \$500 per share, some 30x its level at the start of the month; several brokers suspended purchases of the stock that same day, and at least one hedge fund required an emergency capital injection. Elevated prices and volatility persisted for months. On 24 March the stock fell 34% following the release of earnings and the announcement of a secondary offering, only to recover 53% the very next day. That final pair of days is, in retrospect, the question this dissertation set out to answer: a fall

on genuine fundamental news, reversed almost entirely within twenty-four hours. Was the price responding to new information, or to the order flow that happened alongside?

So, on one side stood the Efficient Market Hypothesis, with its assumptions about how prices incorporate information; on the other stood a security whose price had detached from any plausible fundamental value for weeks. In light of this dichotomy, the motivating question of my PhD research proposal was: where does the truth lie? Is it to be found in the pages of the seminal [Fama \[1965\]](#), or in the more recent econophysics accounts of price formation, in which order flow is treated as endogenous? Starting from that question, my research interests evolved considerably over the course of the PhD, beginning with news-based trading strategies and progressively deepening into high-frequency trading and market microstructure. In particular, I developed a sustained interest in the dynamics of price formation in limit order books around the arrival of new information. The overarching theme of this dissertation is therefore two-fold: *i*) to understand the mechanisms through which new information is digested by market participants and translated into trading activity; and *ii*) to establish whether that information, once absorbed, exerts a causal influence on prices, or whether observed price movements are better explained by the order flow they coincide with.

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Statement of Contributions

This dissertation comprises three research papers. Paper 1 is co-authored; Papers 2 and 3 are single-authored. The table below sets out the nature and extent of each author’s contribution to each paper, following the CRediT taxonomy of contributor roles.

Contribution	Paper 1	Paper 2	Paper 3
Conceptualisation	T.S.	T.S.	T.S., F.B.
Literature review	T.S.	T.S.	T.S.
Data collection and curation	T.S.	T.S. T.A., S.B.	T.S., A.P.
Methodology design	T.S., S.P.	T.S., T.A., S.B.	T.S.
Software and programming	T.S., J.S.	T.S.	T.S.
Formal analysis and results	T.S.	T.S.	T.S.
Writing - original draft	T.S.	T.S.	T.S.
Writing - review and editing	T.S., J.S., F.B., S.P.	T.S.	T.S.
Supervision/Co-Supervision	F.B., S.P., J.S.	F.B., T.A., S.B.	F.B., A.P.
Funding acquisition	S.P.	-	-

Key: T.A. = Prof. Tomaso Aste (host supervisor, University College London). S.B. = Prof. Silvia Bartolucci (host co-supervisor, University College London) F.B. = Prof. Flavio Bazzana (supervisor, University of Trento); S.P. = Prof. Sandra Paterlini (University of Trento); A.P. = Andrea Pramplini (industry supervisor, Head of Financial, Digital & Markets Tech at Intesa Sanpaolo) T.S. = Tesfaye Salarin (PhD candidate); J.S. = Dr. Jacopo Staiano (University of Trento);

Paper 1 corresponds to the working paper Salarin, Staiano, Bazzana and Paterlini, *Examining Intraday Stock Market Response to ESG News Classified by AI: An Event Study Approach*, on which the candidate is first author. Papers 2 and 3 are the candidate’s sole work; parts of Paper 2 were developed during a research visit to the Department of Computer Science, University College London, under the supervision of Prof. Tomaso Aste. The score-driven modelling framework applied in Paper 2 is the published methodology of Campigli, Bormetti and Lillo (2022); those authors were not involved in this research, and the contribution of Paper 2 lies in its extension and application to heterogeneous news events.

Dissertation Abstract

This thesis asks how information enters prices at intraday horizons, and it approaches the question through three essays that move progressively from the reaction of prices to news toward the mechanism by which that reaction is produced.

The first essay studies the intraday processing of firm-specific ESG news for Dow Jones Industrial Average constituents. News items are classified by pillar and sentiment using a large language model, and abnormal returns and trading volumes are measured by event study at the minute frequency. The results are heterogeneous across pillars, and the muted return responses admit two readings that a minute-level design cannot separate: rapid absorption, or limited statistical power.

The second essay moves from returns to the limit order book itself. Using message-level data for six NASDAQ-listed stocks of differing relative tick size, I estimate a score-driven model in which instantaneous price impact is a time-varying parameter. Impact proves news-dependent (earnings releases generate the largest peaks, FOMC announcements sharper but more moderate ones), and it changes sign, which a static specification cannot produce at all.

The third essay asks whether the association between order flow and returns is causal. For five stocks across sixteen FOMC announcements, I decompose risk-adjusted returns into an adverse selection component and an order flow imbalance component, and I instrument the latter. Adverse selection is negative and significant throughout; order flow imbalance, significant under OLS, loses significance once simultaneity is corrected. Cross-sectionally, the adverse selection effect is strongest in large-tick stocks.

Taken together, the three essays follow a single question into finer and finer detail: how a piece of news becomes a price change. The first measures the size of the reaction, the second tracks how the impact of each trade shifts while the news is being absorbed, and the third asks how much of that impact carries information and how much is mechanical. Where the answer rests on an assumption I cannot test directly, I say so.

Introductory Remarks

Human beings are at the core of every phenomenon involving social life. Since the earliest forms of organized exchange, they have coordinated, competed and cooperated in the pursuit of their own interests, thus creating systems whose features constitute a far more interesting, nuanced, and dynamic framework than any single individual action could ever suggest. Clearly, markets are inherently a human phenomenon, and financial markets are, perhaps, the most vivid and data-rich example of such systems.

At their most primitive level, a financial market is the ecosystems in which buyers and sellers coordinate to determine an equilibrium price at which they are willing to transact. Now, as for any other aspect involving human interaction, this coordinated activity does not arise effortlessly. Precisely, for this non-trivial matching to occur efficiently, a set of rules and institutions are required, and have thus been developed over centuries of market activity. Without a clear structure, buyers would seek to minimise the transaction price while sellers, on the contrary, would seek to maximise it, thus generating a strategic *impasse* that would prevent a trade from occurring. In this regard, economists have long endeavoured to resolve this 'tension'. One early solution was through *Walrasian auctions*,¹ in which buyers would communicate bids (i.e., the maximum price at which they are willing to buy), and sellers would post offers or asks (i.e., the minimum price at which they are willing to sell). Notwithstanding their theoretical elegance, Walrasian auctions often led to dissatisfaction on one, if not both sides of the market, as they did not allow buyers and sellers to coordinate dynamically on a mutually acceptable price. When both parties were greedy, few to no trades occurred. When only one side was, large volume imbalances are more likely to arise. Interestingly, most financial markets operated 'within a neighbourhood' of this framework for a long time. Surprisingly, the era of crowded trading pits on exchange floors gathering hundreds of traders waiting for the opening bell before erupting in hand-signals, body gestures and shoutings is only a few decades behind us. In an attempt to overcome these limitations, most markets introduced a new institutional figure: *the specialist*, or *designated market-maker*. Market-makers are legally required to maintain a fair and orderly market [Bouchaud et al., 2018] through continuous quoting (i.e., providing binding bid and ask prices at all times) and trade clearing, thereby ensuring that incoming orders are matched and executed at a price that minimises the number of unsatisfied participants. This institutional innovation marked the first formal rule of organised trading and remains central to the functioning of modern financial markets to this day, just in a completely different form.

The complexity of financial markets, however, runs far deeper than the institutional challenge of matching buyers with sellers. It is deeply rooted in the nature of the participants themselves. Unlike the particles of a physical system, market participants are strategic agents, in the sense that they hold private information, form expectations about the future,

¹Named after the French mathematical economist Léon Walras (1834-1910), who proposed it as a mechanism for understanding how prices reach equilibrium such that supply matches demand.

observe and respond to each other's behaviour, and continuously adapt their strategies in light of what they learn. Undeniably, 'no two market participants' are identical in their beliefs, risk tolerance, time horizon, or information set. And crucially, their interactions take non-linear forms, meaning that the action of one agent changes the environment faced by all others, which in turn alters their behaviour, which feeds back into prices, which shapes future actions. This intricate, recursive, miscellaneous, and adaptive loop is what continuously unfolds in financial markets, and it is precisely this chain of events that detaches financial markets from merely being mechanical equilibrating engines condensing private beliefs into a single signalling price.

In light of this re-interpretation of financial markets, we are inadvertently entering the realm of complexity science and its financial counterpart, *econophysics*, that is the interdisciplinary field that applies various models and concepts associated with the physics of complex systems (e.g. statistical mechanics, condensed matter theory, self-organized criticality, microsimulation, etc.) to model economic phenomena [Rickles, 2007; Mantegna and Stanley, 2000]. In the language of complexity science, financial markets are regarded as *adaptive complex systems*. Allow me to gently pause the reader's attention on a seemingly subtle but essential distinction toward a more rigorous understanding. When we use the term *complex*, we refer to something more than merely 'complicated'. Most activities in our daily lives are complicated, but not all are *complex*. Although no universally accepted definition exists, Herbert Simon offered what remains one of the most cited and pragmatically useful characterisations:

By a complex system I mean one made up of a large number of parts that interact in a nonsimple way. In such systems, the whole is more than the sum of its parts, not in an ultimate, metaphysical sense, but in the important pragmatic sense that, given the properties of the parts and the laws of their interaction, it is not a trivial matter to infer the properties of the whole.

[Simon, 1981, p. 4]

In the absence of a precise definition—rather than seeking a top-down definition- most researchers would deem it reasonable to reverse the logic : first identifying the properties that characterise well-established complex systems, and then examining whether financial markets exhibit them. To this end, I hereby present a *kernel* of such features : many heterogeneous interacting subunits; non-linear interactions between them that generate higher-order properties not deducible from any individual constituent; and feedback mechanisms through which the system's aggregate behaviour reshapes the behaviour of its parts. When we add the *adaptive* attribute in front, we are implicitly emphasising that individuals (i.e., the subunits) accumulate memory along the unfolding of these dynamics, in so doing reshaping their properties and strategies as the surrounding environment evolves, and eventually generating new systemic properties that in response to such changes [Rickles, 2007; Johnson et al., 2003].

Interestingly, financial markets meet all of these criteria: they are composed of large numbers of heterogeneous agents (e.g., buyers and sellers, market-makers, algorithmic traders, etc.) whose non-linear interactions generate aggregate properties that cannot be straightforwardly deduced from the behaviour of any individual participant [Simon, 1981; Rickles, 2007], and those buyers and sellers differ in their information sets, risk aversion, time horizons, strategies, and expectations about the future state of traded assets. They observe each other, adapt, and respond -not to some abstract equilibrium price-, but also to the actions and inferred intentions of other participants. The result is a system whose 'whole' genuinely

matters more than the sum of its parts: one capable of generating structure, regularities, and disorder entirely from within.

The institutional architecture described above defines the 'arena' in which these participants interact. But understanding how prices emerge from their interactions requires us to look beyond its structural features and ask a deeper question: who are these participants, and upon what basis do they make their trading decisions? The answer to that question, and in particular the role assigned to information, has evolved considerably over the past century.

For decades, neoclassical economics has been the dominant framework governing both the theoretical development and empirical analysis of market dynamics. The neoclassical paradigm posits the existence of fully rational agents whose aggregated behaviour drives the price of any good, service, or financial asset toward its so-called '*fundamental value*' (i.e., a latent quantity that emerges from the interaction of opposing forces and is not directly observable). In this view, an asset's true value arises from the interaction of agents individually solving the central economic problem: maximising expected utility subject to a set of binding constraints (i.e., wealth endowments, short-sale restrictions, transaction costs, and institutional mandates). Crucially, this optimisation problem carries a somewhat paradoxical implication for the behaviour of asset prices. Under this perspective, rational agents, in the relentless attempt to exploit any detectable pattern in price movements, will buy when prices appear cheap and sell when they appear expensive². Yet, in doing so, they inadvertently undermine the very patterns they are attempting to exploit, as buying pressure raises prices while selling pressure compresses them. In addition, the apparent 'informational advantage' that motivated the trade is revealed to the market and simultaneously impounded into prices. Exploitable patterns therefore cannot persist by neoclassical construct.

The logical endpoint of this argument is the *martingale* property of prices, positing the best forecast of an asset's future price is its current price, since that price already fully reflects all known information. Price changes, by this logic, can only be caused by the arrival of new, unpredictable information and, since such information is by definition unforeseeable, prices must follow a random walk. This insight has deep historical roots. The random walk model of asset prices traces back to the doctoral thesis of Louis Bachelier [Bachelier, 1900], and was later formalised into what is now considered the backbone of modern quantitative finance: the geometric Brownian motion model.³ Building on this foundation, [Black and Scholes, 1973] and [Merton, 1973] derived the renowned Black-Scholes-Merton (BSM) option pricing model. Though often recognised as an early example of econophysics⁴, BSM stems from the core neoclassical assumptions outlined above and their direct advancements, that is : current prices as optimal predictors of future prices, log-normality of returns, and balanced buying and selling pressures in equilibrium. Indisputably, the influential formalisation of the neoclassical view is the *Efficient Market Hypothesis* (EMH), which states that asset prices at any given time fully reflect all available information [Fama, 1970a]. More precisely, the EMH posits that the market acts as a decentralised aggregator of dispersed private estimates, synthesising the beliefs, forecasts, and valuations of heterogeneous participants into a single publicly observable price. The concept of an 'efficient price' is therefore to be understood through the principle of *no-arbitrage*, whereby no trader, regardless of the relevance of the information they possess, can systematically earn returns above the risk-

²The archetypical *buy-low sell-high* trading strategy

³Johannes Voit [Voit, 2005] refers to this as "the standard model of finance": the view that the logarithm of a stock's price performs a random walk.

⁴Also known as *statistical physics*, econophysics is a branch of physics that studies complex systems in the economics and finance field. In short, they are the utmost rival of neoclassical economics, mainly positing that empirical data is best explained in terms flowing out of statistical physics rather than pure economics.

adjusted benchmark. Any exploitable pattern, once identified, would be immediately traded away, enforcing efficiency as the outcome of a *self-sustaining* equilibrium.

Yet this elegant framework has faced persistent empirical challenges. In fact, the literature of empirical finance documents numerous anomalies resistant to neoclassical explanations, collectively termed -somewhat euphemistically- *financial puzzles*, and conceptually relevant for the introduction to this dissertation. First, the *excess volatility puzzle* [Shiller, 1981], which reveals that prices fluctuate far beyond what fundamental news can justify, with many large price jumps occurring in the absence of any identifiable exogenous trigger [Cutler et al., 1989a]. If the EMH was correct, then every significant price movement would have a somewhat 'traceable' informational cause; yet, careful post-hoc examinations of major market crashes frequently reveal no explainable exogenous cause in fundamentals [Bouchaud et al., 2004a]. In short, prices move too often and by too much to be consistent with the geometric Brownian motion model that underpins the neoclassical framework. Secondly, the *excess trading puzzle*, questioning why rational agents trade so frequently if prices already reflect all information [Bouchaud et al., 2018]. Third and last, the *trend-following puzzle* documents the persistence of predictable price continuations over time. [Jegadeesh and Titman, 1993] detect momentum over shorter horizons of three to twelve months in US equities, while [Lempérière et al., 2014] extend this evidence to longer time scales of weeks to months, across virtually all asset classes and persisting for at least two centuries. Both findings constitute a direct violation of the EMH martingale property, under which past prices should hold no predictive power for future returns.

These conundrums are not isolated curiosities. Rather, they are systematic, persistent, and robust patterns across markets, asset classes, and time periods. Beyond these puzzles, financial time series are known to exhibit a family of *prima facie* non-trivial statistical regularities known as *stylized facts*. A set of universal properties of asset returns that persists independently of idiosyncratic details of financial instruments. May the reader allow me once again to point to [Cont, 2001] for a comprehensive and noteworthy meta-review of stylized facts, as he documented eleven of them. Key examples that I deem relevant for setting the ground to the upcoming dissertation include: fat-tailed return distributions, where tail probabilities substantially exceed Gaussian predictions; volatility clustering, wherein high-volatility periods follow one another as do calm periods; and long-memory volatility with slowly decaying autocorrelations [Bouchaud et al., 2018]. In an adaptive complex system, neither the puzzles nor the stylized facts are considered as 'market anomalies', but rather expected features of a world where aggregate behaviour emerges from the non-linear interactions of heterogeneous, boundedly rational agents [Rickles, 2007]. Hence, the neoclassical framework built on representative rational agents solving well-defined optimisation problems in frictionless markets, is structurally ill-equipped in face of such emergent, heterogeneous dynamics.

The evidence presented above collectively suggests that financial markets are better understood as *adaptive complex systems* rather than as efficient equilibrating machines. In a complex system, the aggregate response to an informational shock is an emergent outcome of heterogeneous agents processing the same signal through different lenses, at different speeds, with different prior beliefs and different liquidity constraints. It should be noted that this heterogeneity in response is not to be seen as a 'failure' of the market in its information-processing mechanism, rather as a systemic structural feature intrinsic to the functioning of financial markets. While adjudicating between these two views lies outside the scope of this dissertation, the distinction is consequential: it is precisely because markets are complex and agents are heterogeneous that the same piece of information can produce fundamentally different responses across contexts, participants, and market conditions.

Central to all of the above is the concept of *information* and the role it plays in shaping asset prices. In its most fundamental characterisation, a price is a compressed, publicly visible representation of the collective beliefs of market participants about the future value of a tradable security. When new information arrives, it triggers belief revision and price adjustment [Grossman and Stiglitz, 1980]. What remains deeply contested, however, is the mechanism through which this adjustment occurs, its speed and completeness, and, mostly, whether information is the *proximate* driver of price changes, or whether it operates indirectly through the intermediate channel of trading activity. Allow me to argue that the question of ‘mechanism’ is not merely academic. Understanding whether -and how- price changes reflect exogenous information or endogenous liquidity dynamics is among the most consequential questions in modern financial markets, where trading activity is continuously flooded with external inputs of varying informativeness. At least up to a certain extent, this problem is *tantamount* to separating ‘signal’ from ‘noise’.

In this regard, practitioners have long embraced and recognised the practical implications of the seminal works by [Kyle, 1985] and [Glosten and Milgrom, 1985]. Undoubtedly, they form the theoretical backbone of modern algorithmic market-making and optimal execution strategies. Together, their research established that information reaches prices not instantaneously or mechanically, but rather through a strategic process of informed trading and market-maker inference. In the former’s framework, an informed trader (i.e., one who possesses private information of an asset’s true value) exploits this informational advantage, gradually, by disguising their trades within the flow of uninformed liquidity demand. Subsequently, the market-maker (unable to distinguish informed from uninformed order flow) adjusts prices in the direction of net buying or selling pressure, thus progressively impounding information into prices as it is revealed through the pattern of trades. In the latter’s framework, the bid-ask spread itself becomes an instrument of information extraction: market-makers widen spreads in the presence of suspected informed trading to protect themselves against *adverse selection*⁵. [Hasbrouck, 1991] later provided the first rigorous econometric framework for measuring the permanent price impact of individual trades, decomposing order flow into its informational and transitory components, a methodology that constitutes a direct intellectual precursor to the empirical strategy of the second and third papers of this dissertation. In short, these models establish a critical insight: *order flow* is the transmission mechanism through which private information becomes public price. Hence, in this view, information does not move prices directly, but rather does so by inducing trades, which, ultimately, drive observed price changes.

Crucially, information in financial markets is not a homogeneous quantity. It spans a wide spectrum: from hard, verifiable, and precisely timestamped public releases (e.g., FOMC⁶ announcements [Andersen and Bollerslev, 1998], earnings releases, and quarterly reviews, whose timing is known in advance and whose content can be objectively quantified), to soft, interpretive, and often unscheduled firm-specific disclosures (e.g., ESG, political news, or controversial corporate announcements), whose subjective interpretation inherently amplifies market uncertainty and whose price impact is correspondingly harder to anticipate [Tetlock, 2007]. This heterogeneity has important implications: different types of information arrive through different channels, reach different participants at different speeds, and are processed through different cognitive and algorithmic frameworks [Hendershott et al., 2011]. And it is precisely this heterogeneous nature of information, and the consequent heterogeneity of the market’s response to it, that the three papers of this dissertation set out to investigate.

⁵The risk of trading at a disadvantageous price with a counterparty who knows more than they do.

⁶Federal Open Market Committee.

Needless to say, the empirical investigation of these questions would have been unfeasible had such a remarkable technological developments not taken place. Physical presence on trading floors was replaced by co-location servers. Instead of hand signals and *open-outcry*, nowadays a *plethora* of algorithms is responsible for submitting, modifying, and cancelling thousands of orders per second. And, most importantly, the opacity of the trading pit was substituted with the publicly visible *Limit Order Book* (or *LOB*), in which, as opposed to quote-driven systems where prices are set by *designated market-makers*, price formation is a *self-organised* process governed by the continuous submissions and cancellations of orders. Modern electronic markets are today *ubiquitous* across all asset classes and trading venues, operating as continuous-time double auctions in which buyers and sellers submit limit orders, each specifying a price p and a quantity Q , which are then stored in a publicly visible queue sorted by price-time priority.⁷ In modern electronic LOBs, every outstanding bid and ask is recorded with microsecond precision, buy orders accumulate on the bid side and sell orders on the ask side. Platforms such as LOBSTER⁸ [Huang and Polak, 2011] (the primary data source for Papers 2 and 3 of this dissertation) reconstruct the full queue of outstanding orders at every instant, allowing the researcher to observe not just transaction prices but the complete state of the book up to 40 levels of depth, timestamped to the microsecond. This transition did not merely accelerate existing market processes, it generated an unprecedented empirical source of data. The granularity of high-frequency limit order book data is both a quantitative and qualitative improvement over daily or weekly data. As a matter of fact, at daily horizons, the causal relationship between information arrival and price adjustment often remains empirically indistinguishable. Meanwhile, at intraday frequencies, the absorption process unfolds in real time and becomes empirically tractable. Intriguingly, high-frequency data also reveal what happens in the *absence* of identifiable news, where endogenous price fluctuations are purely driven by order flow dynamics, therefore completely disconnected from any exogenous informational 'trigger'.

This last observation underlies the theoretical tension that accompanies the reader throughout this dissertation. On one side stand the classical market microstructure models from Kyle [1985] and Glosten and Milgrom [1985] through to Hasbrouck [1991], which treat order flow as the *carrier* of information: order-flows move prices primarily because they reveal private information, and purely uninformed order flow should, in principle, exert no permanent impact on prices [Kyle, 1985; Glosten and Milgrom, 1985]. And, somehow The EMH is the macro-level expression of this view : prices move because, and only because, information moves them. On the other side, a more recent and provocative challenge to this orthodoxy has been formalised by [Gabaix and Koijen, 2021a] in the *Inelastic Market Hypothesis* (IMH) sits at the opposite corner. They argue that financial markets are sufficiently inelastic for the order flow to move prices regardless of its informational content. As they frame it, a unit of buying pressure raises prices by a roughly fixed amount, independent of whether it reflects genuine private information or pure liquidity demand. Under this view, information does not directly cause price changes, but rather acts as a *coordinating signal* that synchronises the trading activity of heterogeneous agents, and it is this coordinated flow that constitutes the real driver of price changes. The implications are extremely significant: if the IMH is correct, then when one observes a price jump around a news announcement, she cannot automatically attribute that jump to the content of the news. Part of it may simply be the mechanical consequence of the coordinated buying or selling pressure that the news triggered, amplified by the inelasticity of the market. Disentangling the informa-

⁷Price-time priority is the most common rule, though pro-rata priority also exists, primarily in futures markets.

⁸LOBSTER Data.

tional and the mechanical channels requires a combination of high-frequency data, causal identification, and microstructural tools that this dissertation 'brings to bear'.

In light of the extensive and thorough introductory background outlined above, the present dissertation aims at investigating the role of information in explaining short-term price dynamics across three complementary empirical studies. Chapter 1 examines how information alone shapes intraday price responses and trading volume around firm-specific ESG disclosures. Chapters 2 and 3 then progressively widen the lens : the former investigates how time-varying instantaneous price impact responds to the arrival of heterogeneous informational events—thus shedding light on the mechanics of price formation across a broad spectrum of news categories -, while the latter ultimately explores how, and to what extent, high-frequency excess-returns around scheduled macroeconomic announcements can be decoupled into an information and order flow channels.

The first paper studies how financial markets process firm-specific ESG news at intraday horizons. Rather than focusing on long-run valuation effects, the analysis concentrates on the short-term mechanisms through which ESG disclosures are incorporated into the main liquidity metrics (i.e., returns and trading volume). The study combines minute-level data for Dow Jones Industrial Average (DJIA) constituents with precisely timestamped ESG news releases. To address the semantic heterogeneity and measurement challenges inherent in ESG disclosures, the paper adopts a classification framework based on a Large Language Model (LLM), identifying both the ESG pillar (i.e., Environmental, Social, Governance) and the sentiment conveyed by each news item. Using an intraday event-study methodology, the paper evaluates abnormal returns and trading volumes following ESG news announcements. Overall, the results reveal a miscellaneous framework across ESG pillars and differences in the speed and nature of market adjustment, with cases of absent or statistically insignificant intraday reactions. Whether these reflect genuinely rapid market absorption of information, or simply the limited power of the tests in the smallest subsamples simply cannot be determined from the minute-level design alone.

Given that the first paper documented a heterogeneous—and sometimes muted—short-term market reaction to firm-specific ESG news, the second paper shifts the question to what the full structure of the limit order book reveals about liquidity and price formation dynamics⁹ around a heterogeneous spectrum of news events, spanning scheduled macroeconomic announcements, political news, and controversial corporate disclosures. Building on Campigli et al. [2022], the paper employs a Score-Driven Aggregated Modified Hasbrouck (SDAMH) model in which instantaneous price impact evolves as a time-varying parameter driven by the score of the predictive likelihood, combined with high-frequency LOBSTER data for five stocks classified by relative tick size over 2023. The paper first documents the main stylized facts inherent to LOBs, showing that their intensity and timing systematically vary across news categories. Secondly, the results of the SDAMH estimation reveal news-dependent price impact dynamics: earnings announcements trigger the largest instantaneous impact peaks across all stocks, consistent with firm-specific uncertainty resolution through a concentrated burst of directed order flow; FOMC announcements produce sharp but comparatively moderate market-wide responses, while controversy news yields heterogeneous dynamics contingent on the nature and timing of each disclosure. Crucially, the time-varying specification uncovers sign reversals in instantaneous price impact, indicat-

⁹The distinction between price formation and price discovery is as slippery as it is necessary. The former refers to the process through which information and order flow are gradually translated into observable prices; the latter refers more specifically to the efficiency with which prices converge to their true fundamental value. The distinction matters particularly at high frequencies, where formation dynamics are often more tractable than discovery.

ing shifts between buy- and sell-side informed flow regimes that static models would miss entirely.

Finally, after documenting how liquidity reacts differently to heterogeneous informational events, the third and final paper asks whether, and to what extent, the documented reaction is attributable to the informational content of news or rather to the inherent endogenous order flow dynamics of financial markets. In this respect, FOMC announcements serve as exogenous shocks, being well-established market-moving events with precise, publicly known timing. In a similar fashion to the previous chapter, I employ high-frequency limit order book data from LOBSTER for five NASDAQ-listed stocks across 16 FOMC events spanning 2023-2024. I then use the *adverse selection* as a proxy for the information component from [Glosten and Harris, 1988], and the order flow imbalance as a measure capturing the endogenous liquidity pressure channel. Under OLS, both components are significant predictors of excess returns. Especially, the information component displays a negative impact, consistent with temporary price overshooting following informed trading, while order flow imbalance enters positively, in line with the well-documented contemporaneous relationship between buying pressure and short-horizon returns [Cont et al., 2014]. However, an instrumental variable estimation then reveals a striking, rather controversial, asymmetry: the information component coefficient remains negative and highly significant, growing in magnitude under IV correction, therefore confirming the causal nature of the adverse selection channel. Contrarily, once controlled for endogeneity, order flow imbalance collapses to statistical and economic insignificance, revealing simultaneity with returns rather than a genuine causal price impact.

Nevertheless, the information component effect proves robust stability across all FOMC events in the sample and to be significantly stronger for large-tick stocks, corroborating the findings of [Parlour, 1998; Foucault et al., 2005] on the more abrupt price adjustment dynamics faced by market-makers in queue-driven markets.

As this dissertation reaches its epilogue, an important and honest claim needs to be made. As the reader will notice while navigating this final chapter, the analysis reveals a compelling tension between the model specification and the empirical functional form of the regressor for the order flow imbalance. The linear specification remains the 'theoretically' preferred and more parsimonious choice according to statistical tests, yet the square-root transformation of order flow imbalance yields a superior empirical fit, which, in turns, perfectly reconciles with recent theories on the concavity (i.e., square-root law) of the market impact [Bouchaud et al., 2018]. Certainly, this is a contradiction that a broader and higher-quality data collection would likely help resolve. After all, for as much as a researcher endeavours to specify the most appropriate model, this tension feels more like an empirical confirmation of George Box's celebrated dictum: "*all models are wrong, but some are useful*" [Box, 1976]. Hopefully, in this dissertation, I have come at least close to the second part of this quote.

The remainder of this thesis is organised as follows. Chapter 1 presents the first paper on ESG news processing. Chapter 2 presents the second paper on LOB dynamics around news events. Chapter 3 presents the third paper on causal decomposition of excess returns around FOMC announcements. Chapter 3.6 concludes the thesis.

Chapter 1

Intraday Information Processing of ESG News: Evidence from High-Frequency Event Studies

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1.1 Abstract

This paper examines how financial markets process firm-specific ESG news at intraday horizons. Rather than focusing on long-run valuation effects, the analysis concentrates on the short-term information-processing mechanisms through which ESG news is incorporated into prices and trading activity in an inelastic market environment. The study combines minute-level data for Dow Jones Industrial Average constituents with precisely timestamped ESG news releases. To address the semantic heterogeneity and measurement challenges inherent in ESG disclosures, the paper adopts a dual classification framework based on a large language model, which identifies both the ESG pillar (Environmental, Social, Governance) and the sentiment conveyed by each news item. We use an intraday event-study method to evaluate abnormal returns and trading volumes following ESG news announcements. The results reveal heterogeneity across ESG pillars and differences in the speed and nature of market adjustment. In several cases, muted or statistically insignificant intraday reactions are consistent with rapid information absorption or anticipation effects, rather than with the absence of informational content, though the design cannot rule out limited statistical power in the smallest pillar-sentiment subsample.

1.2 Introduction

Environmental, Social, and Governance (ESG) issues have become central to modern finance, influencing both investors' decisions and firms' value creation [Capelle-Blancard and Petit, 2019]. At the same time, the rapid expansion of automated and algorithmic trading

has increased the speed at which markets process information, while advances in Artificial Intelligence (AI) and Natural Language Processing (NLP) have enabled the extraction of financial signals from unstructured textual data such as corporate news, disclosures, and media releases [Ersan et al., 2021].

Over the past decade, Corporate Social Responsibility (CSR) and ESG practices have attracted growing attention from regulators and market participants. Financial research has long examined how asset prices respond to non-financial information, often within the debate between the Efficient Market Hypothesis (EMH) and behavioral finance. According to the EMH, all publicly available information should be instantly reflected in security prices [Fama, 1970b], yet empirical evidence reveals frequent underreactions and overreactions to new information. Behavioral finance offers an alternative explanation by emphasizing cognitive biases and investor sentiment as key drivers of such anomalies [Barberis and Thaler, 2003; Kahneman, 2003; Shefrin, 2008]. Qualitative and emotionally charged information, such as ESG news or corporate announcements, can therefore trigger heterogeneous interpretations among investors, producing short-term mispricing and temporary inefficiencies.

A growing number of studies confirm that non-financial and text-based information, from media to ESG disclosures, significantly affects investor behavior [Ángeles López-Cabarcos et al., 2020; Nyakurukwa and Seetharam, 2023a]. At the same time, the rise of sustainable investing has placed ESG considerations at the center of financial decisions, supported by regulatory initiatives and societal expectations [Compact, 2004; Schmidt, 2019]. While ESG ratings by agencies such as MSCI, Bloomberg, and FTSE Russell have become widespread, they rely heavily on firm-reported data and differ substantially across providers, raising concerns about transparency and comparability [Duque-Grisales and Aguilera-Caracuel, 2021; Chatterji et al., 2016]. To overcome these limitations, recent research has focused on ESG-related news as a more immediate and objective information source. Unlike ESG scores, typically updated on an annual or semi-annual basis, ESG news offers real-time signals about firm behavior and stakeholder perception. Empirical evidence shows that ESG news can influence short-term market activity: positive environmental or social developments often generate mild price appreciation, whereas negative governance news tends to provoke stronger adverse reactions [Flammer, 2013; Capelle-Blancard and Petit, 2019]. However, most existing studies operate at a daily frequency and therefore cannot capture the rapid adjustment dynamics typical of high-frequency trading environments.

This study advances the literature in three main directions. First, it extends research on the financial relevance of ESG information by focusing on real-time news rather than low-frequency ESG scores, which may suffer from self-reporting bias, methodological inconsistency, and delayed disclosure. This complements prior evidence showing that ESG activities affect firm-level financial outcomes [Gregory, 2022]. Second, it contributes to behavioral finance by examining whether investors react asymmetrically to positive and negative ESG news, thereby clarifying the role of sentiment in short-horizon price formation. Third, it introduces an intraday, AI-based analytical framework that combines large language models with high-frequency market data, providing a more timely perspective on how non-financial information is incorporated into prices, an aspect not explored in prior studies such as [Capelle-Blancard and Petit, 2019].

To address these questions, we analyze the immediate market response to firm-specific ESG news using an intraday event-study methodology. ESG headlines for the 30 Dow Jones Industrial Average (DJIA) firms are classified with GPT-4 Turbo by ESG pillar¹ and

¹Pillar-level effects need not be homogeneous: [Staněk Gyönyör and Horváth, 2024] show that Environmental and Social scores can reduce market dependence during downturns. In contrast, Governance can push in the opposite direction. This divergence supports decomposing ESG signals by pillar when studying

sentiment, allowing us to evaluate whether ESG disclosures generate measurable abnormal returns and abnormal trading volumes within minutes of their release. This empirical design enables a direct assessment of whether ESG information is incorporated efficiently or whether short-term frictions affect the speed and direction of price adjustments.

The theoretical foundation for this approach is provided by the Inelastic Market Hypothesis (IMH), which posits that short-term price movements reflect not only new information but also order-flow imbalances in markets where supply and demand are not perfectly elastic [Gabaix and Koijen, 2021a; Bouchaud, 2022b]. In such environments, even trades unrelated to fundamentals can move prices because investors adjust portfolios only partially, and liquidity replenishes slowly. Latent Liquidity Theory [Donier et al., 2015; Benzaquen and Bouchaud, 2018] complements this view by showing that investors revise their reservation prices continuously in response to news and price changes, making realized prices the outcome of aggregated order flow rather than purely fundamental valuation.

This microstructural perspective provides a natural rationale for why ESG disclosures, often qualitative, incomplete, or ambiguous, can nonetheless generate short-term price and volume reactions. News releases act as coordination signals that induce heterogeneous interpretations and trigger algorithmic or institutional trading, producing temporary liquidity imbalances. These dynamics, consistent with IMH predictions, imply that market reactions to ESG news reflect both informational content and structural inelasticity². This interpretation aligns with recent findings that ESG headlines can generate reputational effects with short-term market consequences [Zhou and Ge, 2025].

Within this framework, we examine whether short-lived trading imbalances following ESG disclosures are associated with abnormal returns and abnormal volume at the minute level. By classifying each headline by both ESG dimension and sentiment, we capture differential market responses across types of ESG events and across the polarity of the underlying information. This dual-classification approach provides a more granular understanding of how ESG-related announcements influence trading behavior in an inelastic market environment.

This study is not intended to test whether ESG considerations affect asset prices in general, a question that has been extensively addressed in the existing literature. Rather, it focuses on the intraday information-processing mechanisms through which firm-specific ESG news is incorporated into prices and trading activity. By exploiting minute-level data and precise timestamps of news releases, the analysis isolates short-horizon market adjustments. It distinguishes between rapid absorption, delayed reaction, and effective irrelevance across ESG dimensions. The contribution of the paper, therefore, lies in identifying how quickly and through which market channels ESG-related information is processed, rather than in documenting long-run valuation effects.

The remainder of the paper is structured as follows: Section 1.3 presents a review of the relevant literature and develops our main hypotheses. Section 1.4 outlines the research design, data sources, and methodology. Section 1.6 presents the empirical results. Section 1.7 discusses the findings. Finally, Section 1.9 concludes the paper.

market reactions.

²ESG news provides a particularly suitable setting for this analysis, as it often conveys soft, qualitative, and non-standardized information, making short-horizon price adjustment especially sensitive to market depth, trading frictions, and inelasticity.

1.3 Literature Review and Hypotheses Development

In recent decades, the growing accessibility of online information and the rapidly evolving financial landscape have driven an unprecedented demand for data. Consequently, market participants have progressively shifted from relying solely on traditional financial data to incorporating information from news announcements. Since the early 1990s, news volume has increased significantly, and the work of [Mitchell and Mulherin, 1994] stands out as one of the first to test the relationship between daily news frequency and aggregate market activity. Recent technological developments have allowed scholars to extend traditional performance measures by integrating insights from linguistics, machine learning, and behavioral finance. This interdisciplinary approach has supported including non-financial information, such as firm-specific ESG-related news, as a relevant driver of firms' financial performance.

Investors have increasingly acknowledged the role of non-financial information in influencing stock market dynamics. Interest in media coverage has grown, emphasizing its relevance in investment decisions. According to the *information view* and the *salience view*, media coverage plays a key role in shaping investor behavior [Nyakurukwa and Seetharam, 2023b]. The former suggests that the media reduces information costs, leading to better investment decisions and lower asymmetry. At the same time, the latter highlights that media-exposed stocks attract more attention and, consequently, higher demand [Solomon et al., 2014]. The salience view challenges the EMH, suggesting that investors can act on media information before it is fully incorporated into prices.

At the same time, ESG issues reflect a firm's engagement with stakeholders, employees, customers, communities, and the broader environment and society [Xu et al., 2022]. As a result, ESG factors have become integral to firms' long-term strategies and performance metrics. Effective communication of ESG commitments has become crucial to improve ESG performance and investor perception. [De Vincentiis, 2024] suggest that ESG-related disclosure may be strategically timed in response to market conditions: firms are less likely to disclose negative ESG news when return volatility is high, and may modestly increase disclosure of positive ESG news in such periods. This disclosure-timing channel implies that the observed market reaction to ESG news can vary systematically with short-horizon market states.

Professional data providers such as Thomson Reuters, Bloomberg, MSCI, and FTSE have developed ESG scores to assess ESG practices. However, these metrics have faced criticism, mainly due to their reliance on firm-disclosed data, which can introduce self-reporting biases [Xu et al., 2022]. Several studies point out the lack of consistency among rating agencies, which often assign divergent scores to the same company [Avramov et al., 2022; Berg et al., 2022; Billio et al., 2021]. [Liu et al., 2024] show that ESG rating disagreement is associated with higher idiosyncratic return volatility, consistent with greater uncertainty and heterogeneous beliefs around sustainability signals. Consequently, ESG news has emerged as a more objective and timely alternative for assessing ESG performance [Xu et al., 2022]. Recent evidence further highlights that ESG-related headlines can influence investor perception through media-driven reputational channels, shaping short-term market reactions [Zhou and Ge, 2025]. Related research shows that ESG activities also have measurable consequences for firms' internal financial outcomes, such as cash flow generation [Gregory, 2022].

Many ESG studies rely on annual ESG scores, which are more suitable for long-term investment perspectives. ESG news, by contrast, is available at higher frequencies and provides more immediate signals. Yet, institutional and retail investors have only partially exploited ESG-related news in investment processes. While this literature provides extensive evidence on the pricing and trading implications of ESG information at daily or longer

horizons, relatively little is known about how such news is processed at intraday frequencies. In particular, existing studies offer limited insight into the timing, speed, and channels through which ESG-related disclosures are incorporated into prices and trading activity immediately following their release.

Today’s investment environment is increasingly shaped by the high speed at which information circulates. Investors often revise their positions instantly, potentially reacting irrationally to rapidly emerging news [Schmidt, 2019]. This behavior aligns with both the EMH [Fama, 1965] and insights from behavioral finance [Tetlock, 2007; Carretta et al., 2011], which recognize the impact of information flows on asset prices. Although the literature presents mixed findings, various studies support a significant influence of non-financial news on firm valuation. For instance, [Klassen and McLaughlin, 1996] investigate NYSE and Amex firms between 1985 and 1991 and observe positive returns (0.82%) after favorable environmental events and negative returns (1.5%) following adverse ones, especially in firms with weak environmental management.

Similarly, [Wang et al., 2014] find positive (negative) stock price reactions to positive (negative) ESG events. [Krüger, 2015] report a pronounced negative response to negative ESG news, while the reaction to positive news is weaker. [Capelle-Blancard and Petit, 2019] analyzes ESG news items related to 100 listed companies between 2002 and 2010, using data from *Covalence EthicalQuote*. Their findings indicate a drop in market value (about 0.1%) following negative announcements but no systematic gain after positive news. By contrast, [McWilliams and Siegel, 2000] argue that the costs and benefits of ESG actions may offset each other in equilibrium, leading to no net effect on stock returns.

Analyzing ESG news by its three main components, Environmental, Social, and Governance, yields mixed evidence. These differences stem from varying investor sensitivity to each category. For instance, [Pedersen et al., 2021] conclude that governance news has the strongest market impact, challenging [Andrews and Caren, 2010], who highlight broader attention to environmental and social issues. [Ionescu et al., 2019] show asymmetric effects of ESG scores: social scores correlate positively with market capitalization, while governance scores show a negative relationship; environmental scores have limited significance.

Further studies examine the impact of ESG news sentiment. While several authors confirm the market’s sensitivity to negative ESG news [Krüger, 2015], positive news yields more inconsistent results. [Xu et al., 2022] report limited or conditional impact of positive news, depending on stakeholder engagement. [Serafeim and Yoon, 2023] find that only financially material ESG news generates significant price reactions, positive if the news is favorable. Conversely, [La Torre et al., 2020] test 46 Euro STOXX 50 companies and find no evidence that ESG objectives improve stock performance.

Methodologies used to assess the market’s reaction to ESG news vary widely. [Serafeim and Yoon, 2023] employ firm-day panel regressions; [Sabbaghi, 2022] use EGARCH models to assess the impact of ESG news on volatility. Additional evidence from intraday-oriented volatility studies confirms that ESG news can generate measurable high-frequency effects in financial markets [Sabbaghi, 2022]. Our approach aligns with the stream of research using the intraday event-study methodology, including [Capelle-Blancard and Petit, 2019] and [de Vincentiis, 2023].

Event studies remain a foundational tool in empirical finance, stemming from the seminal works of [Fama et al., 1969] and [MacKinlay, 1997]. While many studies adopt daily or monthly horizons, longer windows (e.g., annual) are also common [Kothari and Warner, 1997]. However, one limitation of this approach is the potential for confounding events to bias results [De Jong and Naumovska, 2016]. Shortening the event window mitigates this issue, especially when using intraday data, which has become increasingly available.

Since the 1980s, the literature has applied intraday event studies to returns, volumes, and spreads. For instance, [Patell and Wolfson, 1984] observe that price reactions to earnings and dividend announcements occur primarily within the first 30 minutes post-release. [Lee, 1992] examine abnormal trading volumes by investor type, and [Lee et al., 1993] find that bid-ask spreads widen before announcements and persist afterward, though this effect diminishes when controlling for volume. Similarly, [Brooks et al., 2003] use 15-minute intervals to study unanticipated events, finding elevated spreads and volumes lasting over an hour.

1.3.1 Hypotheses Development

As anticipated in the previous section, this paper focuses on the impact of firm-specific ESG-related news on trading volumes and stock returns. The main reason is that, based on more than 60 years of literature, trading volume activity has been confirmed to play a crucial role in financial markets, as it can inform investors about future stock prices and price volatility. According to [Atiase and Bamber, 1994], who explore the relationship between trading volumes and stock prices, the former is more powerful in uncovering the market's reactions to accounting disclosures. Regarding trading performance, [Blume et al., 1994] detect more profitable trades when market operators employ trading volume in their technical analysis. Surprisingly, even observing changes in trading volumes can provide valuable information about future volatility [Lamoureux and Lastrapes, 1990]. Therefore, as mentioned in the previous section, though it is true that investors' beliefs are directly unobservable, they can be gauged by inspecting changes in trading volume activity as traders revise their beliefs according to the information contained in trading volume changes. In particular, the closer the announcement time, the wider the gap generated by the information asymmetry. Consequently, the unit increment in the information retrieved becomes more valuable.

The literature proposes a dual approach, depending on the nature of the announcement: scheduled or unscheduled.³ Overall, the general finding is that trading volume increases (decreases) before scheduled (unscheduled) news. In the case of scheduled announcements (e.g., earnings announcements), scholars detect a decrease in trading volume because of adverse selection between informed and uninformed traders [Foster and Viswanathan, 1990; Krinsky and Lee, 1996; Chae, 2005]⁴, with a higher rate of decrease in trading volume, the closer we get to the announcement time. [Wang, 1994] finds corroborating results and explains that private information loses value as the announcement date approaches, given that it becomes public, thus eroding the advantage of informed traders. In contrast, in the case of unscheduled announcements, [Bamber, 1987], [Kim and Verrecchia, 1991], and [Chae, 2005] assess an increase in trading volume, since uninformed traders are not aware of the private information held by informed traders, and therefore keep trading with them.

Other authors exploit other factors to explain the response of trading volume to news announcements: some propose investors' confidence and reliance on private information as the drivers of changes in trading activity around earnings disclosure [Liang, 2003]; others support that trading activity is influenced by macroeconomic variables (e.g., interest rates, GDP announcements, or unemployment rate disclosure), and fluctuates during weekdays and holidays [Chordia et al., 2000].

³The difference between scheduled and unscheduled announcements lies in public disclosure: if the release time is publicly known, then the news is categorized as scheduled. Otherwise, it is considered unscheduled. In the case of unscheduled news, every bit of information acquires greater value, and informed traders benefit from the asymmetric information in the market.

⁴Uninformed traders postpone their trading activity to avoid suffering losses to informed traders, who instead increase their trading demand.

Considering the extensive works present in the literature, and since firm-specific ESG news can be regarded as both scheduled and unscheduled news, it is reasonable to question the impact of such news on trading volume. Hence, the first hypothesis of our study reads as follows:

HP. 1: *Firm-specific ESG news significantly impacts the stock’s trading volume over the intraday investment horizon.*

Once we tested the existence of a relationship between traded volumes in correspondence with ESG news announcements, our focus switched to the investigation of the direction of such trades. In recent years, the growing salience of ESG factors has contributed to increasing the popularity of ESG news [Capelle-Blancard and Petit, 2019]. Numerous academic papers have examined stock market reactions to information in ESG news over the last decades. Notwithstanding, it remains an unresolved research question. On one side, several quantitative methodologies (focusing mainly on event-study methods) have been proposed but have resulted in mixed outcomes. Conversely, qualitative methods, such as surveys (with meta-analysis), have also been applied, highlighting a positive relationship between positive CSR and firm performance [Wang et al., 2015].

For this reason, the second hypothesis we test is as follows:

HP. 2: *Firm-specific ESG news significantly impacts the stock’s financial performance over the intraday investment horizon.*

Previous studies in behavioral economics have extensively analyzed how markets react to ESG news and its sentiment. More in-depth, in behavioral finance, a famous proclamation by two renowned academics in the field states *losses loom larger than gains* [Kahneman and Tversky, 1979]. This concept, known as “loss aversion”, posits that experiencing the loss of a specific monetary amount is psychologically almost twice as impactful as gaining an equivalent amount. Framing this concept around the realm of news reflects the emotional impact of adverse events, offsetting and outweighing the emotional impact of positive events. Therefore, in the context of news announcements, it is reasonable to presume that negatively classified ESG-related news exerts a more significant effect than positive news.

Thus, our third hypothesis refers to the asymmetric impact of news depending on the sentiment classification:

HP. 3: *Negatively classified firm-specific ESG news has a larger impact on a firm’s financial performance than positively classified firm-specific ESG-related news.*

While HP2 relates to intraday price discovery following ESG news, HP3 focuses on trading activity, liquidity adjustment, and the sentiment expressed in the news coverage, not merely the fact that an ESG event was reported, thereby capturing complementary, though not necessarily coincident, dimensions of short-horizon market response.

1.4 Research Design

In this section, we present the standard and the alternative event study methodology proposed in this study. In general, this approach is used to test the reaction of a metric of interest (i.e., trading volume and stock prices) to specific events (i.e., firm-specific ESG news in our case). For clarity, we first present the classical event study approach, even though it relates to our second and third hypotheses, and subsequently discuss the alternative trading volume event study, which instead relates to our first hypothesis.

1.4.1 Standard Event Study Methodology

To understand the intraday stock market dynamics (i.e., trading volume and stock returns) in response to the publication of firm-specific ESG-related news, this study grounds the approach of [Calomiris and Mamaysky, 2019] and extends the event study methodology using intraday data. The event study methodology investigates stock price reactions around corporate events (i.e., news, earnings announcement, dividends announcement, M&A, divestitures, disinvestment, or new product launches) since the publication of the seminal work by [Ball and Brown, 2013] and [Fama, 1970b]. Event studies have become a predominant method to evaluate the effect of a news announcement on stock market prices [Kothari and Warner, 2007]. The advantages of event studies methodology are that: *i*) it evades issues of manipulation of financial performance based on accounting data [Endrikat, 2016]; *ii*) it overcomes endogeneity problems proper of multivariate analysis, thus facilitating the study of the causality between occurrences and market responses [Endrikat, 2016], and *iii*) it lowers reverse-causality in the correlation between CSR and accounting measures of firm performance [Capelle-Blancard and Petit, 2019]. Moreover, the event studies approach is also used to assess the relationship between environmental performance and a firm's financial performance [Endrikat, 2016; Fisher-Vanden and Thorburn, 2011; Jacobs et al., 2010]. Grounded on the efficient market hypothesis [Fama, 1970b], event studies aim to assess shareholders' reaction by testing how much the actual return deviates from the expected value over a specific period (*event window*), which is known as *abnormal return* (AR).

Thus, abnormal returns are calculated as follows:

$$AR_{jt}^i = R_{jt} - \mathbb{E}[R_{jt}] \quad (1.1)$$

where AR_{jt}^i measures investors' reaction to the event i related to firm j at time t , R_{jt} is the return of stock j at time t , and $\mathbb{E}[R_{jt}]$ is the observed return (or expected return) calculated using the market model [MacKinlay, 1997], which is the most widely adopted method [Ball and Brown, 2013; Fama, 1970b]. In the market model, firm returns (R_{jt}) are regressed on stock market index returns at time t (i.e., R_{mt}) to get the estimates of the desired parameters $\hat{\alpha}_j$ and $\hat{\beta}_j$ using OLS as follows :

$$\mathbb{E}[R_{jt}] = \hat{\alpha}_j + \hat{\beta}_j R_{mt} \quad t = 1, 2, 3, \dots T \quad (1.2)$$

where $\hat{\alpha}_j$ is the estimated intercept term, representing the average stock return that is not explained by market movements, and $\hat{\beta}_j$ is the estimated slope term, corresponding to the sensitivity of the j -th stock return to the market return, R_{mt} .

The parameters α_j and β_j are estimated over a 390-minute window [-410,-20] using Ordinary Least Squares (OLS), starting from 20-trading minutes before each event. We opt for this estimation window to have a 1-day beta estimation to control for the presence of asynchronous beta or missing estimations. We propose seven event windows, four of them considering the pre-and post-event period (i.e., [-10, 10] minutes, [-20, 20] minutes, [-20, 40] minutes, [-20, 60] minutes), and three including the post-event period starting from the event itself (i.e., [0, 10] minutes, [0, 30] minutes, and [0, 60] minutes). These intervals allow for robustness tests.

Then, according to Eq. 1.1 and Eq. 1.2, abnormal returns are estimated as follows :

$$AR_{jt}^i = R_{jt} - \hat{\alpha}_j - \hat{\beta}_j R_{mt} \quad (1.3)$$

Since the effect of a particular event can propagate along different periods following the event date, we aggregate abnormal returns over the usual $(2N + 1)$ -minute periods

around the event, thus calculating the Cumulative Abnormal Returns (CAR^i), to evaluate the cumulative effect of an event over a specific event window ranging from $[-N, +N]$. This procedure allows us to detect the presence of possible information leakage (for the N minutes before the event), as well as the minutes of and after it. Calculating the Cumulative Abnormal Returns ($CARs$), we assess the total impact of an event over a time frame, that is, the event window:

$$CAR_{jt}^i[-N, +N] = \sum_{\tau=t-N}^{t+N} AR_{j\tau}^i \quad (1.4)$$

To test HP2 and HP3, we first separate firm-specific ESG news into Environmental, Social, and Governance, and then again into Positive and Negative firm-specific ESG-related news. Then, for each category of pillar and sentiment, we compute the Cumulative Average Abnormal Returns (CAAR) across the events to find the average impact of each firm's related events over a time interval as follows :

$$CAAR_{[-N, +N]} = \overline{CAR^i}[-N, +N] \quad (1.5)$$

where $\overline{CAR^i}[-N, +N]$ is the average CAR value over all the events ($i = 1, \dots, K$)

One of the main weaknesses of traditional event studies when testing the hypothesis is the assumption of normality in the data distribution and the absence of serial correlation. Since we use intraday data, the distribution of returns tends to deviate from the normality assumptions, instead exhibiting features such as fat tails and skewness. To overcome these issues and safely test the hypotheses of CAAR significant from zero, we use the Wilcoxon-Signed Test, a non-parametric test for medians in non-normality samples [Kaspereit, 2022].

1.4.2 Trading Volume Event Study Methodology

Research in finance has extensively explored the reaction of trading volume to the inflow of new information in the market. Volume event studies apply the general principles of the standard event study methodology to a time series of trading volumes and typically employ a single-index market model borrowed *mutatis mutandis* from abnormal returns event studies. Subsequently, log turnover of trading volume is calculated to explain individual securities' log trading volume turnover, and predicted residuals are used to test for zero mean abnormal trading volume within some specified event windows [Karafiath, 2009]. In favor of volume event studies, [Ajinkya and Jain, 1989] report that “*Although a well-developed economic theory such as Capital Asset Pricing Model (CAPM) for returns does not seem to exist for trading volume, several theoretical papers linking trading volume to information releases can be used to motivate a trading volume market model.*” [Ajinkya and Jain, 1989, p. 334].

[Chae, 2005] employs an event study to evaluate trading volume changes around announcements. In the US, volume event studies are increasingly used in security fraud litigation cases. The main difference between volume and return event studies is that returns are replaced by the log-transformed relative volume per firm [Campbell and Wasley, 1996], namely:

$$V_{jt} = \log \left(\frac{q_{jt} \cdot 100}{S_{jt} + 0.000255} \right), \quad (1.6)$$

where q_{jt} is the number of shares traded for firm j on minute t and S_{jt} is the outstanding shares of firm j on the whole trading day⁵. The constant 0.000255 in the above equation is

⁵Since we cannot access intraday data on shares outstanding, we use daily data for this measure

added to avoid a log transformation of zero in the case of zero trading volume on a given day. The resulting measure is the trading volume turnover, which controls for the company size effect by scaling the raw volume (q_{jt}) by the number of shares outstanding (S_{jt}) of the specific trading day. This adjustment enriches the robustness of our estimates, thus validating the interpretation of outcomes.

Once again, we use the market model to estimate the expected trading volume over the estimation window. Because the market-model involves more demanding data requirements, we adopt the mean-adjusted model as the default, in line with [Campbell and Wasley, 1996].

The market-model abnormal trading volume is therefore defined as

$$ATV_{jt}^i = V_{jt} - (\hat{\alpha}_j + \hat{\beta}_j V_{mt}), \quad (1.7)$$

where V_{jt} denotes the trading volume of stock j at time t , while V_{mt} being the average market volume of all traded firms. More specifically,

$$V_{mt} = \frac{1}{M} \sum_{j=1}^M V_{jt} \quad (1.8)$$

where M is the number of stocks traded on day t . This equal-weighted average follows provides an estimate of the latent market-wide trading activity factor. Note that, given the presence of strong right-skewness and heteroskedasticity of trading volume, we then consider the logarithmic version of trading volume for the rest of the analysis.

To test HP1, we first separate firm-specific ESG-related news into Environmental, Social, and Governance. Then, for each pillar, we compute the Average Abnormal Trading Volumes (hereafter AATV) per minute across events of each pillar to find the average impact of each firm's related events over a time interval. For each minute, AATV across all events is computed as follows:

$$AATV_\tau = \overline{ATV^i}[-N, +N] \quad (1.9)$$

Both event study analyses are carried out using the statistical software *STATA 18* using the *eventstudy2* package developed by [Kaspereit, 2022].

1.5 Language Model Selection

Language modeling dates back to the 1950s, with Claude Shannon's first application of information theory to human language. Over the decades, interest in large language models (LLMs) has evolved through several waves [Das et al., 2025], culminating in the current large-scale adoption of Large Language Models. Early research and development efforts on Statistical [Jurafsky and Martin, 2000] and Neural Language Models [Bengio et al., 2003] resulted in, albeit performant, narrow and task-specific practical applications. The introduction in 2017 of a novel deep learning architecture, the Transformer [Vaswani et al., 2017], along with the release of Transformer-based pre-trained language models (PLMs) such as BERT [Devlin et al., 2018] and GPT [Radford et al., 2018] marked clear performance improvements across-the-board benchmarks [Wang et al., 2019].

The current phase, often referred to as the era of Large Language Models (LLMs), is enabled by the scaling of such Transformer-based models in terms of size (from millions to billions of parameters) and the development of sophisticated data curation and post-training techniques. With such a scaling, LLMs are found [Kaplan et al., 2020] to be able to perform

very well in the so-called *zero/few-shot* instruction-following scenario: that is, they do not require explicit, supervised adaptation on a specific downstream task; instead, the task itself can be specified as input during inference, along with no (*zero-*) or some (*few-*) examples (*shots*).

Thus, in preliminary experiments, we evaluated two language modeling approaches for our scenario of interest (ESG pillar classification from financial news headlines): *i*) the first, using a publicly available BERT-based PLM, namely FinBERT [Huang et al., 2023] finetuned for ESG pillar classification,⁶ which can conveniently be run on consumer-grade hardware; *ii*) the second, resorting to a large proprietary LLMs (namely, OpenAI’s GPT-4)⁷ to perform the classification task relying on its instruction-following capabilities; in particular, we instruct GPT-4 to output both pillar and sentiment in a single inference pass. [Costola et al., 2023] use machine-learning sentiment extraction to study how complex news flows map into market reactions. In particular, ML-based sentiment measures applied to COVID-19 news explain cross-sectional variation in stock-market responses, highlighting the value of systematic text-based measurement in high-noise environments.

To identify the best-performing classification approach for our scenario, we built a manually annotated dataset by asking human experts to assign labels to text following predefined annotation guidelines. Although this method is labor-intensive, time-consuming, and subject to potential human bias, it remains the most reliable approach for small datasets and tasks requiring high semantic precision. To this end, we select a stratified 10% random subsample from the full dataset (138 news headlines out of 1,381). After that, the annotation process is conducted by two trained research assistants with prior exposure to ESG concepts as follows: each annotator was independently requested to read the headline of every article and assign each one of them to one (and one only) out of the following four mutually exclusive labels: *E* (Environmental), *S* (Social), *G* (Governance), or *O* (Other). The annotation guidelines include label definitions as well as illustrative examples. Finally, disagreements are resolved through discussion and, when necessary, adjudicated by the annotators. This technique produces high-quality and internally consistent labels that serve as the ground-truth benchmark for preliminary evaluation of model performance.

We evaluate the inter-annotator agreement using Cohen’s κ coefficient [Cohen, 1960; Artstein and Poesio, 2008], which quantifies the extent to which two annotators agree beyond what would be expected by chance; its range is in the interval $[-1, 1]$, with the extremes -1 and 1 indicating complete disagreement and complete agreement, respectively, and a value of 0 indicating agreement purely due to chance. The resulting Cohen’s κ from our data annotation effort is 0.44, indicating moderate agreement.

We then evaluate the performance of GPT-4 and FinBERT, analysing the obtained F1 performance on the annotated data; F1 is defined as the harmonic mean between *precision* and *recall*, and is preferred to accuracy due to its robustness to data imbalance. More in detail, *Precision* is defined as the proportion of positive identifications that are actually correct (True Positives, TP, as opposed to False Positives, FP), and it is computed as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (1.10)$$

Conversely, *Recall* is defined as the proportion of actual positives that were correctly identified (as opposed to False Negatives, FN, indicating actual positives that were erro-

⁶Publicly available at <https://huggingface.co/yiyanghkust/finbert-esg>.

⁷Available under checkpoint id `gpt-4-1106-preview`.

Model	F1					Precision					Recall				
	E	S	G	O	μ	E	S	G	O	μ	E	S	G	O	μ
Majority	0	0	.6	0	.26	0	0	.43	0	.18	0	0	1	0	.43
FinBERT	.71	.53	.42	.32	.46	.73	.39	.95	.29	.64	.69	.82	.27	.35	.47
GPT-4	.82	.64	.81	.52	.7	.7	.68	.72	1	.77	1	.61	.92	.35	.73

Table 1.1: Performances for GPT-4 and FinBERT on ESG pillar classification; Majority indicates the baseline; μ reports the weighted average for the corresponding metric. The numbers in bold indicate the best performance result among the three models.

neously predicted as negatives), and it is computed as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (1.11)$$

Thus, the *F1 score* is computed as

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1.12)$$

Table 1.1 reports the results comparing the performance of each LLM-based classifier against the human-annotated ground truth. As a baseline, we include a majority classifier that always predicts the most frequent class in the training data. GPT-4 shows the strongest overall performance, achieving consistently high F1 across ESG pillars (ranging within [.522, −.821]). Conversely, FinBERT performs moderately well but exhibits inconsistent behaviour across categories.

Given these results, we select GPT-4 for subsequent analyses, as it not only demonstrates superior performance but also allows us to obtain both ESG Pillar Classification and Sentiment labels in a single inference pass.

Within the context of this study, the use of GPT-4 should be interpreted primarily as a measurement device rather than as a technological contribution *per se*. ESG news is inherently noisy, semantically heterogeneous, and prone to classification disagreement, a well-documented issue in the ESG literature. The dual-classification framework adopted in this paper, distinguishing ESG content and sentiment, leverages the language model to extract a consistent latent representation of ESG-related information from unstructured text, thereby reducing subjective coding and measurement error. From an empirical perspective, this approach allows ESG news to be treated as a structured informational shock suitable for high-frequency event-study analysis. [Cini and Ferrari, 2025] demonstrate that machine-learning methods can meaningfully improve ESG measurement, predicting future ESG rating classes from accounting and risk variables with high accuracy. This supports the broader view that computational approaches can reduce noise in sustainability signals and improve empirical identification.

1.6 Experimental Setup

1.6.1 The Sample

The dataset we use comprises two separate data collections: one for extracting firm-specific ESG-related news headlines and another for retrieving DJIA components’ financial data on intraday stock prices. Our data extraction pipeline is implemented in Python.

First, we collect the headlines of ESG-News announcements from January 1st to March 31st, 2023. These headlines are about the 30 listed firms in the Dow Jones Industrial Average

1.6. Experimental Setup

GICS Sector	Frequency
Basic Materials	1
Energy	1
Communication Services	2
Consumer Cyclical	3
Consumer Defensive	3
Industrials	4
Financial Services	5
Healthcare	5
Technology	6
Total	30

Table 1.2: Sample firms by economic sector according to the GICS Classification Standards. Data are provided by LSEG Eikon for the 30 DJIA constituents over the period January 1st- March 31st, 2023.

index, and are obtained via the News Monitor App provided by the *LSEG Eikon* platform using the selected period at the exact date-time of their release (with seconds precision). The Eikon financial tool engine allows filtering for the specific asset classes, periods, languages, and topics; therefore, the following query is entered: `R:[STOCK NAME] AND LAN:ENGLISH AND Topic:ESG`, where `R` is the standard way to redirect the software towards a specific list of assets coming after it, `LAN:ENGLISH` is used only to get news in English. In contrast, `Topic:ESG` is needed to get news and research relating to a company’s Environmental, Social, and Governance standards used by ethical investors to screen their investments. The available news sources span from News Wires to Global Press and Web News, including Breaking News Alerts, Research, Video, Filings, Transcripts, and Press Releases.

In total, the extraction outputs 1,791 ESG-related news headlines. Duplicate news related to the same event, poorly specified items, and news co-occurrences for the same stock on the same date are dropped from the sample, leaving 1,381 ESG-related news headlines (77% of the initial sample). This means roughly 460 ESG-related news items per month or, on average, 15.34 daily news items per firm. The total number of news per firm ranges between 2 and 163, with a mean of 47.2 and a standard deviation of 44.17.

Data on the financial performance of the 30 firms listed in the Dow Jones Industrial Average Stock Market Index (DJIA) and the raw trading volumes and outstanding shares also come from the LSEG Eikon platform. We collect data from the LSEG Eikon platform through the Eikon API Key for the selected period. This dataset contains evenly spaced intraday trading volumes, the 30 DJIA components stock prices, and the Stock Market Index of reference (the DJIA). Our study focuses only on “open-market sessions” of each trading day in the selected period, thus excluding transactions from the “pre-market” or “after-market” trading hours.

The DJIA consists of a price-weighted average of 30 actively traded blue-chip stocks, primarily industrials, covering stocks also traded on the NYSE. The study is limited to the DJIA Index, a valuable indicator of the largest US companies’ share performance.

Table 1.2 displays the selected sample of firms according to the GICS Sector Classification by S&P Dow Jones Indices and MSCI. Despite comprising only 30 stocks, the DJIA is a value-weighted, diversified stock market index, with Technology (6), Healthcare (5), and Financial Services (5) among the sectors with the highest share. This indicates that such sectors significantly influence the DJIA’s performance during the period. Meanwhile, Consumer Cyclical (3), Consumer Defensive (3), and Industrials (4) share a moderate presence in the DJIA. Sectors with the lowest representation include Communication Services (2), Basic Materials (1), and Energy (1).

Recently, several classification techniques have been proposed in the literature. On the

Sentiment	ESG Type				Frequency
	(E)	(S)	(G)	(O)	
Positive	167	142	56	20	28.46%
Negative	18	109	403	7	37.92%
Neutral	40	32	271	132	33.62%
Total	225	280	720	156	1,381
Frequency	16.29%	20.28%	52.14%	11.30 %	100.00%

Table 1.3: Number of ESG news classified as Positive, Negative, or Neutral for DJIA constituents from January 1 - March 31, 2023. Data are from LSEG Eikon’s “News Monitor App”, and the GPT-4 Turbo LLM performs the classification

one hand, [Zhao et al., 2018] classify ESG News based on a scoring method involving 38 aspects extrapolated from CSR and ESG reports published by firms. On the other hand, [Nugent et al., 2021] use “word distribution” to identify 3-5 topics for each ESG category [Xu et al., 2022].

Our study employs GPT-4 Turbo for classification purposes, a renowned, cutting-edge generative LLM from the field of NLP. Specifically, we instruct the model to classify our news sample as either Environmental (*E*), Social (*S*), Governance (*G*), or Other (*O*), as well as belonging to Positive (*P*), Negative (*N*), or Neutral(*NE*) sentiment. More in detail, the following prompt is used to instruct the LLM:

You will be provided with ESG news headlines. Your task is to classify whether the article falls within an ESG category, namely Environmental (class E), Social (class S), Governance (class G), or Other (O), with the associated score (ESG_Score). Also, you should classify the headline sentiment as Positive, Neutral, or Negative, with the associated score (Sentiment_Score). Your output should be in JSON format exclusively with the following keys: “ESG_Class”, “ESG_Score”, “Sentiment”, and “Sentiment_Score”. Both scores should range between 0 and 1 for the selected class.

An example of the classification by GPT-4 Turbo is in Table A.3 in Appendix A. Table 1.3 presents the distribution of ESG news headlines by ESG category (Environmental, Social, Governance, and Other) and sentiment (Positive, Negative, or Neutral).

Hence, the first dataset about ESG-related news headlines contains information about each ESG announcement’s date and time, the stock it refers to, the ESG category, and its associated sentiment. Table 1.3 provides the distribution of ESG news per ESG category and Sentiment. Regarding the ESG pillar, most news is classified as Governance (52.14%), while Environmental and Social news share similar frequency distributions. Meanwhile, the classification of firm-specific ESG-related news by sentiment depicts a reasonably uniform distribution of the news.

Our second dataset contains two financial metrics with intraday frequency: *i*) trading volumes and *ii*) stock prices. The choice to use a modified version of the trading volume instead of the raw measure lies in the extreme kurtosis (35,205.94) and extreme positive skewness (134.90), which indicate that the time series of raw volume is non-normally distributed. For this reason, the raw volume turnover is replaced by the logarithmic transformation, thus leading to the Trading Volume Turnover, whose calculation is extensively discussed in the previous section. This correction exhibits a reduced kurtosis of 8.57 and a reduced skewness of 2.15. Intraday log returns for each stock are used for the empirical analysis.

Table 1.4 and Table 1.5 provide summary statistics for the trading volume turnover and

1.6. Experimental Setup

Sector	Obs	Mean	Std. Dev.	Min	Max
Basic Materials	24,112	-7.7025	0.4961	-8.2737	-0.2988
Communication Services	48,112	-7.8767	0.3833	-8.2741	-0.7728
Consumer Cyclical	72,319	-7.9062	0.4031	-8.2739	-1.3541
Consumer Defensive	72,336	-8.0311	0.3125	-8.2741	-1.7210
Energy	24,118	-7.8576	0.3323	-8.2734	-3.4796
Financial Services	118,590	-7.8591	0.4601	-8.2740	-0.5130
Healthcare	120,376	-7.8085	0.4794	-8.2740	-0.1796
Industrials	96,326	-7.7464	0.4896	-8.2736	-0.4923
Technology	145,055	-7.4856	0.7587	-8.2741	0.4226

Table 1.4: Summary statistics of Trading Volume Turnover by GICS Sector.

Sector (GICS Classification)	Obs	Mean	Std. Dev.	Min	Max
Basic Materials	23,134	0.000281	0.0891	-3.0090	2.1614
Communication Services	47,479	0.000206	0.1007	-9.6793	6.9682
Consumer Cyclical	61,298	-0.000124	0.1015	-8.3778	7.9481
Consumer Defensive	71,385	-0.000026	0.0646	-7.1846	6.2832
Energy	23,833	-0.000323	0.0905	-3.8439	2.8789
Financial Services	92,124	-0.000170	0.1026	-7.3413	7.4511
Healthcare	106,100	-0.000094	0.0693	-2.2209	2.3356
Industrials	77,964	-0.000776	0.0963	-5.6362	2.7015
Technology	143,014	0.000663	0.0985	-11.2292	13.5374

Table 1.5: Summary statistics of stock log-returns by GICS Sector.

for the logarithmic returns by sector classification.⁸

Therefore, the final dataset consists of: *i*) a dataset containing ESG-related news headlines, *ii*) the relative change in intraday stock price for each firm in the DJIA Stock Market Index, and *iii*) the intraday Trading Volume Turnover ratio for the whole period of analysis (January 1 - March 31, 2023), matched at the precise minute they occurred. A potential source of problems might be that news can occur outside “open-market sessions”. To control for this issue, we exploit the *shift(maximum_event_date_shift)*⁹ parameter from the *eventstudy2* package in STATA and set the value to 390. This way, we allow news during closed market sessions to delay their impact on trading volume and stock prices relative to the following trading day¹⁰.

At this stage, the analysis moves into applying the event study methodology. Standard approaches in economics and finance research identify the event with a specific occurrence in time. Classical setups for event-study analysis require a 120-day estimation period, beginning 130 days before the event to avoid overlapping with the event window [Sprenger et al., 2014].

In this study, the release of ESG-related news indicates the specific event date for each firm. Consequently, the event window is built considering the $[-20, 60]$ minute interval around each firm-specific event date. We compute Abnormal Returns for each event over the different periods, as reported in the methodology section of this paper. To calculate the expected returns for each firm’s stock, we estimate the market model’s α and β parameters over an estimation window specified as $[-410, -20]$ backward trading minutes starting 20 minutes before each firm-specific event window. The DJIA Stock Market Index is used as a

⁸The reader can find the same statistics by stock in Table A.1 and Table A.2, in Appendix A respectively).

⁹If an event date does not coincide with any of those dates, it is shifted to the next date on the date line. *maximum_event_date_shift* specifies the maximum number of calendar days an event date is shifted forward. The default value is 3.

¹⁰In the resulting sample, some news items overlap with others in the time interval of the empirical analysis. We believe that this, also due to the not particularly high number of cases, should not impact the results. In any case, as a robustness analysis, the results of the empirical study with the sample reduced to only news items without overlap are given in the 1.8. The qualitative results obtained do not change even with the reduced sample.

t	N. of Events	AATV	PatellADJ	Corrado
-14	124	.0897	**	*
19	124	-.0687	**	*
44	124	-.0683	*	*

Table 1.6: Intraday Abnormal Turnover around Environmental News—Open market sessions. Only the AATV with significance in both tests is shown. *, **, *** indicate statistical significance at 10%, 5% and 1% level respectively.

proxy for the USA market returns. Cumulative Average Abnormal Returns encompass the following event windows: *i*) CAAR[−10, 10], *ii*), CAAR[−20, 20], CAAR[−20, 40] and *iii*) CAAR[−20, 60], CAAR[0, 10], CAAR[0, 30], CAAR[0, 60].

1.7 Empirical Results

This section presents the empirical results from testing hypotheses HP1, HP2, and HP3. First, we show the intraday change in cross-section average abnormal trading volume around the publication of ESG news by E (Table 1.6), S (Table 1.7), and G (Table 1.8) pillars. Secondly, we discuss the intraday change in cross-sectional cumulative average abnormal returns around the release of ESG news, both by ESG pillar and ESG Sentiment (Table 1.9 and Table 1.10). Lastly, we propose a similar analysis but consider a double-classification by the ESG pillar and ESG sentiment around the publication of ESG news (Table 1.11).

1.7.1 Impact of ESG news on intraday Average Abnormal Trading Volumes (AATVs)

To investigate HP1, we evaluate the cross-sectional average abnormal trading volume turnover over the event window [−20, +60] around the publication of Environmental, Social, and Governance news (see Table 1.6, Table 1.7, and Table 1.8, respectively).

The statistical significance of results is evaluated using two tests: a parametric (i.e., Adjusted Patell Test, hereafter Patell ADJ) and a non-parametric (i.e., the Corrado Rank Test with cross-correlation adjustment, hereafter Corrado test). The choice of these tests is motivated by the characteristics of the data. First, the transformed trading volume series departs substantially from normality, making reliance on standard parametric assumptions problematic. Second, to account for potential dependence across events, we relax the assumption of independence by allowing for cross-correlation. For instance, the trading activity of one stock may influence that of another, which, if ignored, could bias the estimation. By incorporating cross-correlation of abnormal trading volumes across ESG news events, we mitigate this bias and ensure more reliable inference.

Environmental News Table 1.6 provides the AATV reaction to the publication of firm-specific ESG-related news classified by the GPT-4 Turbo LLM as environmental for DJIA companies analyzed over the 81-minute event window [−20, +60]. Overall, the AATV shows a negligible market reaction.

The slight, or mostly absent, significance over the whole event window indicates no significant deviation of abnormal trading volume from zero over the selected period, both when relying on the Patell ADJ and Corrado tests. There are only sporadic significant deviations from zero before or after the event, more precisely, at minutes -14, 19, and 44. This suggests that market participants did not react strongly to environmental news within the DJIA securities during this period. A plausible explanation is that, for well-established

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t	N. of Events	AATV	PatellADJ	Corrado
8	152	.0437	***	**
29	152	.0586	**	*
30	152	.0567	**	**
43	152	.0329	*	*

Table 1.7: Intraday Abnormal Turnover around Social News—Open market sessions. Only the AATV with significance in both tests is shown. *, **, *** indicate statistical significance at 10%, 5% and 1% level respectively.

firms, such news may have been largely anticipated, either through prior disclosures or because these firms follow predictable environmental policies.

Furthermore, environmental news often includes initiatives implying long-term strategies, which may not lead to immediate changes in investors' beliefs, reflected in sporadic and intermittent significant changes in trading activity. Another explanation could be that investors prioritize other types of news, such as financial or governance-related information, which directly impacts a firm's short-term performance. The almost complete absence of significant AATVs, particularly in the critical minutes right before and after the news release, suggests that, despite the increased relevance, environmental factors may not be perceived as impactful in trading volumes during the observed period.

Social News The cross-sectional AATV over the 81-minute event window $[-20; +60]$ for the case of ESG news classified as social is shown in Table 1.7: most of the AATVs are not significant, with only sporadic significance in the post-announcement period.

Compared to environmental news, social news generates slightly higher, though still modest, trading volume activity. The AATV shows minor spikes mostly in the minutes following publication, suggesting little pre-announcement effect. These spikes, observed at minutes 8, 29, 30, and 43, indicate that some specific events triggered abnormal trading volumes, a finding partially supported by the Adj Patell test. However, the impact is neither widespread nor persistent enough to be statistically significant overall. This limited effect likely reflects the qualitative nature of social news, often related to CSR, labor relations, or scandals, which markets may not view as critical drivers of trading activity. Moreover, DJIA firms are well-established companies with largely predictable social policies, reducing the surprise element for investors.

Governance news Governance news, by contrast, shows a much more evident impact on trading volumes (Table 1.8). The AATV reveals significant deviations from the expected trading volume during non-event windows. As opposed to the previous two event studies on environmental and social news, the Adj Patell and Corrado tests confirm the statistical significance of positive changes in trading volumes in the pre-event period, as well as in the post-event period, thus signaling a robust reaction of trading volume activity to the release of ESG news classified as governance. Interestingly, AATV shows a positive trend, with only relatively few time intervals of insignificant changes in trading volume activity (i.e., $[1,5]$, $[18,31]$, $[47,51]$, and $[53,60]$).

A possible explanation is that governance news has direct and technical implications for a firm's management, strategy, and financial outlook. Such news is typically clearer and easier to interpret than environmental or social disclosures, and institutional investors closely monitor DJIA firms. Events like leadership changes, board decisions, or litigation can trigger rapid reassessments of firm value, explaining the stronger and more persistent AATV response. Unlike E and S news, trading volume reactions to governance announcements are

t	N. of Events	AATV	PatellADJ	Corrado
-16	307	.0336	**	*
-10	307	.0479	***	**
-8	307	.0592	***	***
-7	307	.0353	**	*
-6	307	.0305	*	*
6	307	.0457	***	**
7	307	.0539	***	***
10	307	.0298	**	*
14	307	.0416	**	**
15	307	.0314	***	
39	307	.0578	***	***
41	307	.0529	***	**
44	307	.0869	***	***

Table 1.8: Intraday Abnormal Turnover around Governance News—Open market sessions. Only the AATV with significance in both tests is shown. *, **, *** indicate statistical significance at 10%, 5% and 1% level respectively.

not confined to the immediate post-event window, underscoring their potential to drive critical shifts in corporate management.

1.7.2 Impact of ESG news on intraday Cumulative Average Abnormal Stock Returns (CAARs)

To test HP2 and HP3, we first evaluate the overall impact of ESG-related news headlines classified by GPT-4 Turbo, and then perform a deeper analysis by further categorizing the news by sentiment. The results are reported in three separate tables: Table 1.9 provides the Cumulative Average Abnormal Returns (CAARs) around ESG news announcements by ESG pillar. Table 1.10 illustrates a breakdown of the CAARs around the publication of ESG news by ESG sentiment. Lastly, Table 1.11 exhibits CAARs around the release of ESG news separated by ESG class and ESG sentiment. In all cases, the null hypothesis states that CAAR equals zero over the event window, implying that any significant deviation reflects a short-term impact of ESG news on firm value. To test this, we assess the significance of CAAR values using the non-parametric Wilcoxon Signed-Rank test.

Out of the 1,381 news from the final originated dataset, 1128 news (81.7%) are classified by the GPT-4 Turbo LLM in terms of sentiment: 334 as Positive (29.6%), 484 as Negative (42.9%), and 310 as Neutral (27.5%); while in terms of ESG category, 207 are classified as Environmental (18.4%), 264 as Social (23.4%), and 657 as Governance (58.2%). Thus, from Table 1.9, 1128 news articles are employed to perform the event study. The remaining 18% is dropped from the analysis due to unmatched observations or missing returns data.

News Impact by ESG Pillar

Table 1.9 reports the results of the event study methodology applied to the firm-specific ESG-related news classified into three pillars: Environmental, Social, and Governance. More specifically, the table analyzes how each pillar impacts the intraday stock returns of companies in the Dow Jones Industrial Average (DJIA) over various event windows. Environmental news reveals mixed outcomes: *i)* the CAAR for the shortest event window $[-10; 10]$ is positive and statistically significant at the 5% level. This denotes a general positive intraday market reaction to the release of environmental news. This might be the case of investors reflecting optimism about a company’s environmental initiatives or performance in a decade where individuals have increased their attention to environmentally related topics. Nonetheless, such impact tends to vanish over longer event windows, as reported in the following

1.7. Empirical Results

CAAR	Environmental	Social	Governance
[-10; 10]	.0219**	.0203	-.0654
[-20; 20]	-.0098	.0086	-.0607
[-20; 40]	-.0077	-.0356	-.0958
[-20; 60]	-.0099	-.0498	-.0964
[0; 10]	.0106	.0124	-.0675
[0; 30]	-.0434	-.0113	-.0778
[0; 60]	-.0291	-.0638*	-.0976
Observations	217	269	694

Table 1.9: Impact of firm-specific ESG-related news categorized by ESG class on the firm’s market value. This table illustrates the average response of stock returns around the Environmental, Social, and Governance news publication using the non-parametric Wilcoxon Signed-Rank test. CAAR[-10; 10], CAAR[-20; 20], CAAR[-20; 40], CAAR[-20; 60], CAAR[0; 10], CAAR[0; 30], and CAAR[0; 60] are the Cumulative Average Abnormal Returns over 21, 41, 61, 81, 10, 30, and 60 minutes around the publication of the ESG news. The estimation window is [-411, -21] (390 minutes or one trading day). Figures are in percent. ESG news headlines are retrieved from the “News Monitor App” from the LSEG Eikon service. The sample is composed of the 30 stocks listed in the DJIA between January 1 and March 31, 2023. *, **, *** indicate statistical significance at 10%, 5% and 1% level respectively. 85.45% (=1,180) of total extracted news (=1,381) are processed in the event study.

rows of Table 1.9.

On the other hand, social news tells a different story: the interval [0; 60] shows a significant negative CAAR of -0.0638%. This suggests that market operators may perceive social news adversely over a longer horizon. A possible explanation lies in the nature of such news: market participants might struggle to quantify its immediate implications, as issues related to corporate social responsibility, employee relations, or labor disputes are harder to translate into short-term financial outcomes. Nevertheless, social initiatives can generate meaningful long-term benefits, implying that investors may eventually view them as a source of value creation.

Governance news proves to be statistically insignificant and therefore does not provide further insights into short-term market dynamics. Although the CAAR values are consistently negative across all event windows, ranging from -0.0654% over the 21-minute interval [-10, +10] to -0.0963% over the 81-minute interval [-20, +60], these deviations are not significant. This suggests that, despite the frequent association of governance issues with management quality, corporate decisions, compensation schemes, and business stability, such news does not trigger a measurable market reaction in the short run when considered independently of sentiment.

The study’s findings support HP2 in only a third of the cases. Specifically, environmental news classified by the GPT-4 Turbo LLM shows a modest but significant positive impact on firm value, although this effect dissipates beyond the 21-minute event window. In contrast, social news exhibits a negative effect over the [0, 60] interval, while governance news shows no significant effect across any window. Overall, the results suggest that ESG news exerts only a limited influence on intraday stock returns, with environmental news driving short-term gains, social news generating longer-horizon negative effects, and governance news remaining largely inconsequential.

A possible explanation for this outcome is the unique relevance that environmentally sustainable practices have gained in the last decades. Most companies have lately been influenced by a commonly recognized global warning call to care about their environmental footprints, as well as advancing sustainable initiatives aimed at reducing the negative impact of their operations on the environment. Therefore, news regarding environmental practices

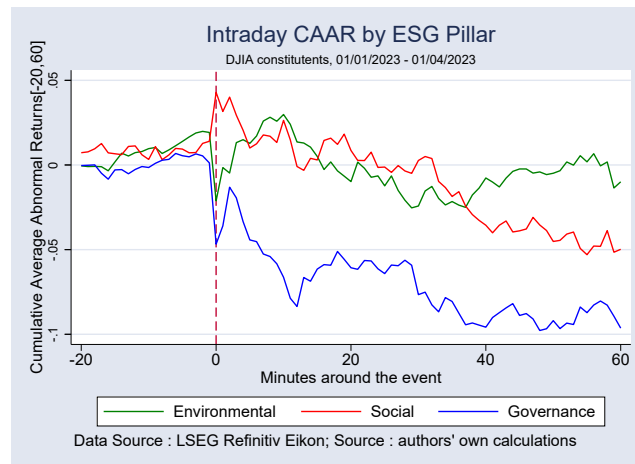


Figure 1.1: Cumulative Average Abnormal Returns around ESG-related news by ESG Pillar

carries some value for investors. Even so, it is interesting that ESG news, taken independently as a single pillar, displays low and inconsistent statistical significance. This could be attributed to several factors: *i*) heterogeneity in ESG news since it involves several topics, and not all environmental, social, or governance news has the same impact. This means that some news might be highly material to investors, while others might be less relevant, leading to variability in the significance of the results; *ii*) market indifference, since investors might be indifferent to ESG news, thus limiting or muting their reaction to the publication of such news; *iii*) market anticipation, market operators might anticipate the arrival of specific announcements, albeit ESG news is generally considered an unscheduled event; or *iv*) insufficient statistical power, given the limited number of events in several pillar-sentiment pairs, which limits the capacity of the tests to detect economically meaningful effects.

News Impact by ESG Sentiment

After assessing the impact of ESG-related news headlines on a firm's market value over three windows of a short-term investment horizon, we want to test HP3 about whether negatively classified news has a more significant impact than its positive counterpart. We then propose a further breakdown into the three Sentiment categories (Positive, Negative, Neutral) for each news class (E, S, or G) to inspect such asymmetric influence.

Table 1.10 presents the results of the second analysis: an event study providing insights into how different types of ESG-related news are classified according to their sentiment impact on stock returns.

Similar to the previous analysis on ESG news separated by ESG pillar, also in this analysis, we detect significance, though at 1% level this time, only for positive news in the shortest event window $[-10, 10]$ and $[0, 10]$. This indicates that, in general, positive ESG news leads to favorable CAARs, estimated at 0.0336% in the $[-10, 10]$ event window and at 0.0176% in the 10-minute event window $[0, 10]$. These results emphasize the quick reward of the market for firms releasing positive firm-specific ESG-related news. Positive ESG news spans from announcements about successful sustainable initiatives, positive social impact, or favorable regulatory decisions. These are good examples of how ESG news enhances a company's reputation. These are the positive ESG news items to which investors attribute a positive sentiment. However, as is the case for ESG news by ESG pillar, CAARs eventually decline over longer event windows, though not showing statistical significance. This indicates

1.7. Empirical Results

CAAR	Positive	Negative	Neutral
[−10; 10]	.0336***	-.0244	-.0649
[−20; 20]	.0188*	-.0357	-.0666
[−20; 40]	-.0099	-.0621	-.0887
[−20; 60]	-.0008	-.0768	-.0863
[0; 10]	.0175***	-.0184	-.0731*
[0; 30]	-.0207	-.0303	-.0958
[0; 60]	-.0295	-.0635	-.1011
Observations	365	513	449

Table 1.10: Impact of firm-specific ESG-related news categorized by ESG Sentiment on a firm’s market value. Using the non-parametric Wilcoxon Signed-Rank test, this table illustrates the average response of stock returns around Positive, Negative, and Neutral news publications. CAAR[−10, 10], CAAR[−20, 20], CAAR[−20, 40], CAAR[−20, 60], CAAR[0; 10], CAAR[0, 30], and CAAR[0, 60] are the Cumulative Average Abnormal Returns over 21, 41, 61, 81, 10, 30, and 60 minutes around the publication of the ESG news. The estimation window is [−411, −21] (390 minutes or one trading day). Figures are in percent. ESG news headlines are retrieved from the “News Monitor App” from the LSEG Eikon service. The sample is composed of the 30 stocks listed in the DJIA between January 1 and March 31, 2023. *, **, *** indicate statistical significance at 10%, 5% and 1% level respectively. 96.01% (=1,327) of total extracted news (=1,381) are processed in the event study.

that initial optimism may be moderated as the market processes and digests the news.

In contrast, negative news leads to negative CAARs across all selected event windows. This pattern indicates that the market anticipates unfavorable outcomes following negative ESG news, such as poor environmental performance, social controversies, or news about initiated litigation or corporate misconduct. However, none of them shows statistical significance. Despite the adverse direction of the market reaction, the absence of statistical significance might imply that the strength of the response can vary depending on the specific context of the news, along with its perceived severity. As expected, the consistent negative CAARs around the publication of negatively classified ESG news reflect the higher risk investors perceive about a wide range of topics (e.g., regulatory penalties, loss of consumer trust, increased operational costs, etc.). The CAARs display a consistently downward trend, which, though insignificant, would be consistent with investors expecting negative ESG news to harm future performance. Nevertheless, as none of the estimates is statistically significant, this remains qualitative rather than quantitatively established. Figure 1.2 illustrates the Cumulative Average Abnormal Returns over the 81-minute event window around the publication of ESG-related news classified as Environmental, Social, or Governance, using the sentiment score.

Interestingly, ESG news classified as neutral also tends to result in negative CAARs, with the [0; 10] window showing a statistically significant negative CAAR of -0.0732%. This suggests that even news that might be deemed neutral by the market could lead to uncertainty or slight pessimism among investors of a single firm. This is because, different from positive and negative firm-specific ESG news, it is qualitatively more complex to assign the neutrality of a news item.

News Impact by ESG Pillar and Sentiment

Table 1.11 presents the empirical results corresponding to the test of HP3. To do so, we show the change in CAARs in response to independently classified ESG news by ESG pillar and sentiment. We notice that results partially corroborate with HP3, as it is true that, after classifying ESG news by pillar and sentiment, at least one of the pillar’s sub-categories denotes great significance, namely, governance news. Nonetheless, we have to reject HP3 in

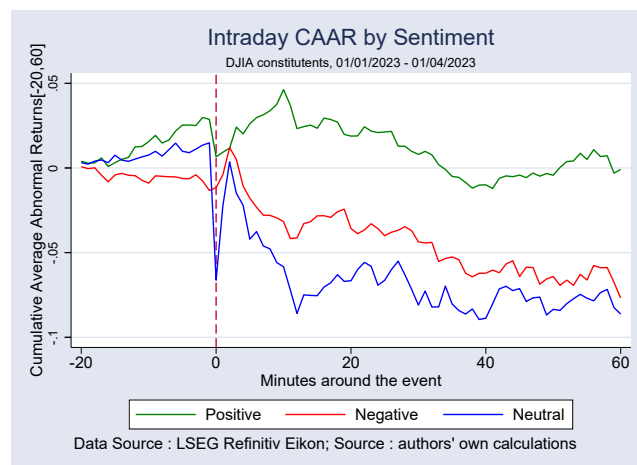


Figure 1.2: Cumulative Average Abnormal Returns around ESG-related news by ESG Sentiment

CAAR	Environmental			Social			Governance		
	Pos.	Neg.	Neu.	Pos.	Neg.	Neu.	Pos.	Neg.	Neu.
$[-10; 10]$	.0608*	-.0931	-.0924	-.0095	.0426	.0825	.0476**	-.0458	-.1151
$[-20; 20]$	.0332	-.1918	-.1094	-.0495	.0302	.2058	.1301***	-.0534	-.1066
$[-20; 40]$	.0232	-.1108	-.0926	-.1278	-.0146	.3257	.1465***	-.0779	-.1668
$[-20; 60]$	.0411	-.2795	-.1021	-.1151	-.0529	.2699	.1175**	-.0762	-.1654
$[0; 10]$	.0441**	-.1443	-.0605	-.0246	.0418	.0823	.0123	-.0352	-.1298**
$[0; 30]$	-.0156	-.2350	-.0719	-.0816	-.0047	.2973	.0623**	-.0322	-.1708*
$[0; 60]$	.0129	-.3214	-.0710	-.1244	-.0703	.2460	.0315	-.0512	-.1898
Observations	162	18	37	137	103	29	48	385	261

Table 1.11: Impact of firm-specific ESG-related news categorized by Sentiment on the firm’s market value. This table illustrates the average response of stock returns from least-squares regressions around Positive, Negative, and Neutral ESG news publications. $CAAR[-10, 10]$, $CAAR[-20, 20]$, and $CAAR[-30, 30]$ are the Cumulative Average Abnormal Returns over 21, 41, and 61 minutes around the publication of the ESG news. The estimation window is $[-120, -70]$. Figures are in percent. ESG news headlines are retrieved from the “News Monitor App” from the LSEG Eikon service. The sample is composed of the 30 stocks listed in the DJIA between January 1 and March 31, 2023. *, **, *** indicate statistical significance at 10%, 5% and 1% level respectively. 81.1% of total extracted news was correctly processed in the event-study

favor of its opposite, as in the case of the previous event study on ESG sentiment. Indeed, the only significant CAARs are related to positive environmental and governance news.

The left side of Table 1.11 examines the impact of firm-specific environmental news on stock market returns categorized by sentiment. Among the three sentiment classes, positive environmental news generates significant short-term market reactions, with $CAAR[-10, 10] = 0.0608\%$ (p-value $\leq 10\%$) and $CAAR[0, 10] = 0.0441\%$ (p-value $\leq 5\%$). This suggests that environmental initiatives are perceived as enhancing long-term sustainability and reputation, thereby improving firm value in the eyes of investors. However, this effect is confined to the immediate period following the news and does not persist over longer event windows. The middle side of the table investigates social news by sentiment, but does not reveal any statistically significant effects.

The right-hand side of the table presents the impact of governance news by sentiment. Results show a robust positive effect of positive governance news, with CAARs consistently increasing across several event windows, including $CAAR[-10, 10] = 0.0476\%$,

1.8. Robustness

$CAAR[-20, 20] = 0.1301\%$, $CAAR[-20, 40] = 0.1469\%$, $CAAR[-20, 60] = 0.1175\%$, and $CAAR[0, 30] = 0.0623\%$. This corroborates earlier findings that positive news exerts a stronger influence on intraday returns, while highlighting that governance-related disclosures receive heightened investor attention. Interestingly, neutral governance news also exhibits significant negative reactions, with $CAAR[0, 10] = -0.1298\%$ and $CAAR[0, 30] = -0.1707\%$, suggesting that ambiguity around governance events may raise investor concerns and depress firm value. By contrast, negative governance news does not yield significant results across any window.

These findings highlight the critical importance of classifying ESG news not only by type (Environmental, Social, Governance) but also by sentiment. While the analysis by the ESG pillar alone provides only limited evidence of significant effects, the additional breakdown by sentiment reveals meaningful relationships, such as the short-term positive impact of positive environmental and positive governance news, and the significant negative impact of neutral governance news. This layered classification proves essential for uncovering the nuanced ways in which markets process ESG information, showing that investor reactions depend not just on the category of news but also on its tone.

1.8 Robustness

In this section, we present the results of the empirical analysis by reducing the sample to only those news items that do not overlap with others in the reporting period. They confirm the results reported in the main text.

t	N. of Events	AATV	PatellADJ	Corrado
19	87	-.0768	**	*
44	87	-.0707	*	**

Table A: Intraday Abnormal Turnover around Environmental News—Open market sessions & Non-Overlapping Events. Only the AATV with significance in both tests is shown.

t	N. of Events	AATV	PatellADJ	Corrado
20	95	-.0483	**	**
33	95	.0546	***	**
43	95	.0383	*	*
55	95	-.0475	***	***
58	95	-.0240	*	**

Table B: Intraday Abnormal Turnover around Social News—Open market sessions & Non-Overlapping Events. Only the AATV with significance in both tests is shown.

t	N. of Events	AATV	PatellADJ	Corrado
-11	140	.0763	**	*
-7	140	.0600	***	**
-2	140	.1077	***	***
0	140	.0814	***	*
6	140	.0567	**	*
7	140	.0450	*	*
8	140	.0816	***	**
9	140	.0631	***	*
10	140	.0544	**	**
14	140	.0739	***	***
16	140	.0366	**	*
17	140	.0756	***	**
19	140	.0592	***	**
39	140	.0657	**	**
44	140	.0540	*	**
55	140	.0414	**	*

Table C: Intraday Abnormal Turnover around Governance News—Open market sessions & Non-Overlapping Events. Only the AATV with significance in both tests is shown.

CAAR	Environmental	Social	Governance
[−10; 10]	.0476*	.0319	-.0564
[−20; 20]	.0517	.0367	-.0606
[−20; 40]	.0398	.0156	-.0489
[−20; 60]	.0509*	.0399	-.0764
[0; 10]	.0222*	.0242	-.0597
[0; 30]	-.0196	.0239	-.0474
[0; 60]	-.0045	.0153	-.0833
Observations	135	159	214

Table D: Impact of non-overlapping firm-specific ESG-related news categorized by ESG class on a firm’s market value. This table illustrates the average response of stock returns around the Environmental, Social, and Governance news publication using the non-parametric Wilcoxon Signed-Rank test. CAAR[−10, 10], CAAR[−20, 20], CAAR[−20, 40], CAAR[−20, 60], CAAR[0, 10], CAAR[0, 30], and CAAR[0, 60] are the Cumulative Average Abnormal Returns over 21, 41, 61, 81, 10, 30, and 60 minutes around the publication of the ESG news. The estimation window is [−411, −21] (390 minutes or one trading day). Figures are in percent. ESG news headlines are retrieved from the “News Monitor App” from the LSEG Eikon service. The sample is composed of the 30 stocks listed in the DJIA between January 1 and March 31, 2023. *, **, *** indicate statistical significance at 10%, 5% and 1% level respectively. The sub-sample comprises 43.87% (=592) of the total extracted news (=1,381), where 85.81% (=508) of the whole sub-sample was correctly processed in the event study.

1.8.1 LLM Validation and Reproducibility

Since both the ESG and the Sentiment classification are LLM-generated, they inherently constitute a source of measurement error which naturally propagates throughout all downstream tests, raising valid questions regarding the validation and reproducibility of this analysis.

Building on 1.5, model performance is evaluated against a stratified 10% subsample independently labeled by two trained annotators, yielding an overall inter-annotator agreement of $\kappa = 0.44$ based on the [Landis and Koch, 1977] scale. Furthermore, any residual disagreement is resolved by the author at a later stage with reference to the written annotation guidelines.¹¹ As reported in Table 1.1, GPT-4 attains a weighted F_1 of 0.70 against 0.46 for FinBERT and 0.26 for the majority-class baseline, and dominates on every

¹¹I note this explicitly: this could potentially be a source of ‘expectancy’ bias, though mitigated here by the fact that labels are assigned on the basis of headline text alone, independently of any market outcome, and that the guidelines were fixed before annotation began and never modified.

1.8. Robustness

CAAR	Positive	Negative	Neutral
[-10, 10]	.0482**	.0111	-.0302
[-20, 20]	.0608**	-.0356	.0003
[-20, 40]	.0403	-.0060	-.0129
[-20, 60]	.0480*	-.0377	.0177
[0, 10]	.0269*	-.0030	-.0319
[0, 30]	.0039	-.0150	-.0115
[0, 60]	-.0036	-.0456	-.0018
Observations	220	171	189

Table E: Impact of non-overlapping firm-specific ESG-related news categorized by ESG Sentiment on a firm’s market value. Using the non-parametric Wilcoxon Signed-Rank test, this table illustrates the average response of stock returns around Positive, Negative, and Neutral news publications. CAAR[-10, 10], CAAR[-20, 20], CAAR[-20, 40], CAAR[-20, 60], CAAR[0, 10], CAAR[0, 30], and CAAR[0, 60] are the Cumulative Average Abnormal Returns over 21, 41, 61, 81, 10, 30, and 60 minutes around the publication of the ESG news. The estimation window is [-411;-21] (390 minutes or one trading day). Figures are in percent. ESG news headlines are retrieved from the “News Monitor App” from the LSEG Eikon service. The sample is composed of the 30 stocks listed in the DJIA between January 1 and March 31, 2023. *, **, *** indicate statistical significance at 10%, 5% and 1% level respectively. The sub-sample comprises 43.87% (=592) of the total extracted news (=1,381), where 97.97% (=580) of the whole sub-sample was correctly processed in the event study.

pillar. Its weakest category is *Other* ($F_1 = 0.52$), which displays high precision but low recall. The interpretation of this result is that the model is conservative in assigning the *Other* class, and therefore tends to classify genuinely non-ESG headlines into an ESG pillar rather than the reverse. This deserves particular attention, as the perimeter of ‘*Other*’ is precisely what constitutes sample inclusion. In misclassifying, the model does not merely reallocate observations across pillars, rather it admits non-ESG items into the event sample. The consequences of misclassification ultimately depend on its relationship with the outcome variable. Notice that the classifier only reads headline text, without access to prices, trading volumes, or dates. There is, therefore, no mechanism by which its errors could be related to the magnitude of the subsequent market reaction, which supports the assumption that the measurement error is *non-differential* with respect to the dependent variable. In the econometrics of measurement error, non-differential misclassification in a grouping variable attenuates estimated effects toward zero rather than artificially manufacturing them, as demonstrated by [Aigner, 1973].

In terms of reproducibility, three features support replication. First, the model is accessed at a pinned checkpoint (**gpt-4-1106-preview**) meaning that identical requests are served by identical weights throughout the sample period and in any subsequent replication. Second, generation is performed at a *temperature* parameter = 0, which makes the mapping from headline to label effectively deterministic and removes run-by-run variation. Third, the prompt is reproduced *verbatim* in Section 1.6, together with the JSON output schema, so the procedure can be re-executed with ease. Nonetheless, two limits persist: the checkpoint is proprietary and may be withdrawn, in which case a subsequent model would require re-validation against the same annotated benchmark; secondly, I have not conducted a systematic prompt-perturbation analysis, so sensitivity of the labels to prompt design is not quantified here.

Hence, two critical conclusions emerge : first, any statistically significant effect reported throughout this chapter is -if any- understated relative to the true magnitude that “perfect” labels would reveal. Second, the muted or insignificant responses documented across several pillar-sentiment subsamples cannot be attributed solely to *efficient absorption* by the market, since attenuation bias from labelling error offers an equally sound explanation for the

absence of significant outcomes.

1.9 Conclusions

Our study advances a further understanding of how non-financial information, that is, firm-specific ESG-related news, affects intraday stock market dynamics. The observed sample includes companies in the Dow Jones Industrial Average over the period January-March 2023. Employing GPT-4 Turbo, a state-of-the-art LLM from the rising field of NLP, we apply an intraday event-study methodology across several event windows to capture both immediate and slightly delayed intraday market reactions of trading volumes and stock returns. The selected event windows include intervals that capture both potential pre-announcement activity (i.e., $[-10, 10]$, $[-20, 20]$, $[-20, 40]$, $[-20, 60]$) as well as post-announcement adjustments (i.e., $[0, 10]$, $[0, 30]$, and $[0, 60]$). This design allows us to assess whether firm-specific ESG news is rapidly incorporated into trading activity as well as prices, or whether market responses unfold with delay. Our findings offer several insights for investors, regulators, and researchers. Market participants may use the results to devise new short-term strategies (or to refine already in-use ones) to better incorporate practitioners' behavior around disclosures of several types of non-financial information. Regulators, on their side, might draw from our results to monitor speculative behavior or potential information leakage. Finally, scholars can further explore the mechanisms linking information disclosures, market microstructure, and investor sentiment to investigate further the underlying mechanisms driving short-term price discrepancies by intertwining behavioral finance with the latest machine learning models.

The event study methodology provides us with two sets of results: firstly, we investigate the reaction of Average Abnormal Trading Volume Turnover (AATV) to ESG news clustered by ESG pillar over the 81-minute event window; secondly, we examine the response of stock returns to ESG news classified by pillar, sentiment, and both by pillar and sentiment together. On the one hand, our first analysis on intraday trading activity reveals a heterogeneous scenario: evidence shows that environmental and social news generate little to no systematic change in abnormal trading volume across the event window, with only sporadic deviations from expected activity. By contrast, we observe a greater significance around the release of governance news, even in the pre-event period, showing a clear and persistent increase in intraday trading volume before and after the announcement. This is interesting, as the pre-event activity might signal the presence of information leakage or market anticipation. At the same time, the strong post-event reaction highlights the greater relevance of governance disclosures.

On the other hand, the evidence suggests partial market efficiency, as it displays a moderate ability to incorporate almost all publicly available information quickly, though it does not constitute solid evidence for it.. Notably, in the first classification of ESG news by ESG pillar, environmental announcements generate a short-lived positive reaction, whereas social news exhibits a mild negative effect over longer horizons. Our results corroborate the findings by [Vu et al., 2024] concerning the short-term persistence of ESG news, even though we do not confirm the significance of governance news as they do. Therefore, we can only partially confirm our second hypothesis.

When incorporating sentiment, we detect the importance of news tone: positive ESG news produces an immediate, though short-lived, price appreciation, and this result holds when we restrict our analysis to post-announcement periods, even though again only for very close minutes relative to the events. Despite showing no significance, negative and

neutral news primarily correspond to negative stock returns throughout the specified event windows. Hence, this result suggests that we have to reject our third hypothesis.

Interestingly, a combined classification by ESG pillar and sentiment uncovers additional nuances: over the 21-minute $[-10, 10]$ and the 10-minute $[0, 10]$ windows, environmental news items labeled as positive are associated with short-term positive reactions. Furthermore, positive governance news exhibits a significant positive impact in both the pre- and post-announcement windows surrounding its release. Unexpectedly, even neutral news shows a mild, statistically significant negative effect, though negligible in terms of persistence, as it dissipates within the first half-hour following the news announcement.

As for market-timing, our results align closely with [Anese et al., 2023], as we also show that investors tend to react more in the *second-10-minutes*, or within 20 minutes from the occurrence of the news. This implies that the propagation of such news takes at least 10 to 20 minutes to flow into the stock prices, hence ruling out the possibility that information diffusion is an instantaneous process.

In terms of theoretical implications, we detect the presence of significant abnormal volumes and price adjustments (although modest) within the first trading hour around the release of a firm's specific ESG-related news. This works as the first evidence that markets do not fully and immediately incorporate ESG-related information, hence challenging the strong and semi-strong forms of the Efficient Market Hypothesis [Fama, 1970b]. Our findings also corroborate the results of [Anese et al., 2023] by detecting the delayed incorporation of ESG news in the market. Furthermore, these identified patterns align with various empirical studies from the behavioral finance literature, where cognitive frictions and heterogeneous interpretation slow down the incorporation of information into the main liquidity measures. This inevitably creates opportunities for arbitrage strategies that exploit short-lived market inefficiencies.

From a practical standpoint, our study entails a twofold contribution: on the one hand, market operators can use it to gauge future market trends and uncover investors' beliefs. This suggests that certain ESG news types, especially governance-related announcements classified by sentiment, can serve as short-horizon trading signals. On the other hand, delayed price adjustments also have implications for risk management and order-execution strategies, as understanding the timing of market reactions helps investors adjust exposure and liquidity decisions.

Notwithstanding, three limitations constraint the interpretation and generalizability of these results. First, the cross-section analysis is restricted to the 30-DJIA constituents, that is, large, blue-chip stocks for which ESG disclosures are most likely to be anticipated through prior reporting. Second, the sample spans a single quarter (January-March 2023), so the findings may reflect some period-specific conditions rather than a stable regime capable of supporting robust, generalizable conclusions. Third, and as discussed in Section 1.8.1, several pillar-sentiment subsamples are small (sometimes even less than 30), which indisputably limits the power of the tests and means that insignificant estimates cannot be read as evidence that no reaction occurred.

Future research could expand the scope of this analysis by considering a broader set of firms beyond the DJIA and incorporating a larger sample of ESG-related news over more extended periods. This would enhance the generalizability of the findings and provide deeper insights into how different markets and investor groups process ESG information.

To be completely honest, the findings of the first paper left me with a sense of productive dissatisfaction. Then I realized that the heterogeneous and sometimes non-existent intraday price responses to ESG news could have been consistent with rapid information absorption or anticipation effects. However, the minute-level event study methodology could

not distinguish between these competing explanations. Each limitation of the first paper maps onto a specific design choice in the second. Observing returns and volumes at the minute frequency meant observing the *outcome* of the adjustment process rather than the process itself; message-level limit order book data make the mechanism visible, since every submission, cancellation and execution is timestamped and signed. Measuring an average reaction over an event window meant assuming that the market’s capacity to absorb a trade is constant across that window; allowing instantaneous price impact to evolve relaxes exactly that assumption. And relying on a single class of unscheduled, firm-specific news meant that scheduling and informational content were confounded; the second paper therefore widens the event set to span scheduled macroeconomic announcements, earnings releases and unscheduled controversies, so that the two can be separated.

Observing returns and trading volume at the minute frequency meant observing the *outcome* of a process whose *mechanism* remained invisible. Then a sense of curiosity sparked inside of me and the urge to dig deeper prevailed. In order to understand how the market actually processes new information, how liquidity responds, how the bid-ask spread evolves, how order flow imbalances develop, I needed to investigate the limit order book for tick-by-tick data granularity. And that is what motivated a research visit to the Department of Computer Science at University College London, where, under the supervision of Prof. Tomaso Aste, I worked directly with high-frequency limit order book data for the first time in my entire life. Genuinely, the rest of the dissertation is firmly grounded on the knowledge I have acquired from that visiting period. To be precise, over the rest of my PhD, I have only worked with L2¹² data. In this perspective, what follows is a deeper treatment of limit order book structure and its key properties, intended to equip the reader for the analysis that the second paper develops.

The Limit Order Book

Following Bouchaud et al. [2018], a limit order book $\mathcal{L}(t)$ is the collection of all outstanding limit orders for a given financial instrument at time t . It sits at the nucleus of modern electronic trading: a self-organising, continuously evolving record of all publicly visible buying and selling interest, updated in real time as orders are submitted, executed, or cancelled. For this reason, the LOB is commonly regarded as the decentralised provider of visible liquidity in financial markets.

Each order x residing in the book is characterised by four properties: a direction $\epsilon_x \in \{+1, -1\}$ (where $+1$ denotes a buy order and -1 a sell order), a limit price p_x , a quantity Q_x , and a submission time t_x . Formally:

$$x := (\epsilon_x, p_x, Q_x, t_x). \quad (1.13)$$

As one would expect, a typical LOB is composed of two sides: buy limit orders $\mathcal{B}(t)$ (for which $\epsilon_x = +1$) and sell limit orders $\mathcal{A}(t)$ (for which $\epsilon_x = -1$). Orders on each side

¹²L2 data refers to an LOB that provides the complete state of the order queue at multiple price levels on both sides of the market, timestamped to the microsecond and updated with every order submission, cancellation, and execution event. This contrasts with L1 data, which captures only the best bid and ask prices and last transaction price, and with L3 data, which additionally identifies individual order IDs and trader-level information, useful for High-Frequency Trader identification as shown in [Goudarzi and Bazzana, 2023]. The LOBSTER platform I used [Huang and Polak, 2011] reconstructs L2 data for NASDAQ-listed stocks at up to 40 levels of depth on each side of the book.

accumulate at discrete price levels¹³, forming a queue of pending interest that a trade-matching engine continuously monitors for executable pairs.

Key Properties

The four fundamental quantities derived from the LOB at any instant t are the bid price, the ask price, the mid-price, and the bid-ask spread:

$$b(t) := \max_{x \in \mathcal{B}(t)} p_x \quad (\text{best bid})$$

$$a(t) := \min_{x \in \mathcal{A}(t)} p_x \quad (\text{best ask})$$

$$m(t) := \frac{1}{2} [a(t) + b(t)] \quad (\text{mid-price})$$

$$s(t) := a(t) - b(t) > 0 \quad (\text{bid-ask spread})$$

By definition, $a(t) > b(t)$ at all times: were this not the case, outstanding buy and sell orders would be immediately matched and removed from the book. The bid price $b(t)$ is therefore the highest price at which it is immediately possible to sell; the ask price $a(t)$ is the lowest price at which it is immediately possible to buy. The bid-ask spread $s(t)$ measures the cost of immediate execution and serves as the primary indicator of market liquidity, as [Glosten and Milgrom \[1985\]](#) showed. In addition, it also reflects the adverse selection risk faced by market-makers in the presence of informed traders.

Order Types and Price priority rules

Traders interact with the LOB through two broad categories of orders. A *limit order* (or passive order) specifies oth a price and a quantity, entering the queue to await execution at the chosen price or better. A *market order* (or aggressive order) demands immediate execution at the best available price, consuming liquidity from the opposite side of the book. This dichotomy reflects a fundamental trade-off: limit orders offer price improvement but bear execution uncertainty, while market orders guarantee immediacy at the cost of paying the spread. On top of those basic order types, several other forms of order serve specific trading purposes. *Iceberg orders* allow large traders to conceal the full extent of their interest by disclosing only a fraction of the total quantity. This marks an important consideration for adverse selection, since full disclosure of a large order would immediately reveal complete private information, which could potentially move prices before execution. *Stop orders* trigger a market order automatically when the price reaches a specified threshold. Its usage is primarily confined to loss-limitation or profit-protection.

When multiple limit orders reside at the same price level, a *priority rule* determines execution sequence. The most common rule is *price-time priority*: orders are ranked first by price (best-priced orders execute first) and ties are broken by submission time (earlier orders execute first). An alternative, used primarily in futures markets, is *pro-rata priority*, which distributes execution proportionally to order size at the best price.

Together, the existence of multiple order types and the establishment of priority rules have significant implications for observed market behaviours. For what concerns the former aspect, it underscores a point central to this dissertation, as LOBs do not merely record

¹³All price-changing arriving orders are required to have a price that is a multiple of a tick - θ -, which is the minimum admissible price increment between consecutive price levels in the book

passive liquidity, but rather provide for a strategic environment for market participants to reveal their private information. In terms of the latter instead, logically, under price-time priority (i.e., the majority of modern electronic trading platforms), traders have a strong incentive to submit limit orders early and at competitive prices, generating the intraday liquidity patterns whose response to heterogeneous information the second paper analyses in detail.

Tick Size and its Implications

A feature of LOBs that will prove particularly discussed throughout the second and third papers is the *tick size* (i.e., the minimum admissible price increment between consecutive price levels in the book). Tick-size fundamentally shapes the strategic environment of the LOB. A large spread relative to the exchange tick¹⁴ (a *large-tick* stock) creates wide, crowded queues at each price level, where queue position becomes a valuable commodity and price competition is effectively suppressed. A small tick size relative to price (a *small-tick* stock) produces thin queues and intense price competition, with market-makers frequently undercutting one another by a single tick. This distinction has been shown to generate systematically different price impact dynamics [Bouchaud et al., 2018], and motivates the classification of stocks by relative tick size adopted in the second paper.

From ESG News to a Heterogeneous Information Taxonomy

The first paper concentrated on a single category of firm-specific news (i.e., ESG disclosures) and documented heterogeneous, and sometimes absent, intraday responses across the different ESG pillars. The natural follow-up question is whether this heterogeneity is specific to ESG news or whether it reflects a more general property of how limit order book dynamics respond to different types of informational events. For this reason, the paper you are about to read broadens the taxonomy of information considered, spanning scheduled macroeconomic announcements, political news, and unscheduled corporate controversy disclosures. This broader taxonomy allows for a systematic comparison of LOB dynamics across the full spectrum from hard, anticipated information to soft, surprise disclosures.

¹⁴This quantity, θ , is set to 0.01\$ (or 1 cent) for the exchange of reference in this dissertation, that is NASDAQ.

Chapter 2

Limit Order Book Dynamics around Heterogeneous News

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2.1 Abstract

This paper examines limit order book dynamics around the release of a heterogeneous set of news events spanning scheduled macroeconomic announcements, political news, and unscheduled controversial announcements. Using high-frequency LOBSTER data for six stocks classified by relative tick-size over 2023 - mid-2024, I first document the main stylized facts of intraday LOB behaviour and demonstrate their systematic variation in intensity and timing across news categories. Building on [Campigli et al. \[2022\]](#), I use a Score-Driven Aggregated Modified Hasbrouck (SDAMH) model in which instantaneous price impact $b_{0,t}$ evolves as a time-varying parameter driven by the score of the predictive likelihood. This framework enables a systematic comparison of price formation dynamics across scheduled and unscheduled informational events. The results reveal news-dependent impact dynamics: earnings announcements generate the largest $b_{0,t}$ peaks across all stocks, systematically exceeding FOMC responses, consistent with the firm-specific nature of earnings information. FOMC announcements produce sharp but comparatively moderate market-wide jumps, while controversy news yields heterogeneous responses contingent on the nature and timing of each disclosure. Notably, the time-varying specification uncovers $b_{0,t}$ sign reversals, thus indicating shifts between buy- and sell-side informed flow regimes that static models would miss entirely.

2.2 Introduction

The cornerstone of modern electronic trading is the Limit Order Book (LOB). As stated in [Bouchaud et al. \[2018\]](#), the LOB ‘*is a collection of lower-priced buy orders and higher-priced sell orders, separated by a bid-ask spread. The persistent aspect of limit orders is key for the stability of the LOB state and the market price.*’ At its core, it is precisely this architecture, namely the continuous interaction of patient limit order providers, or *liquidity providers*,

and impatient market order submitters, or *liquidity takers*, that has made the LOB one of the primary objects of study in the market microstructure literature. The research question under investigation is, therefore, how the price impact of trades evolves in real time around the arrival of a heterogeneous range of news. The question matters because standard empirical research of price impact assumes it to be constant over a trading day, whereas the moments of greatest interest (i.e., the minutes surrounding an information release) are precisely those in which the informational environment, and therefore price impact itself, is most likely to be shifting. With that in mind and using message-level limit order book data for six NASDAQ-listed stocks, I first verify that the dataset reproduces the canonical stylized facts of market microstructure, and then estimate a score-driven model in which instantaneous price impact is allowed to vary over time, tracing its trajectory around scheduled macroeconomic announcements, earnings releases, political and unscheduled corporate controversies.

The present study contributes to this strand of literature along two complementary dimensions. On the one hand, it provides a systematic overview of the canonical stylized facts of market microstructure (i.e., intraday U-shaped activity patterns, fat-tailed return distributions, volatility clustering, and the long memory of order flow) across a heterogeneous panel of six NASDAQ-listed stocks, using tick-by-tick data spanning January 2023 to April 2024. This serves both as an empirical validation of the dataset and as a foundation for the main analysis. On the other hand, this study adds to the stream of literature challenging an assumption common to several empirical applications of foundational models: namely that price impact is constant over trading horizons. Crucially, this assumption is more of an econometric convenience than an implication backed by the underlying theory. As a matter of fact, in Kyle [1985] the price impact parameter $\lambda = \sigma_v / 2\sigma_u$ depends on fundamental volatility and noise trading volume, two properties that shift whenever news arrives. Likewise, in Glosten and Milgrom [1985] the bid-ask spread is an increasing function of the probability of trading with an informed agent, which Easley and O'Hara [1992] show to be far from constant over the trading day. Thus, theory implies that λ should fluctuate systematically with the information environment, and news announcements are precisely the kind of event for which that environment changes discretely. Estimating a uniform λ across a trading day is therefore not a competing theoretical claim, but rather a restriction imposed in estimation, as it averages over exactly the variation the theory predicts.

This is precisely the proposition the empirical model of this paper is designed to test. The article applies the Score-Driven Aggregated Modified Hasbrouck (SDAMH) model developed by Campigli et al. [2022] in a high-frequency setting to a broad spectrum of heterogeneous news events -spanning scheduled macroeconomic announcements and unscheduled shocks- with the aim of tracing the real-time evolution of the instantaneous price impact of trades. By allowing this parameter to vary dynamically in response to the arrival of new information, the SDAMH framework reveals how different categories of news generate heterogeneous and asymmetric responses in the price adjustment process, hence shedding new light on the interaction between informed and uninformed traders at the precise moment of information release.

The main findings can be summarised as follows. The estimated impact path is strongly time-varying and its shape differs systematically across news categories. Earnings announcements generate the largest instantaneous impact in the sample, consistent with the firm-specific nature of earnings information, which resolves uncertainty through a concentrated burst of directed order flow in a single stock. FOMC announcements produce sharp but comparatively moderate responses, in line with the market-wide diffusion of monetary policy signals across assets simultaneously. Unscheduled controversy news generates the most

heterogeneous dynamics of all, including pronounced sign reversals in $b_{0,t}$ at precisely times-tamped regulatory events -episodes in which the dominant informed flow switches side, and which a constant-parameter framework would obscure entirely.

The remainder of the paper is organized as follows. Subsection 2.2.3 describes the data and the empirical analyses. Section 2.3 presents the main empirical results, and Section 2.4 concludes.

2.2.1 Literature Review

Among the seminal works, Kyle [1985] develops a model in which informed traders strategically camouflage their orders within the order flow, thus creating permanent price impact proportional to their information advantage and introducing the key parameter λ to measure it. Building on this, Glosten and Milgrom [1985] show that bid-ask spreads arise from adverse selection, as market-makers widen spreads when facing potentially informed traders, while Easley and O'Hara [1992] developed empirical models of the probability of informed trading from trade data. More relevant to the present work, Hasbrouck [1991] provides an empirical framework for decomposing price changes into permanent and transient components using vector autoregressions. Taken together, these foundational models treat price impact as a static parameter, constant across trades and time.

The theoretical foundations of LOBs were laid by Glosten [1994], who showed that a competitive LOB can survive adverse selection and remain open even in the presence of informed traders. Proceeding from this, Foucault [1999], Parlour [1998] and Foucault et al. [2005] characterized the strategic submission decisions of traders facing a trade-off between execution probability and price improvement, predicting that order book depth, the bid-ask spread and order submission rates all respond systematically to shifts in the informational environment. Empirically, Biais et al. [1995] provided one of the first comprehensive characterizations of LOB dynamics on the Paris Bourse, documenting the *U-shaped* intraday pattern in order submission rates and the positive autocorrelation in order flow. These are well-documented stylized facts in the market microstructure literature which I successfully retrieve in the LOB analysis of Section 2.2.6. Extending this line of work, Ranaldo [2004] and Hasbrouck and Saar [2009] showed that order submission strategies are significantly influenced by current LOB conditions, including depth imbalance and recent price movements. Moreover, the relationship between order flow and price dynamics has since become central to the literature: with the advent of high-frequency LOB data, researchers have developed increasingly refined measures of order flow imbalance, and more recent work by Cartea et al. [2018] and Stoikov [2018] has shown that OFI is a leading indicator of price changes over horizons of seconds to minutes.

Concurrently, the impact of news announcements on financial markets has been extensively studied at daily frequency. In terms of scheduled announcements, Cieslak et al. [2019] show that equity returns are systematically higher in the weeks leading up to FOMC meetings, pointing to a broader pattern of pre-announcement information leakage. Fleming and Remolona [1999] and Balduzzi et al. [2001] document large price reactions and volume spikes following scheduled announcements in bond and equity markets. However, though much of this work has revolved around daily horizons, very little research focuses on the intraday dynamics around scheduled and unscheduled news. From a microstructure perspective, Andersen et al. [2003] use a high-frequency framework to show that announcement surprises generate jumps in exchange rates and equity returns, with the direction and magnitude of the surprise being the primary determinant of the price response. Riordan et al. [2013] and Cao et al. [2009] study LOB dynamics around earnings announcements and find that both

the depth and shape of the book change significantly in the minutes before and after the release; specifically, [Cao et al. \[2009\]](#) show that the order imbalance at multiple depth levels predicts short-term returns, particularly around news events. [Lucca and Moench \[2015\]](#) document a significant pre-FOMC equity drift that cannot be explained by standard risk factors, suggesting that informed trading occurs in the hours before the announcement. Studies on the impact of FOMC announcements have seen particular growing interest due to their market-wide impact and the high degree of pre-announcement uncertainty despite the known timing.

The literature draws a fundamental distinction between scheduled and unscheduled news events. Scheduled macroeconomic announcements—FOMC decisions, Non-Farm Payrolls and CPI releases—are characterized by known timing but uncertain content, allowing market participants to anticipate the arrival of information and adjust their positioning accordingly in the pre-announcement window. [Fleming and Remolona \[1999\]](#) and [Balduzzi et al. \[2001\]](#) document large and rapid price reactions following such releases in bond and equity markets, while [Andersen et al. \[2003\]](#) show that the surprise component of the announcement—rather than the announcement itself—is the primary driver of the price response. Unscheduled news, by contrast, such as earnings surprises, geopolitical shocks or analyst rating changes, arrives without warning, leaving market participants with no opportunity to pre-position. The LOB response to these two classes of events differs substantially: scheduled announcements tend to generate a sharp, symmetric withdrawal of liquidity in the pre-announcement window followed by a burst of aggressive order flow at release, whereas unscheduled news produces more disorderly and heterogeneous dynamics. Importantly, even within the class of scheduled announcements, not all news shocks are equal: the direction, the magnitude and the degree of surprise of a release each elicit distinct order book responses. This asymmetry in responses across order types—most strikingly, the contrast between aggressive sell-side visible market orders and strategic buy-side hidden executions documented in this paper—is precisely what a static price impact parameter cannot capture, and what motivates the time-varying framework adopted here.

2.2.2 The Limit Order Book

A Limit Book (LOB) is the set of all active orders (i.e. all limit orders). It is an order-driven market ¹ that is responsible for matching sellers with the buyers firstly according to price priority, and secondly by queue priority. Figure 2.1 is an illustrative example of how the matching of orders occurs in the LOB. An order $z = (p_z, q_z, t_z)$ submitted at time t_z with limit price p_z and size q_z (with $q_z > 0$ for buys and $q_z < 0$ for sells) is a commitment to *buy* (resp. *sell*) up to $|q_z|$ units of the asset at a price no greater than (respectively no less than) p_z [[Bouchaud et al., 2018](#)]. LOBs thus display the available depth at different levels of the order book. Let $V(p)$ denote the total volume of limit orders submitted at price p . Formally,

$$V(p) = \sum_{z \in L : p_z = p} q_z, \quad (2.1)$$

where L is the set of active limit orders, and q_z is the signed size of order z .

¹Traders can choose whether to provide or consume liquidity by submitting different types of orders to a LOB. Liquidity provision is decentralized and becomes an endogenous process. Limit order books are therefore more flexible than quote driven markets

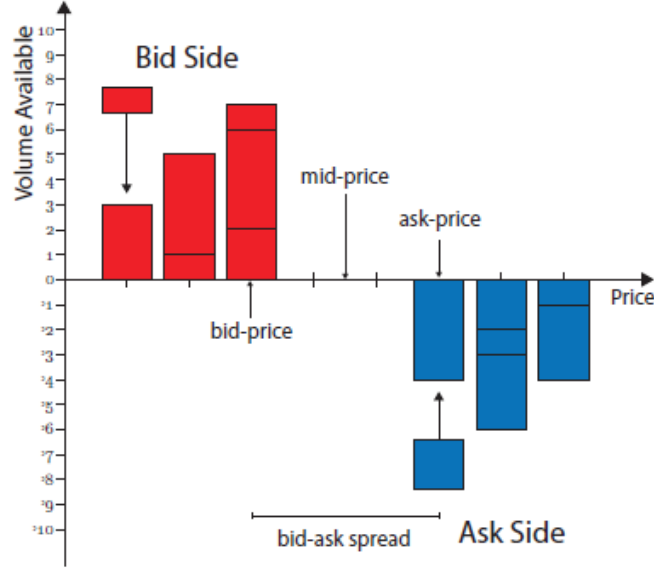


Figure 2.1: Example of a visualization of a LOB, i.e. the collection of limit orders (active orders). The buy orders (positive volume) are displayed in red whereas the sell orders are displayed in blue [Bouchaud et al., 2018]

The limit order book is characterized by several key quantities. First, the *bid price* $b(t)$ represents the highest price at which a buyer is willing to purchase the asset at time t . Formally, among all active buy orders (those with positive quantity $q_z > 0$), the bid is:

$$b(t) := \max_{z \in L: q_z > 0} p_z. \quad (2.2)$$

Symmetrically, the *ask price* $a(t)$ is the lowest price at which a seller is willing to sell, defined as:

$$a(t) := \min_{z \in L: q_z < 0} p_z, \quad (2.3)$$

where $q_z < 0$ denotes sell orders. These top-of-book prices are commonly called the *best bid* and *best ask* to distinguish them from limit orders at inferior prices deeper in the book.

The gap between these prices defines the *bid-ask spread*:

$$s(t) := a(t) - b(t), \quad (2.4)$$

which represents the cost of immediate execution and is a standard measure of market liquidity. The *mid-price* is the average of the bid and ask:

$$m(t) := \frac{1}{2}[a(t) + b(t)]. \quad (2.5)$$

This mid-price serves as a proxy for the true underlying value of the asset in the absence of microstructure frictions.

Finally, market rules impose discrete price grids through the *tick-size* (θ), which specifies the minimum price increment. All limit orders must comply with this constraint:

$$|p_i - p_j| \geq \theta \quad \forall i, j. \quad (2.6)$$

For U.S. equities, the tick-size is typically \$0.01 for stocks priced above \$1. As a final note, limit order books (LOBs) operate with finite resolution parameters: there is typically a minimum tick-size (price increment) and a minimum lot size (trading quantity).

2.2.3 Data

The data cover six stocks: three large-tick stocks, two small-tick stocks, and one medium-tick stock (Table 2.1). All stocks are listed on the NASDAQ Stock Exchange and are among the most actively traded securities in the United States. The sample spans January 2023 - April 2024, totaling 273 trading days of high-frequency limit order book data sourced from LOBSTER (*Limit Order Book System – The Efficient Reconstructor*²), which reconstructs the full order book from NASDAQ message data. For each stock-day pair, LOBSTER provides two source files: a *message file* recording every order-level event, and an *order book file* recording the state of the LOB after each event up to a user-specified depth level. Figures 2.2 and 2.3 show an example of each file structure, as provided by the LOBSTER documentation.

message file.

Time (sec)	Event Type	Order ID	Size	Price	Direction
⋮	⋮	⋮	⋮	⋮	⋮
34713.685155243	1	206833312	100	118600	-1
34714.133632201	3	206833312	100	118600	-1
⋮	⋮	⋮	⋮	⋮	⋮

Figure 2.2: Example of a LOBSTER message file. Source: [LOBSTER Data Structure](#).

order book file.

Ask Price 1	Ask Size 1	Bid Price 1	Bid Size 1	Ask Price 2	Ask Size 2	Bid Price 2	Bid Size 2	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1186600	9484	118500	8800	118700	22700	118400	14930	...
1186600	9384	118500	8800	118700	22700	118400	14930	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Figure 2.3: Example of a LOBSTER order book file. Source: [LOBSTER Data Structure](#).

The message file contains the following fields for each event: Timestamp (in seconds since midnight, with nanosecond precision), Event Type, Order ID, Order Size, Price, and Trade Direction. The order book file records the bid and ask prices and quantities at each of the L specified levels of the book. To be precise, for the LOB dynamics part, I used $L = 10$, retaining the ten best bid and ask price levels at each point in time.³

Following [Briola et al., 2024], in Table 2.1 I classify stocks into small-, medium- and large-tick categories. The classification is based on the average spread ($\langle\sigma\rangle$) measured in

²A high-frequency LOB dataset derived from NASDAQ's Historical TotalView-ITCH feed.

³Note that, for the second part (i.e., the SDAMH model) instead, I only used $L = 1$ instead, as in that case I am only interested in the best bid/ask prices and sizes. Moreover, while I consider the whole trading day in the first part of the LOB analysis where I prove the canonical microstructural stylized facts, in the SDAMH section of the empirical analysis I only retain observations occurring between 09:35 and 15:54 to account for the well-documented microstructure noise associated with the opening auction and the final minutes of trading.

2.2. Introduction

units of the relative exchange tick-size (θ), denoted by $\langle\sigma\rangle/\theta$ and defined as follows: small-tick stocks are defined as those stocks for which $\langle\sigma\rangle \geq 3\theta$, medium-tick stocks satisfy $1.5\theta \lesssim \langle\sigma\rangle \lesssim 3\theta$, whereas large-tick stocks are characterized by $\langle\sigma\rangle \lesssim 1.5\theta$.

Table 2.1: The sample : for each stock, I report the ticker, the extended name, the sector according to the GICS⁴ taxonomy, and the Tick class following the classification proposed in [Briola et al., 2024].

Ticker	Stock Name	Sector	Tick Class
MSFT	Microsoft Corporation	Information Technology	Medium-tick
IBM	International Business Machines Corporation	Information Technology	Small-tick
AMZN	Amazon.com, Inc.	Consumer Discretionary	Large-tick
BAC	Bank of America Corporation	Financials	Large-tick
GS	The Goldman Sachs Group, Inc.	Financials	Small-tick
CSCO	Cisco Systems, Inc.	Information Technology	Large-tick

For illustrative purposes, Figure 2.4 reports the distinction between a small-tick and a large-tick stock by displaying the distribution of depth (in dollar value) across $L = 10$ price levels through a snapshot of the LOBs for two contrasting cases of randomly selected stocks (one per main category): in the upper panel (i.e., the small-tick stock, GOOG), the order book exhibits extreme sparsity, with a spread/tick-size ratio of ≈ 29.62 units on June 21, 2012, and liquidity concentrated at only a handful of discrete price points near the mid-price (\$574.4 - \$576.00). Most price levels contain zero depth, creating large gaps in the distribution. The few populated price points hold substantial notional value (almost to \$200,000 on the bid side), but the overall distribution is highly discontinuous. In contrast, in the lower panel, INTC exhibits a spread/tick-size ratio of ≈ 1.23 units for the same date. Its order book is significantly denser, with depth distributed across numerous price levels spanning \$27.1 to \$27.450. Both bid and ask sides show substantial liquidity at many prices. The distribution is more continuous and symmetric around the mid-price (\$27.34).

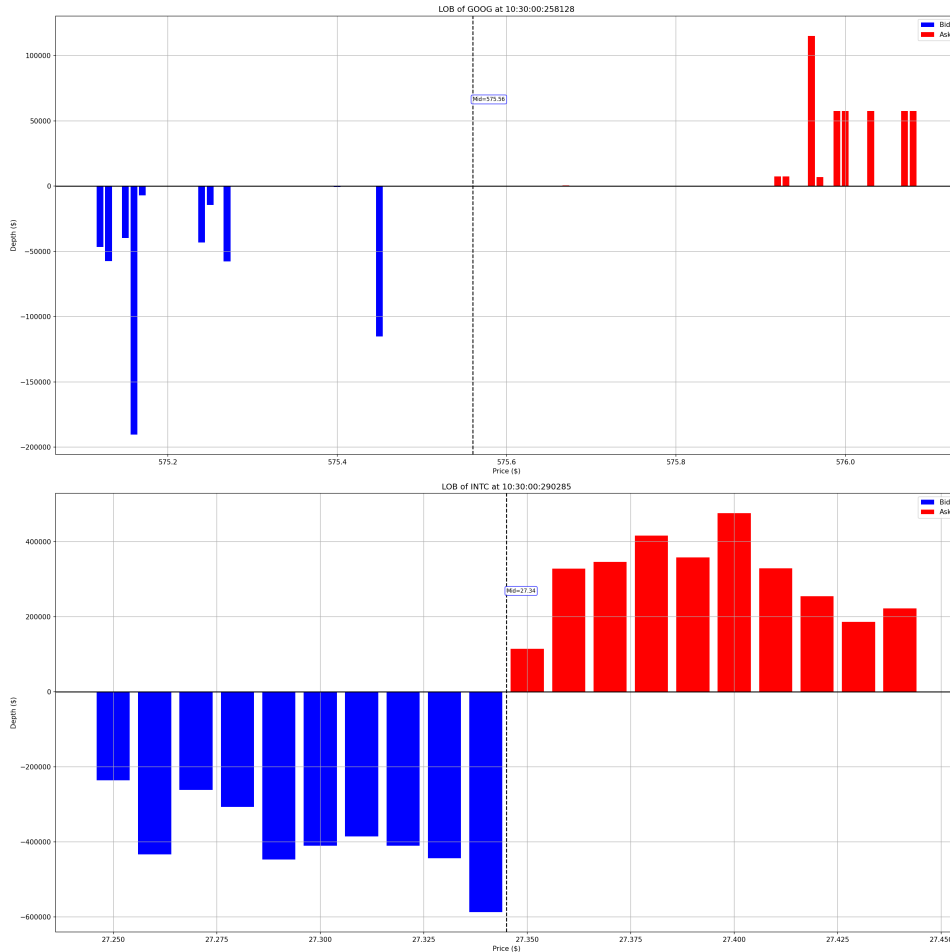


Figure 2.4: Snapshot of the LOB, zoomed into a region around the mid price, for GOOG (small tick: ABOVE) and INTC (large tick: BELOW) around 10:30 am on June 21, 2012. The typical LOB of small tick stocks is very sparse and the bid-ask spread is a multiple of one tick. The typical LOB of large tick stocks is very dense around the mid-price and the typical bid-ask spread is equal to one tick.

2.2.4 Summary Statistics

In this section, I present the main summary statistics and briefly discuss them. Table 2.2 shows that stocks span a wide range of price levels, from BAC at 34.35 to GS at 352.86, reflecting heterogeneity in nominal prices across the sample. Notice that the Average bid-ask spreads (in dollars) vary considerably: AMZN, CSCO, and BAC exhibit spreads close to 1 tick ($0.014/0.01 = 1.14$, $0.011/0.01 = 1.11$ and $0.012/0.01 = 1.22$ respectively), thus classifying them as large-tick stocks in the sense of [Dayri and Rosenbaum, 2015]. In contrast, GS and IBM show an average spread of 17.8 and 3.33 ticks respectively, placing them firmly in the small-tick regime. Finally, MSFT occupies an intermediate positions as a medium-tick stock.

In terms of Average Volume at Best—the total quantity available at best bid and best ask—large-tick stocks occupy the first positions, followed by medium- and small- tick stocks.

2.2. Introduction

This is not surprising, as market-makers tend to post larger quantities when the spread offers sufficient compensation. As opposed, GS and IBM (i.e., small-tick stocks) show the thinnest books at the top of the queues.

Hidden orders use is generally low for most stocks, with hidden volume fractions ranging from 20% (MSFT) up to 66.83% for GS. This high value for GS reflects the thinner nature of its order book : when dealing with severely small-tick stocks, traders have greater incentive to conceal order size.

Considering the distribution of orders placed, Limit order submissions average between 31.7 (GS) and 783.1 (AMZN) thousand per day, with cancellations tracking submissions closely (cancellation-to-submission ratios are around 80%) across all stocks, consistent with the well-documented high cancellation rates in modern electronic markets. For what concerns order flow, one can immediately spot that the fraction of buy market orders hovers around 50% for all stocks, consistent with the absence of directional bias over the sample period.

Table 2.2: Average price (\$), Average Spread (in dollars), Average Daily Volume at Best, Average Hidden Volume, Average Hidden Volume fraction, Average Daily Limit Orders, Average Daily Cancellations, Average Buy Limit Orders fraction, Average Daily Market Orders, Average Market Orders fraction. All estimates are computed using intraday data over the trading days of January 2023. Red, green, blue and green colors indicate, respectively, large-, medium-, and small-tick stocks.

	AMZN	CSCO	IBM	MSFT	BAC	GS
$\langle Price \rangle$	93.844	47.988	141.252	236.641	34.352	352.864
$\langle Spread(\$) \rangle$	0.014	0.011	0.033	0.022	0.012	0.178
$\langle VolumeAtBest \rangle$	108.047	203.338	32.566	81.215	229.502	51.916
$\langle TotalDailyVolume \rangle$	673.539	112.628	64.749	1156.022	90.890	99.128
$\langle HiddenVolume \rangle$	162.543	30.259	17.857	239.514	34.213	68.634
$\langle HiddenVolumeFrac(\%) \rangle$	24.170	26.648	26.517	20.154	37.491	66.803
$\langle LOs \rangle$	783.110	198.048	65.359	748.846	183.656	31.695
$\langle Canc \rangle$	630.827	158.532	48.017	635.018	142.016	21.117
$\langle BuyLOsFrac(\%) \rangle$	49.084	49.487	49.524	50.258	50.871	48.646
$\langle MOs \rangle$	76.568	19.256	10.227	95.814	14.845	8.539
$\langle BuyMOs \rangle$	52.202	51.441	49.672	51.000	49.669	47.452

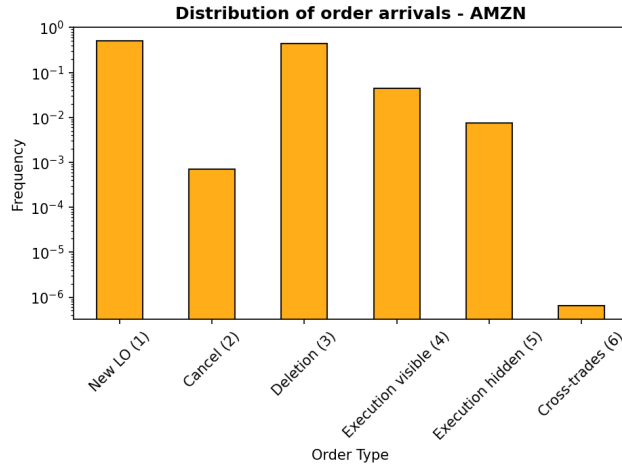
2.2.5 Event Types and Trade Direction

LOBSTER classifies each order-level event into one of six types, summarized in Table 2.3. The analysis focuses on Types 1 through 5, which together account for the vast majority of observed events (see Figure 2.5)⁵. In line with expectations, one can immediately notice how Limit Orders and Deletions clearly dominate (both $\approx 50\%$ of all events). Market Orders (4) and Hidden Orders (5) come second, while cancellations are rather rare, and cross-trades are essentially negligible ($\approx 10^{-6}$)

⁵For the sake of completeness, the reader is redirected to Appendix B where analogous results are reported for all other equities.

Table 2.3: LOBSTER Event Types

Event Type	Label	Description
1	New Limit Order	Submission of a new limit order to the book
2	Cancellation	Partial cancellation of an existing limit order
3	Deletion	Full removal of an existing limit order
4	Execution (Visible)	Execution of a visible limit order against an incoming market order
5	Execution (Hidden)	Execution of a hidden (iceberg) limit order
6	Cross Trade	Off-book cross trade; excluded from the analysis

**Figure 2.5:** Distribution of order arrivals for AMZN on March 7,2023 on a log scale.

The *TradeDirection* field takes values $+1$ (buy limit order) and -1 (sell limit order). For execution events (Types 4 and 5), LOBSTER follows the direction of the resting limit order (rather than the aggressive market order): a *TradeDirection* of $+1$ indicates that a resting *buy* limit order was executed, which corresponds to an incoming *sell* market order, and vice-versa. This is crucial both for the construction of the dataset as well as for the interpretation of results. Following this convention, throughout this paper, I therefore reverse the sign for execution events to align with the standard convention of buyer- and seller-initiated trades, such that $+1$ denotes a buyer-initiated market order and -1 a seller-initiated one.

2.2.6 LOB Analysis

Before proceeding to estimate the SDAMH model, I first demonstrate the presence of financial regularities that are well-known to relate to most financial time series (i.e., the stylized facts), and of the main documented microstructural empirical properties of LOBs.⁶ Recall that, since many features of an LOB state follow a strong intra-day pattern, it is good practice -and in line with the related financial literature- to apply standard data filters in order to remove potentially erroneous observations. For this reason, I *(i)* exclude the first and last five minutes of trading (09:30–09:35; 15:55–16:00) to remove the noisy market-open period during which the LOB is being populated, and *(ii)* drop events with missing or zero prices and sizes. Notice that this procedure is carried out both when proving stylized facts

⁶See Figures from B.5 to B.10 in Appendix B for the canonical stylized facts of financial markets.

as well as when estimating the SDAMH. Instead, while when analyzing the main market microstructure properties of LOBs, I instead account for the whole trading day.

Microstructural Properties

One of the most common stylized facts in market microstructure is that buyer-initiated orders tend to push prices upward, while seller-initiated orders exert downward pressure. Figure 2.6 illustrates this for Bank of America (BAC) over January 2023⁷. In a similar fashion to [Bouchaud et al., 2018], I consider both limit and market orders. I count time t in *order-event* time, in which time t increments by 1 every time a new limit/market order arrives. Consequently, b_t and a_t denote the values of the bid- and ask-prices immediately before the arrival of the n^{th} limit or market order. In this regard, several features are worth noting : first, buy and sell market orders exhibit a strong, permanent price response that plateaus at approximately $+0.006$ and -0.0035 respectively, consistent with the canonical result that aggressive orders are the primary driver of price discovery. Second, limit orders display a qualitatively similar but quantitatively weaker response, plateauing at roughly $+0.002$ and -0.0018 .

For completeness, in Figure 2.7 I show the shape of the response function $\mathcal{R}(\ell)$ for my six stocks. The sharp rise at $\mathcal{R}_\infty = \mathcal{R}(\ell \rightarrow \infty)$, which is 2-10x larger than the initial value $\mathcal{R}(1)$, is due to the autocorrelation of market orders, which tends to increase the price for a certain fraction of time. Figure 2.8 displays the intraday average values of Market- and Limit Order Volumes (in millions of \$), the Volume at Best (in shares), as well as the Spread (in \$) for a large- (left) and small- (right) tick stock over a sample month (Jan 3 - Feb 14, 2023). In Panels 1 and 2 of Figure 2.8, I display the average total volume of aggressive orders (MOs) and resting orders (LOs) sampled every two-minute throughout each of the first 30 trading days of 2023 for BAC. As in [Bouchaud et al., 2018], one can notice that, in both large- and small-tick stocks, the trading activity exhibits a *U-shape* profile, or better a “*J-shape* profile” with asymmetric peaks at the beginning and at the end of the day and a minimum at around midday. More in depth, the activity during the busiest periods is about four 6/7x greater than activity during the quietest periods, the reason for this being that—at every market opening—it takes a considerable amount of minutes/hours before the market can fully digest company news/global events revealed during the overnight period, which causes more intense spikes right at or shortly after the opening. Analogously, the sharp increase toward the close is likely driven by inventory management considerations, as both traders and market-makers rush to unwind intraday positions to minimize overnight exposure. The third panel, which reports the volume path, provides additional support to the LO and MO dynamics. Volume is comparatively low at the open, in line with the reduced order activity documented above, and exhibits an approximately sevenfold increase near the close. Panel 4 shows the spread dynamics⁸ : notice how it gets thinner throughout the trading day, proportionally more markedly for small-tick stocks, while more moderately for large-tick stocks. According to [Bouchaud et al., 2018], the narrower the bid–ask spread, the lower the degree of the adverse selection faced by liquidity providers, as overnight news gets progressively digested by market participants. Figure 2.8 suggests that liquidity is much more scarce at the open than it is at the close, which indicates higher average cost of

⁷The results of this preliminary statistical analysis of LOBs has been performed for all stocks. Results are available upon request.

⁸To ensure tractability, the bid–ask spread is computed at randomly selected event-time observations. Alternative definitions include sampling in calendar time or measuring the spread immediately before trade execution. Since these approaches rely on different sampling schemes, they can produce systematically different estimates.

immediacy (i.e. of submitting a market order) in the morning than later in the day. Lastly, recall that a wider spread forces market orders to penetrate deeper in the book to match their requested volume.

Figure 2.9 illustrates the distribution of spread (in ticks) for BAC over the selected period: for large-tick stocks, the spread distribution is highly concentrated at the minimum spread (which is equal to one tick, θ), with infrequent moments when the spread widens to 2θ or, in extreme cases, 3θ (or larger). Notice how the right tail of the distribution approximately exhibits an exponential decay.

Figure 2.10 shows the empirical cumulative density functions (ECDFs) of Limit Orders and Market Orders' sizes (in lots). While the most frequent order sizes are relatively small (a few thousands of dollars), large trades occur with non-negligible probability. Note that the upper tail of the ECDF appears to decay approximately according to a power law, with an exponent scattered around $-5/2$ (as represented by the dotted line). Overall, this evidence confirms that order sizes are not clustered around some average sizes, rather clustered around a representative scale.

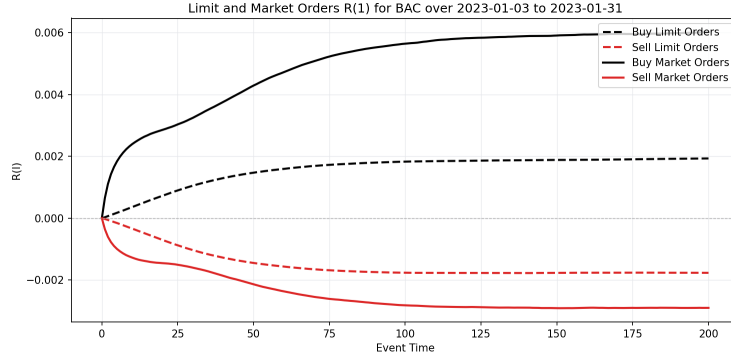


Figure 2.6: Lag-1 Response function for BAC (Jan 3 - Jan 31, 2023)

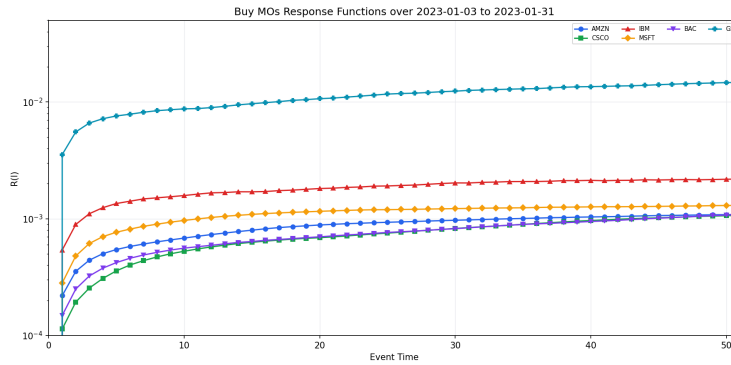


Figure 2.7: Response function $\mathcal{R}(\ell)$ in market order event-time, for (circles) AMZN, (squares) CSCO, (upper-looking triangles) IBM, (downward-looking triangles) BAC, (cross) GS, (diamond) MSFT. (Jan 3 - Jan 31, 2023)

2.2. Introduction

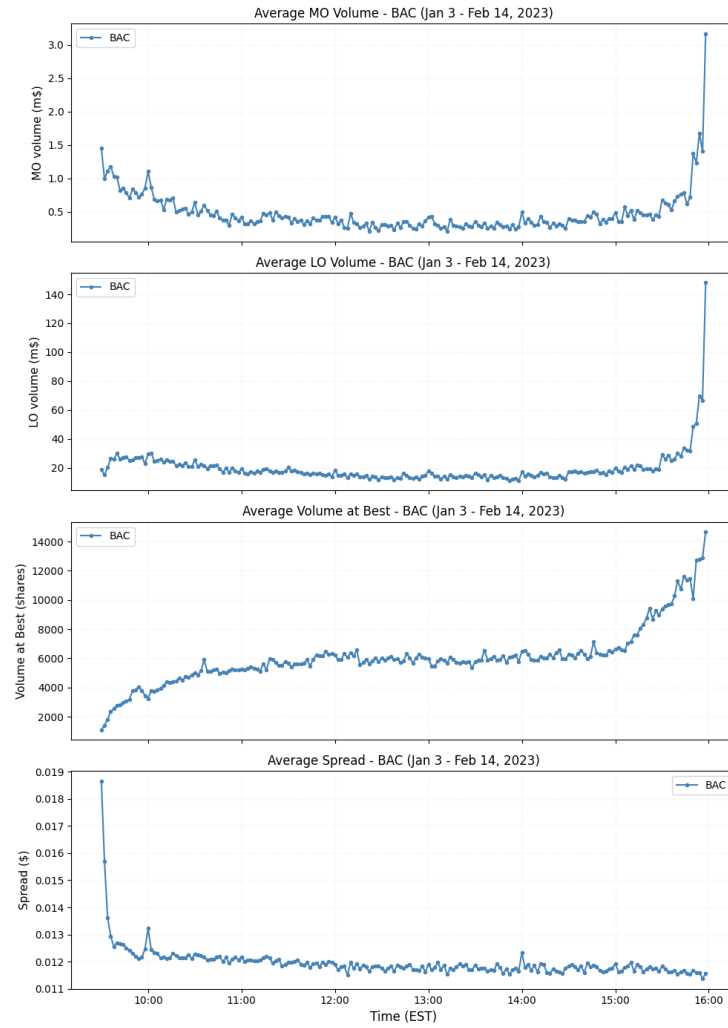


Figure 2.8: Average Limit Orders (LOs), Average Market Orders (MOs), Volume, and Spread for BAC over the sampled period (Jan 3 - Feb 14, 2023)

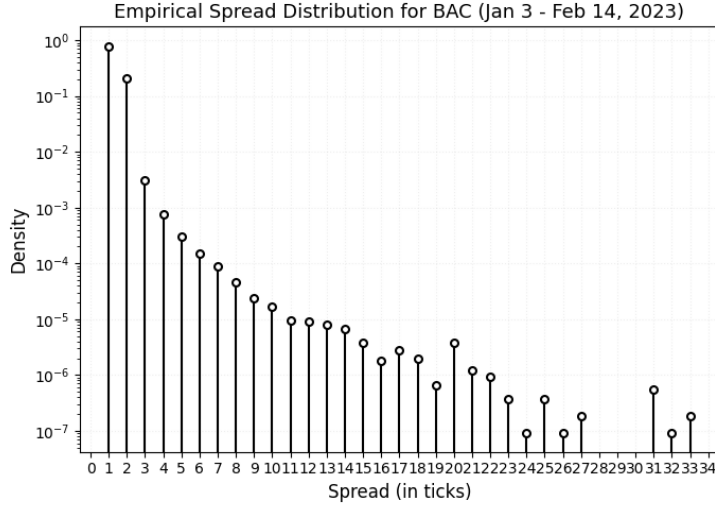


Figure 2.9: Empirical distributions of the bid-ask spread $s(t) = a(t) - b(t)$, measured at the times of events at the best quotes, for BAC

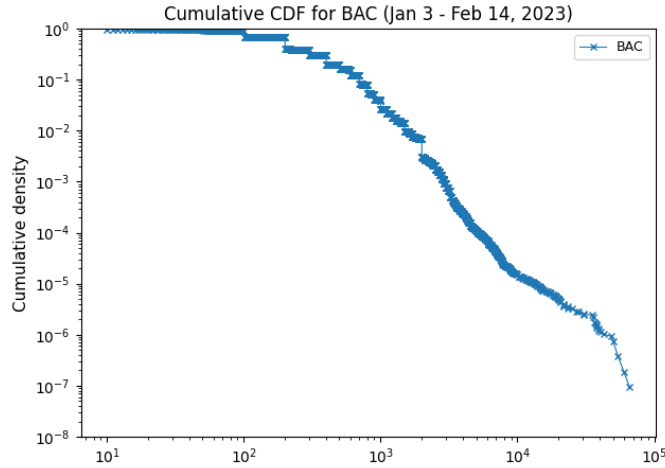


Figure 2.10: ECDF log-log plot for the sizes of market orders and limit orders, expressed in number of shares for BAC.

2.2.7 Score-Driven Aggregated Modified Hasbrouck

To study the relationship between trades and prices around heterogeneous news, I use a novel methodology by [Campigli et al., 2022], who propose a *Score-Driven Aggregated Modified Hasbrouck* (hereafter referred to as SDAMH) model to measure the dynamics of liquidity through a time-varying version of the instantaneous price impact around news events. In this respect, I first introduce the foundational Structural Vector Autoregressive (SVAR) model introduced by [Hasbrouck, 1991], which jointly models mid-quote returns and trade signs to measure price impact and information content:

$$\begin{aligned} r_t &= \mu_1 + \sum_{i=1}^p a_i r_{t-i} + \sum_{i=0}^p b_i x_{t-i} + v_{1,t} \\ x_t &= \mu_2 + \sum_{i=1}^p c_i r_{t-i} + \sum_{i=1}^p d_i x_{t-i} + v_{2,t} \end{aligned} \tag{2.7}$$

where r_t is the mid-point price return (i.e., the mid-point right after the trade) at event-time t ; $x_t \in \{-1, +1\}$ is a trade sign dummy indicator for orders (sell/buy) at event-time t ; b_0 captures the instantaneous price impact of a trade; $v_{1,t}$ and $v_{2,t}$ are noise terms with finite variances defined, respectively, as σ_1^2 and σ_2^2 ; finally, $Cov(v_{1,t}, v_{2,t}) = 0$

The analytical tractability of this linear model—whose parameters can be estimated using OLS—allows for the possibility of deriving the Impulse Response Function (IRF), as well as the asymptotic (i.e., long-run, cumulative IRF) effect of a trade on prices in a closed-form solution [Lütkepohl, 2006]. In financial terms, this represents the information content of a trade. Nevertheless, the drawback of this model is that it assumes a constant information content of trades. In reality, this parameter is subject to frequent oscillations (i.e., incoming news or hectic intraday trading dynamics). Moreover, the assumption of constant liquidity has long been debated and dismantled by the literature. For this reason, the authors adopt a Dynamic Conditional Score (DCS) framework⁹ from [Creal et al., 2013] and [Harvey, 2013].

Before the formal specification, it is worth stating the intuition in plain terms. The idea of the score-driven approach is to let a time-varying parameter be updated by the score of the model’s own likelihood, so that each revision is proportional to how surprised the model was by the latest observation, scaled by how informative that surprise is. The familiar GARCH model is the special case in which the conditional density is Gaussian and the time-varying parameter is the variance, where the squared residual is precisely the score; the score-driven framework generalises this logic to any parameter and any conditional density. Two simpler alternatives are worth ruling out explicitly. A rolling-window regression would allow b_0 to vary, but weights every observation inside the window equally and discards the model’s own structure in forming the update. A Kalman filter would use that structure, but requires a linear Gaussian state-space representation, which is unavailable here because the trade-sign equation is Bernoulli with a logistic link. The score-driven recursion handles the non-Gaussian density natively and remains observation-driven, so the likelihood is available in closed form and can be maximised directly, without simulation or particle filtering.

Specifically, let $\{z_t\}_{t=1}^T$ be a sequence of observations with $z_t \in \mathbb{R}^M$, and let $P(z_t | f_t)$ denote the conditional density, where $f_t \in \mathbb{R}^W$ is a vector of time-varying parameters. The score is defined as :

$$\nabla_t := \frac{\partial \log P(z_t | f_t)}{\partial f_t'} \tag{2.8}$$

Under the DCS specification, the parameter dynamics evolve according to

$$f_{t+1} = \omega + \beta f_t + \alpha s_t, \quad s_t := \mathcal{I}_{t|t-1}^{-1} \nabla_t \tag{2.9}$$

where $\mathcal{I}_{t|t-1}$ denotes the Fisher information matrix, ω is a W -dimensional vector, and α and β are $W \times W$ matrices of static parameters.

In this framework, the authors assume the parameters are time-varying according to DCS dynamics. To do so, they first modify the existing SVAR model to account for two crucial

⁹DCS are part of GAS models—Generalized Autoregressive Score Models—see an example of these at <https://www.gasmodel.com/>

features: *i*) defining the conditional probability density function of the model $P(z_t | f_t)$; and *ii*) the binary nature of x_t , which entails a complex temporal structure for $v_{2,t}$.

To specify the probability density of the observables, they cleverly replace the second equation in Eq. 2.7 with an inverse logistic map $\mathbb{P}(x_t = +1 | \mathcal{F}_t)$, consistent with the binary nature of a trade and ensuring non-linearity in the model. The model then becomes :

$$r_t = \mu_1 + \sum_{i=1}^p a_i r_{t-i} + \sum_{i=0}^p b_i x_{t-i} + v_{1,t}$$

$$\mathbb{P}(x_t = +1 | \mathcal{F}_{t-1}) := \pi_t = \text{Logit}^{-1} \left(\mu_2 + \sum_{i=1}^p c_i r_{t-i} + \sum_{i=1}^p d_i x_{t-i} \right) \quad (2.10)$$

in which two drift terms (μ_1 and μ_2) have been added to include possible drifts of prices and trade signs. Assuming $v_{1,t} \sim \mathcal{N}(0, \sigma_2^2)$ and $\text{Logit}^{-1}(z) = (1 + e^{-z})^{-1}$ is the inverse logistic map of z .

To address the persistence in order flow documented by [Lillo and Farmer, 2003], [Bouchaud, 2009; Bouchaud et al., 2018] and [Toth et al., 2015], the authors propose an extension of the Hasbrouck model using an aggregation technique from [Corsi, 2009] and readapted to market microstructure by [Hasbrouck, 2021] outlined as:

$$x_t^{(L_1)} = \frac{1}{L_1 - 1} \sum_{i=2}^{L_1} x_{t-i}, \quad x_t^{(L_2)} = \frac{1}{L_2 - L_1} \sum_{j=L_1+1}^{L_2} x_{t-j} \quad (2.11)$$

where and $L_1, L_2 \in \mathbb{N}$ stand for the number of past lags considered for the aggregation. In line with their methodology, I also set $L_1 = 10$ and $L_2 = 100$. The model now becomes:

$$r_t = \mu_1 + a_1 r_{t-1} + b_0 x_t + b_1 x_{t-1} + \bar{b}_{10} x_t^{(10)} + \bar{b}_{100} x_t^{(100)} + v_{1,t}$$

$$x_t = \mu_2 + c_1 r_{t-1} + d_1 x_{t-1} + \bar{d}_{10} x_t^{(10)} + \bar{d}_{100} x_t^{(100)} + v_{2,t} \quad (2.12)$$

Notice how this model specification captures the long-memory in order flow, which is well-documented stylized fact in market microstructure. In addition, it also prevents the model from hitting the 'curse of dimensionality', by ensuring a finite number of lags is employed.

Finally, to capture time-varying liquidity and market impact, the authors extend the AMH model by allowing the instantaneous impact parameter $b_{0,t}$ to evolve according to Dynamic Conditional Score (DCS) dynamics [Creal et al., 2013; Harvey, 2013]. All remaining parameters are held constant at their static level.

The authors first define :

$$\mu_t^{(1)} := \mu_1 + a_1 r_{t-1} + b_{0,t} x_t + b_1 x_{t-1} + \bar{b}_{10} x_t^{(10)} + \bar{b}_{100} x_t^{(100)}$$

$$\mu_t^{(2)} := \mu_2 + c_1 r_{t-1} + d_1 x_{t-1} + \bar{d}_{10} x_t^{(10)} + \bar{d}_{100} x_t^{(100)} \quad (2.13)$$

Then, according to the DCS specification, they assume that $b_{0,t}$ and σ_t^2 follow the dynamics

$$\begin{pmatrix} b_{0,t+1} \\ \log(\sigma_{t+1}^2) \end{pmatrix} = \begin{pmatrix} \omega \\ \varrho \end{pmatrix} + \begin{pmatrix} \beta & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} b_{0,t} \\ \log(\sigma_t^2) \end{pmatrix} + \begin{pmatrix} \alpha & 0 \\ 0 & \gamma \end{pmatrix} \begin{pmatrix} s_{t,1} \\ s_{t,2} \end{pmatrix} \quad (2.14)$$

where the scaled score s_t is defined in Eq. (2.8). The conditional densities are given by

$$p_r(r_t \mid x_t, b_{0,t}, \sigma_t^2, \mathcal{F}_{t-1}, \theta) = \frac{e^{-\frac{1}{2\sigma_t^2}(r_t - \mu_t^{(1)})^2}}{\sqrt{2\pi\sigma_t^2}} \quad (2.15)$$

$$p_x(x_t \mid \mathcal{F}_{t-1}, \theta) = \text{Logit}^{-1}\left(\mu_t^{(2)}\right)^{\frac{1+x_t}{2}} \left(1 - \text{Logit}^{-1}\left(\mu_t^{(2)}\right)\right)^{\frac{1-x_t}{2}}$$

Following the DCS specification, the dynamics of $b_{0,t}$ and $\log(\sigma_t^2)$ are driven by a score-driven recursion of the form described in Eq. 2.9, where the scaled score is derived from the conditional densities of returns and trade signs. The return component follows a Gaussian density centred on $\mu_t^{(1)}$, while the trade sign component follows a Bernoulli distribution parameterized via the inverse logistic transformation.

After some straightforward computations, the log-likelihood function is given by

$$\begin{aligned} \mathcal{L} &:= \sum_{t=1}^T \mathcal{L}_t(r_t, x_t \mid \mathcal{F}_{t-1}, \theta) \\ &= -\frac{T}{2} \log 2\pi - \frac{1}{2} \sum_{t=1}^T \log \sigma_t^2 - \frac{1}{2} \sum_{t=1}^T \frac{(r_t - \mu_t^{(1)})^2}{\sigma_t^2} \\ &\quad + \sum_{t=1}^T \frac{1+x_t}{2} \log \left(1 + e^{-\mu_t^{(2)}}\right)^{-1} + \sum_{t=1}^T \frac{1-x_t}{2} \log \left(1 - \left(1 + e^{-\mu_t^{(2)}}\right)^{-1}\right) \end{aligned} \quad (2.16)$$

thus

$$\nabla_t = \begin{pmatrix} \frac{\partial \mathcal{L}_t}{\partial b_{0,t}} \\ \frac{\partial \mathcal{L}_t}{\partial \sigma_t^2} \end{pmatrix} = \begin{pmatrix} \frac{x_t}{\sigma_t^2} (r_t - \mu_t^{(1)}) \\ \frac{(r_t - \mu_t^{(1)})^2 - \sigma_t^2}{2\sigma_t^4} \end{pmatrix} \quad (2.17)$$

and

$$\mathcal{I}_{t|t-1}^{-1} = \mathbb{E}_{t-1} [\nabla_t \nabla_t^\top]^{-1} = \begin{pmatrix} \sigma_t^2 & 0 \\ 0 & \sigma_t^2/2 \end{pmatrix} \quad (2.18)$$

After computing the score and Fisher information matrix in closed form, the update equations simplify to :

$$b_{0,t+1} = \omega + \beta b_{0,t} + \alpha x_t (r_t - \mu_t^{(1)}) \quad (2.19)$$

$$\log(\sigma_{t+1}^2) = \log(\sigma_t^2) + \gamma \left[\frac{(r_t - \mu_t^{(1)})^2}{\sigma_t^2} - 1 \right] \quad (2.20)$$

The update for $b_{0,t}$ is an autoregressive recursion in which the impact parameter is revised at each event time in proportion to the signed prediction error $x_t(r_t - \mu_t^{(1)})$. In line with

their proposed methodological framework¹⁰, I refer to this final model as the Score-Driven Aggregated Modified Hasbrouck (*SDAMH*) model.

Recall that, consistent with the paper’s focus, only the estimated time-varying instantaneous market impact curve is used in what follows.

Estimation Procedure

I employ a three-stage estimation procedure:

1. **Stage 1 (AH estimation):** Estimate the AH model via OLS and logistic regression to obtain initial parameter values.
2. **Stage 2 (AMH estimation):** Use AH estimates as starting values for joint maximum likelihood estimation of the AMH model via standard Quasi-Newton optimization method. This provides improved estimates under the assumption of constant b_0 .
3. **Stage 3 (SDAMH estimation):** Initialize the SDAMH model with AMH estimates, setting $\alpha = \gamma = 0.01$ as starting values. Optimize the full SDAMH log-likelihood with parameter bounds: $\alpha \in [0.001, 0.1]$ and $\gamma \in [10^{-6}, 0.1]$ to ensure stability.

All optimization uses the L-BFGS-B algorithm with convergence tolerances of 10^{-10} for both function and gradient values. I implement the SDAMH log-likelihood using *Numba just-in-time* (JIT) compilation to achieve computational efficiency for the recursive filtering step.

2.3 Empirical Results

The objective of this paper is to present the time-varying instantaneous impact around a wide set of news items. As an illustrative example, Figure 2.11 displays the high-frequency evolution of the best bid, best ask, and mid-price over the full trading day, with an inset zooming into the one-hour window surrounding the FOMC announcement (13:30 - 14:30) on June 16th, 2021 for AMZN. The sharp mid-price decline at 14:00 reflects the immediate market response to the announcement, followed by sustained volatility as participants reassess their positions in light of the new information.

¹⁰Model selection based on the Bayesian Information Criterion and out-of-sample likelihood confirms that the integrated DCS specification, corresponding to $\omega = 0$ and $\beta = 1$, provides the best fit to the data.

2.3. Empirical Results

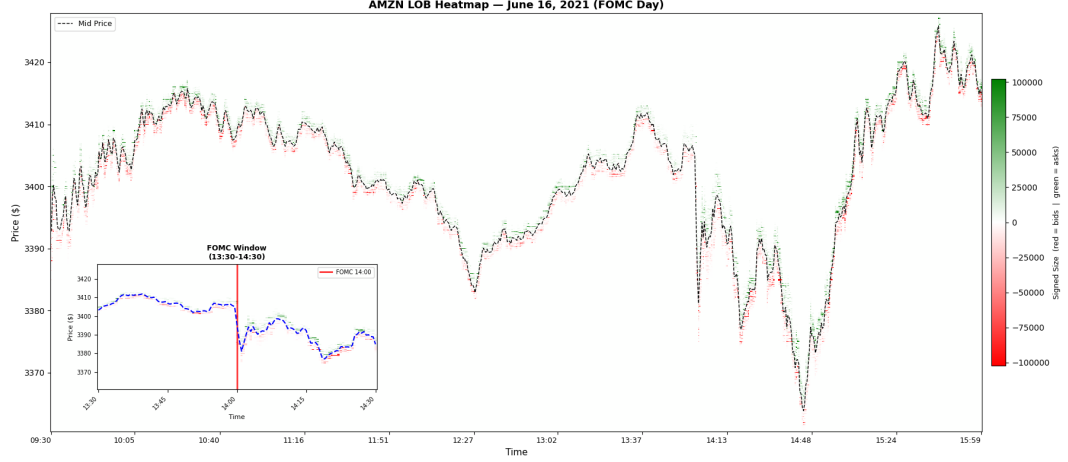


Figure 2.11: Example of a visualization of a LOB, i.e. the collection of limit orders (active orders). Best bid (offer) prices are displayed in red (green)

To complement Figure 2.11, I also employ the Order Flow Imbalance (OFI) measure of [Cont et al., 2014]. For each message event n , the OFI contribution e_n captures the net change in depth at the best bid and ask relative to the previous event:

$$e_n = \underbrace{\mathbf{1}_{P_n^B \geq P_{n-1}^B} q_n^B - \mathbf{1}_{P_n^B < P_{n-1}^B} q_{n-1}^B}_{\text{bid-side contribution}} - \underbrace{\left(\mathbf{1}_{P_n^A \leq P_{n-1}^A} q_n^A - \mathbf{1}_{P_n^A \geq P_{n-1}^A} q_{n-1}^A \right)}_{\text{ask-side contribution}}, \quad (2.21)$$

where P_n^B (P_n^A) denotes the best bid (ask) price at event n , and q_n^B (q_n^A) denotes the corresponding depth. The bid-side contribution increases when the best bid price rises or additional depth arrives at the current best bid, and decreases when the best bid falls or depth is removed. The ask-side contribution exhibits symmetric behavior. Consequently, $e_n > 0$ indicates net buying pressure, while $e_n < 0$ indicates net selling pressure.

The cumulative OFI over a time window $[t-h, t]$ aggregates these event-level contributions:

$$\text{OFI}(t, h) = \sum_{n: t-h < t_n \leq t} e_n. \quad (2.22)$$

For visual purposes, I aggregate OFI over non-overlapping 300-second intervals for examining FOMC event windows. Figure 2.12 illustrates OFI dynamics for Amazon on June 16, 2021, an FOMC announcement day. The upper panel displays instantaneous OFI in 5-minute intervals: green (red) shading indicates net buying (selling) pressure, while the black line (right axis) tracks the mid-price. The lower panel shows cumulative OFI over the trading day. Several worth-commenting patterns are visible: first, instantaneous order flow correlates strongly with contemporaneous price movements. Green spikes coincide with price increases around 09:15-09:30 and 15:15-15:45, while red spikes are followed by declines during 12:00-13:00 and 14:00-14:15. Second, the 14:00 FOMC announcement triggers brief selling pressure, with the mid-price reaching its intraday low of \$3,365 near 14:30. Third, and most strikingly, the final trading hour exhibits extraordinary net buying: cumulative OFI surges from 50,000 to 330,000 shares, driving the mid-price from \$3,370 to \$3,420 ($\approx 1.48\%$, or 148 bp). This closing rally of 280,000 net shares suggests large-scale institutional order execution, likely via time-weighted algorithms. Overall, Figure 2.12 confirms that OFI serves as a strong contemporaneous predictor of price movements, validating its use as a measure of latent order flow imbalance.

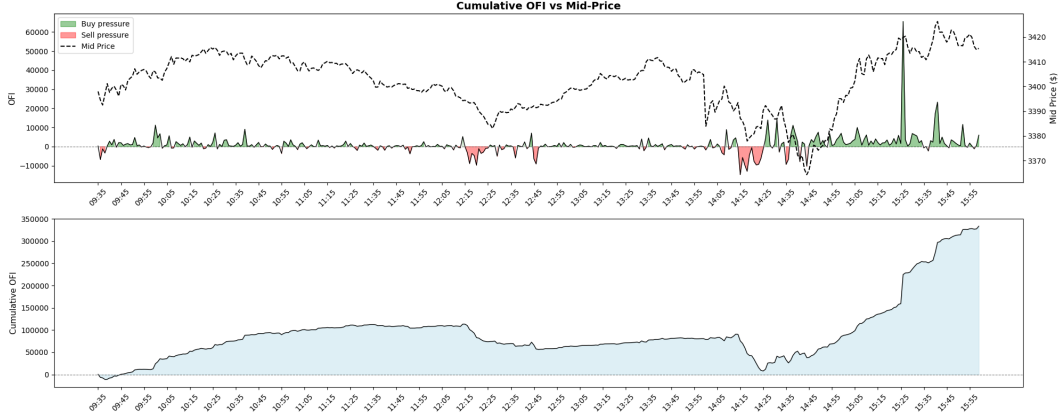


Figure 2.12: Order Flow Imbalance and Price Dynamics for Amazon (June 16, 2021, FOMC Day). *Upper panel:* Instantaneous OFI in 5-minute intervals (green = buying pressure, red = selling pressure) overlaid with mid-price (black line, right axis). *Lower panel:* Cumulative OFI over the trading day. The FOMC announcement at 14:00 triggers selling pressure followed by a massive closing rally (15:00–16:00) with 280,000 shares of net buying flow.

2.3.1 Data calibration

For each stock, the SDAMH model is calibrated separately on each trading day using the full intraday transaction record for that date. This choice is motivated by the event-driven nature of the analysis : since each day constitutes a specific daily occurrence, other data calibration methods (such as a rolling-window) would blur the cross-day variation in $b_{0,t}$, which my framework tries to isolate. Day-by-day calibration, instead, ensures that the static parameters are always estimated on the same information set as the event under study. Notice that the findings are broadly cross-sectionally consistent and stable across trading days, as evidenced by the overall uniformity reported in Table 2.4, which reports the estimated static parameters (*Est*) of the SDAMH model with p-values in parentheses for all six stocks.

2.3.2 SDAMH Estimation Results

The intercept of the return equation, μ_1 , captures any global drift in mid-price returns. It is statistically significant and positive for most stocks, with the exception of GS, for which it is negative, indicating the presence of a global drift. The intercept of the trade sign equation, μ_2 , governs the unconditional frequency of buyer-initiated trades. Consistent with the summary statistics reported in Table 2.2, μ_2 is close to zero and negative for all stocks except GS, corroborating the near-symmetric distribution of buy and sell order flow.

The autoregressive coefficient a_1 is positive for most stocks, indicating positive short-term return autocorrelation, with the exception of BAC and GS, for which the negative sign signals short-term mean reversion in mid-price returns. Regarding the lagged impact of aggregated past trades on future returns, the coefficients b_1 , $\bar{b}_{10}$, and $\bar{b}_{100}$ are overall strongly significant, with isolated exceptions primarily concerning the significance of $\bar{b}_{10}$. Their signs tend to be negative for most stocks, with the exception of BAC, CSCO, and IBM, suggesting that the directional pressure of past order flow on returns partially reverses at longer horizons. Consistent with [Campigli et al., 2022], the absolute magnitude of b_1 exceeds that of $\bar{b}_{10}$ and $\bar{b}_{100}$ across all stocks, confirming that the contemporaneous impact of trades

on returns dominates over lagged contributions. The strong and persistent significance of these coefficients indicates that past market order pressure exerts a non-negligible influence on future price dynamics at the high-frequency scale.

Turning to the trade sign equation, c_1 measures the feedback of past returns on the current trade direction, capturing the impact of past returns on current (or recent) trades. The estimated c_1 is negative and highly significant for all stocks, implying that price increases (decreases) are systematically followed by a higher proportion of sell (buy) orders, consistent with a contrarian response of market participants to short-term price movements. The persistence of trade signs is confirmed by the positive and significant estimates of d_1 , $\bar{d}_{10}$, and $\bar{d}_{100}$ for most stocks, indicating that buy (sell) orders tend to follow preceding buy (sell) orders, in line with the long-memory property of order flow documented in Section 2.2.6. Notably, $\bar{d}_{10}$ is negative for most stocks with the exception of BAC and GS, suggesting a partial reversal in order flow persistence at intermediate horizons. Finally, the DCS parameters α and γ govern the score-driven fluctuations of the instantaneous price impact $b_{0,t}$. Though small, it is but statistically significant across all stocks, providing formal support for the time-varying parameter specification adopted in the SDAMH model.

2.3.3 Time-varying instantaneous price impact on heterogeneous news

Figure 2.13 - Panels (a) and (b) - display the filtered $b_{0,t}$ paths for IBM and BAC on June 14, taken as a representative FOMC date.¹¹ The two stocks exhibit structurally opposite baselines: IBM's $b_{0,t}$ remains positive throughout the session, while BAC's operates entirely in negative territory, reflecting a persistent sell-side dominance in informed order flow for the rate-sensitive financial stock. Despite this difference, both series share the same announcement-window signature. For BAC, $b_{0,t}$ decays monotonically from near zero at the open to a flat pre-announcement floor of approximately -4×10^{-5} , where it remains essentially frozen in the ten minutes preceding the release. At exactly 14:00 ET, the coefficient jumps discontinuously, crossing zero and briefly touching positive territory before settling at a less negative plateau of approximately -1.5×10^{-5} to -2×10^{-5} for the remainder of the event window. This sign crossing is directly interpretable as a resolution of directional uncertainty: the first rate hold after fifteen months of consecutive hikes is received as a relief event for a rate-sensitive institution, momentarily reversing the dominant order-flow imbalance. For IBM, the dynamics are qualitatively similar but $b_{0,t}$ remain in the positive domain: it declines gently from $\approx 1 \times 10^{-5}$ to $\approx 0.8 \times 10^{-5}$ in the pre-announcement window, then jumps sharply to roughly 3.5×10^{-5} at 14:00 ET—a near-threelfold increase - before oscillating at an elevated level throughout the subsequent window. In both cases, the jump is upward and discrete regardless of the policy content, confirming that the filter captures the resolution of uncertainty rather than a directional response to the announcement.

Moving to earnings announcements, Figure 2.13 - Panels (c) and (d) - show the filtered $b_{0,t}$ path for BAC on April 18 and for IBM on October 25, corresponding to the trading days following the Q1 2023 earnings releases. Earnings dates produce the highest absolute $b_{0,t}$ values in the entire sample, exceeding FOMC jumps by a substantial margin. The April 18 opening produces a peak of approximately 1.25×10^{-4} , the global maximum across both stocks and all event categories. Similarly, the October 25 earnings date yields the highest $b_{0,t}$ value across all IBM observations at approximately 8×10^{-5} , exceeding every FOMC date for that stock. The pattern across all earnings dates is consistent: $b_{0,t}$ opens at a large

¹¹Results are qualitatively unchanged for all other six FOMC, earnings and main controversy news dates of 2023. They are available upon request.

positive value reflecting the high information content of after-hours earnings releases, peaks within the first few minutes of trading as the opening auction clears the accumulated pre-market order flow, then decays rapidly as the earnings information is absorbed into prices. For BAC, this decaying pattern is accompanied by a sign crossing: $b_{0,t}$ turns negative within the first hour of trading as the prevailing sell-side regime reasserts itself following the transient earnings-driven positive shock. This intraday sign crossing is unique to BAC earnings dates and reflects the interaction between the firm-specific information event and the stock's structural adverse-selection environment. The finding that earnings generate higher peak instantaneous impact than FOMC announcements is interpretable: while FOMC decisions represent a market-wide signal whose information is diluted across all assets simultaneously, earnings releases are firm-specific shocks whose full uncertainty resolves in a concentrated burst of order flow directed at a single stock.

Before turning to the individual controversy events, it is worth stating what a negative $b_{0,t}$ means, as a sign-reversal carries economic content that a constant-parameter model specification would instead overlook. Normally, instantaneous impact is positive, meaning that a buyer-initiated trade consumes liquidity at the ask and pushes the mid-quote up. As opposed, a negative estimate reverses this, and since $x_t \in \{-1, +1\}$, the statement is symmetric (i.e., buys are followed on average by a falling mid-price, and sells by a rising one). The magnitudes involved in my case are small at the level of the individual trade: with BAC trading near \$29, the value $b_{0,t} \approx -3 \times 10^{-5}$ corresponds to a decline of roughly \$0.0009 in the mid-quote, under a tenth of one tick and below the half-tick resolution of the mid-price itself. The parameter is therefore an average over many thousands of message events rather than the result observable on any single trade, which is precisely why a filtering approach is required to recover it. To the best of my knowledge, the mechanism most consistent with such a sign-reversal is 'informed liquidity provision'. Roşu [2020] shows that informed traders optimally submit limit rather than market orders when the mispricing they perceive is moderate, reserving aggressive orders for large mispricings. The experimental evidence in Bloomfield et al. [2005] points the same way, documenting that informed traders switch from acting as *liquidity-takers* to *liquidity-providers* as their informative advantage progresses, and that their choice evolves as the value of their information changes over the trading session. An informed seller in this position posts on the ask and replenishes the top-price levels as fills (i.e., buy market orders) arrive rather than crossing the spread, both to avoid paying the impact cost of immediacy and to avoid revealing the information they possess. In my case, a regulatory fine (as in the case of BAC) whose direct cost is known but whose reputational consequence is not, sits naturally in that intermediate gray region. However, for a large-tick stock such as BAC, where the spread is pinned at one tick (hence price-competition is unavailable), this necessarily takes the form of queue-competition. So, the aggressive flow arriving against that supply is predominantly uninformed (i.e., traders reading the fine as a '*cheap entry*' and paying the ask for immediacy), therefore the recorded trade sign identifies who was impatient rather than who was informed. As the seller's supply is repeatedly refilled at the ask by patient market-makers, other participants and market-makers infer the presence of one-sided selling interest and cancel or lower their bids. The mid-price therefore declines through 'quote revision' rather than through 'aggressive selling', while buyer-initiated trades continue to be recorded. Thus, one could emphasise that $b_{0,t}$ measures association rather than causation: the aggressive buy does not cause the subsequent decline, but co-occurs with it, since both are generated by the same execution order.

Finally, Figure 2.13 -panels (e) and (f) - illustrate the filtered $b_{0,t}$ path for the two most

2.3. Empirical Results

informative controversy events in the sample. The BAC July 11 CFPB¹² fine produces the clearest discrete event-driven regime change in the dataset: $b_{0,t}$ is mildly positive at the market open, then crosses sharply into negative territory at 09:35 ET and remains persistently negative for the remainder of the session, reaching a trough near -3×10^{-5} around 13:00. This sign reversal at a precisely timestamped regulatory event is directly interpretable as a shift in the dominant informed order flow from neutral to sell-side, driven by participants pricing in the reputational and regulatory implications of the fine beyond its direct financial cost. For IBM, the most interpretable controversy result is the May 15 employee lawsuit dismissal. Given that the *Reuters* headline arrived at 05:35 ET -prior to the market open -the event time is set to 09:35 ET, the first moment of regular trading. The $b_{0,t}$ inset reveals a two-stage pattern: a sharp dip at 09:35 as the opening auction clears the accumulated pre-market order flow, followed by an immediate recovery and sustained elevation through the post-event window. This dip-then-spike is consistent with a positive legal outcome: the opening auction momentarily compresses adverse selection as the favourable news is absorbed, after which residual uncertainty about the broader legal implications drives renewed instantaneous price impact as the regular session begins.

¹²Consumer Financial Protection Bureau

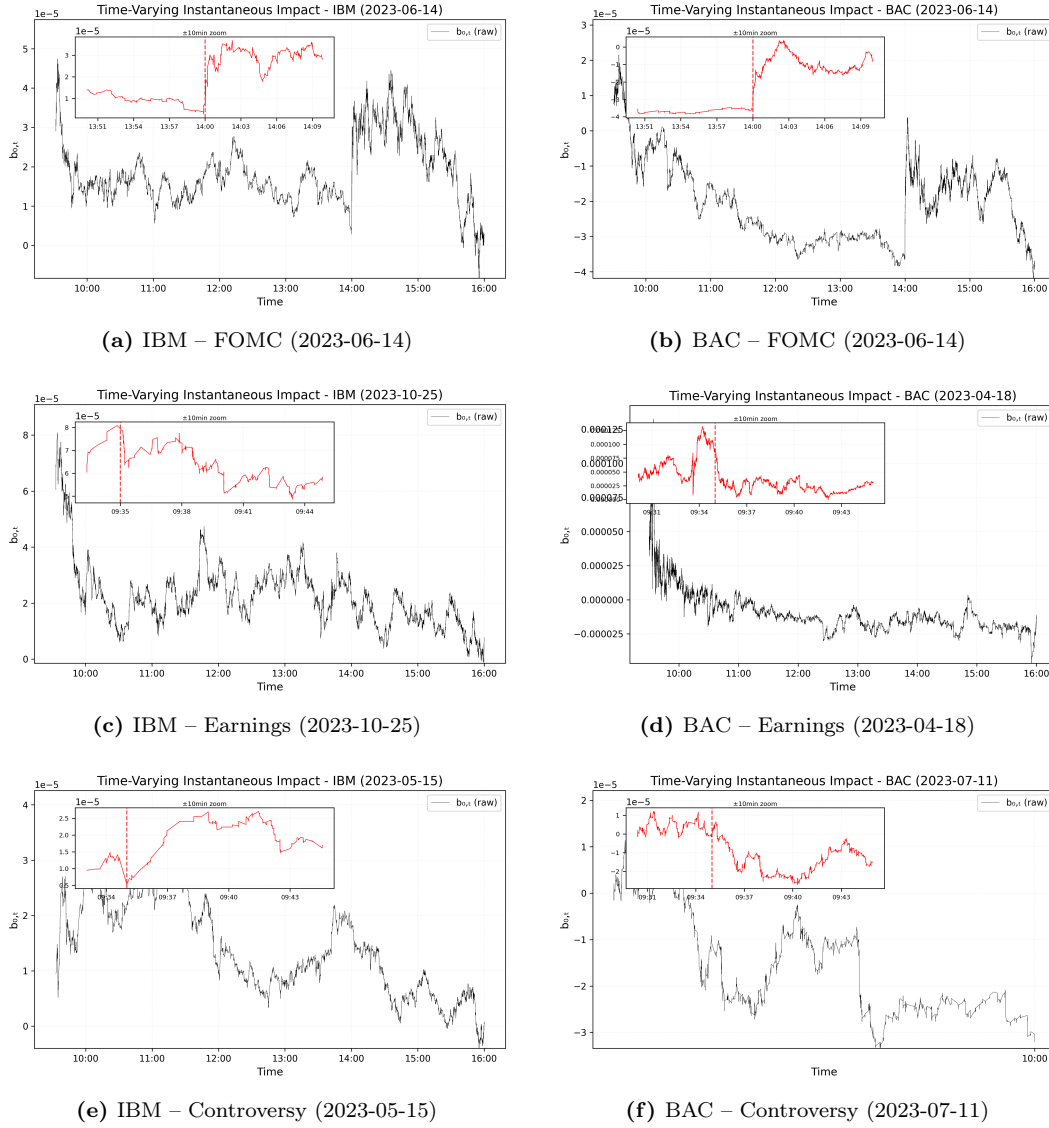


Figure 2.13: Estimates of the time-varying instantaneous market impact function $b_{0,t}$ of the SDAMH model around three categories of news events: FOMC announcements (a, b), earnings releases (c, d), and controversy news (e, f). The inset in each panel zooms into the ± 10 -minute window around the event time (dashed vertical line).

2.4. Conclusions

Table 2.4: Temporal daily average of the coefficients (*'Est'*) on the 252 trading days, and p-value (*p-val*) (in brackets) of the estimates.

Parameter	AMZN		CSCO		IBM		MSFT	
	Est.	p-val	Est.	p-val	Est.	p-val	Est.	p-val
μ_1	2.32×10^{-7}	0.000	4.20×10^{-7}	0.000	4.20×10^{-7}	0.000	7.76×10^{-8}	0.085
a_1	2.19×10^{-3}	0.015	3.64×10^{-3}	0.031	3.64×10^{-3}	0.031	5.20×10^{-4}	0.224
$b_{0,0}$	6.30×10^{-6}	0.000	3.33×10^{-6}	0.000	3.33×10^{-6}	0.000	1.71×10^{-5}	0.011
b_1	1.74×10^{-5}	0.000	2.93×10^{-5}	0.000	2.93×10^{-5}	0.000	7.59×10^{-6}	0.000
$\bar{b}_{10}$	-2.56×10^{-6}	0.000	6.21×10^{-6}	0.000	6.21×10^{-6}	0.000	-2.63×10^{-6}	0.000
$\bar{b}_{100}$	-5.49×10^{-6}	0.000	1.49×10^{-7}	0.152	1.49×10^{-7}	0.152	-3.52×10^{-6}	0.000
μ_2	-1.06×10^{-3}	0.012	-3.44×10^{-4}	0.143	-3.44×10^{-4}	0.143	-4.82×10^{-4}	0.223
c_1	-4905.651	0.000	-4820.021	0.000	-4820.021	0.000	-4460.964	0.000
d_1	1.289	0.000	1.626	0.000	1.626	0.000	1.221	0.000
$\bar{d}_{10}$	-0.029	0.000	-0.011	0.000	-0.011	0.000	-0.034	0.000
$\bar{d}_{100}$	0.699	0.000	0.385	0.000	0.385	0.000	0.730	0.000
σ_0	9.76×10^{-5}	0.000	9.98×10^{-5}	0.000	9.98×10^{-5}	0.000	9.30×10^{-5}	0.000
α	3.92×10^{-3}	0.000	3.00×10^{-3}	0.000	3.00×10^{-3}	0.000	2.68×10^{-3}	0.000
γ	1.15×10^{-2}	0.000	8.16×10^{-3}	0.000	8.16×10^{-3}	0.000	9.32×10^{-3}	0.000

Parameter	BAC		GS	
	Est.	p-val	Est.	p-val
μ_1	8.17×10^{-7}	0.000	-1.62×10^{-6}	0.000
a_1	-2.13×10^{-2}	0.000	-4.17×10^{-2}	0.000
$b_{0,0}$	5.25×10^{-6}	0.000	5.34×10^{-5}	0.000
b_1	4.34×10^{-5}	0.000	2.69×10^{-5}	0.000
$\bar{b}_{10}$	1.25×10^{-5}	0.000	-2.12×10^{-7}	0.452
$\bar{b}_{100}$	1.20×10^{-6}	0.000	-6.39×10^{-6}	0.000
μ_2	-2.49×10^{-3}	0.020	4.21×10^{-3}	0.004
c_1	-2834.954	0.000	-1673.000	0.000
d_1	1.705	0.000	1.396	0.000
$\bar{d}_{10}$	0.046	0.000	0.015	0.000
$\bar{d}_{100}$	0.390	0.000	0.654	0.000
σ_0	1.53×10^{-4}	0.000	2.84×10^{-4}	0.000
α	2.98×10^{-3}	0.000	6.00×10^{-3}	0.000
γ	6.31×10^{-3}	0.000	1.79×10^{-2}	0.000

2.4 Conclusions

This paper has investigated the dynamics of instantaneous price impact around the release of a broad set of heterogeneous news (i.e., FOMC announcements, earnings announcements, and major controversy news) for six stocks (BAC, GS, IBM, MSFT, CSCO) classified by their relative tick-size. To do so, I first reproduce the main stylized facts and microstructural properties of Limit Order Books, to then proceed at developing a SDAMH model to capture the liquidity dynamics around such news events following the methodology proposed by [Campigli et al., 2022]. For what concerns the LOB analysis part, I successfully document the main stylized facts and microstructural properties of the LOB for the six stocks in the sample. The canonical empirical regularities of financial markets are successfully retrieved: return distributions exhibit fat tails relative to the Gaussian benchmark, volatility clustering is evident from the slow decay of the ACF of absolute returns, and raw returns are essentially uncorrelated at the intraday level, consistent with weak-form market efficiency. At the microstructural level, order flow displays strong and persistent autocorrelation consistent with a long-memory process, with the PACF cutting off sharply after lag 1, pointing to an underlying latent process rather than direct multi-lag interactions —

consistent with the well-documented order-splitting behaviour of institutional traders. Intraday activity follows the well-known U-shaped (or more precisely J-shaped) pattern across all stocks, with trading intensity, volume, and volatility peaking at the open and close and reaching a minimum around midday. The bid-ask spread narrows throughout the trading day as overnight information is progressively absorbed, with the effect more pronounced for small-tick stocks. These findings are consistent across all six stocks regardless of tick-size classification, validating both the dataset and the empirical framework before proceeding to the main analysis.

Regarding the SDAMH analysis part, the findings suggest an interesting picture : clearly there's evidence of instantaneous market impact right after FOMC announcements, while there's lighter evidence of a pronounced price impact following the release of both earnings announcements as well as controversy news. In particular, I find that earnings announcements generate the largest peak instantaneous impact values across the entire sample, systematically exceeding those observed around FOMC dates, consistent with the firm-specific nature of earnings information which resolves uncertainty in a concentrated burst of directed order flow. FOMC announcements, by contrast, produce sharp but more moderate jumps in $b_{0,t}$, reflecting the market-wide diffusion of the signal across all assets simultaneously. As expected, controversy news produces the most heterogeneous responses, with the direction and persistence of the impact shift depending heavily on the nature and timing of the disclosure. Across all news categories, the time-varying framework reveals dynamics that a static price impact parameter would entirely obscure, most strikingly the sign reversals in $b_{0,t}$ that signal shifts in the dominant informed order flow between buy- and sell-side regimes. These results highlight the importance of allowing price impact to evolve dynamically in high-frequency settings, and suggest that the informational content of news—rather than its scheduling—is the primary determinant of the magnitude and persistence of the market impact response.

Although the primary limitation of this study lies in its limited cross-section —since six equities cannot fully represent the broader NASDAQ universe, driven primarily by the substantial licensing costs of message-level limit order book data that restrict the feasible sample—, it is important to clarify the aim of the sample selection. That is, rather than being drawn at random, the six stocks were deliberately chosen to span the complete taxonomy of relative tick-size regimes. Consequently, the research design supports valid inferences regarding how instantaneous price impact dynamics vary across these specific tick-size structures, without overgeneralizing those findings to the entire market population.

Similarly, the sample period covers January 2023–April 2024, an event window characterized by elevated interest rate uncertainty. But, despite the stylized facts documented here remain robust and well-established in existing literature, the estimated magnitudes of instantaneous price impact may not directly differ in calmer macroeconomic conditions.

Chapter 3

Information and Order Flow: a Causal-inference Study of Price Dynamics around Macroeconomic News

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Abstract

This paper investigates the causal relationship between informed trading and price dynamics around Federal Open Market Committee (FOMC) announcements, examining whether high-frequency excess returns primarily result from exogenous informational shocks or endogenous mechanical order flow dynamics. Using high-frequency LOBSTER data for five NASDAQ-listed stocks across 16 FOMC events spanning 2023-2024, I decompose risk-adjusted returns into an adverse selection component (IC_{lag}) and an order flow imbalance component (OFI), thereby testing their respective causal contributions to intraday excess returns. The adverse selection component is negative and highly significant across all specifications and error correction methods, consistent with temporary price overshooting following informed trading. Order flow imbalance, while positively significant under OLS, loses significance entirely once endogeneity is corrected via instrumental variables, thus reflecting simultaneity with returns rather than a genuine causal price impact channel. Cross-sectionally, the adverse selection effect is significantly stronger for large-tick stocks, where market-makers face more constrained quote adjustment and greater exposure to informed order flow.

3.1 Introduction

The Efficient Market Hypothesis (hereafter EMH) regards a financial market as a measurement apparatus that aggregates private estimates of an asset’s true (but hidden) value and, after quick and efficient elaboration, provides an output price. At the core of the EMH lies the idea that the market price of an asset may deviate from the asset’s “intrinsic value” [Fama, 1965]; market participants then form private beliefs about this hidden value (either based on evaluation models or conventions) and trade accordingly. In this view, the release of a new piece of information both leads private beliefs to evolve and objectively changes the intrinsic value of the asset.

However, the problematic nature of the EMH has been extensively documented in over half a century of academic and market research [Jensen, 1978a; Shiller, 1980; Cutler et al., 1989a; Barber and Odean, 1999a; Lempérière et al., 2014]. According to this strand of the literature, such inconsistencies of EMH may possibly be traced back to somehow ‘overlooked and post-hoc’ social traits in the assumed underlying forces driving price changes, where EMH appears in retrospect as a theory of informational advantage held by “sophisticated traders” [Fama, 1965] and belief coordination among agents. The very fact that insider dealing is criminally prosecuted is tantamount to the notion that private information holds genuine predictive power. Unsurprisingly, insider trading is central to Kyle’s seminal work [Kyle, 1985] on the informational content of prices, where informed traders strategically incorporate private signals into prices.

Historically, classical empirical tests of EMH have investigated whether securities prices “fully reflect” delimited information sets [Fama, 1970a]: past prices for *weak-form* efficiency, public news and announcements for *semi-strong form*, and both public and private information for *strong form*. At its core, EMH posits that asset prices may deviate from intrinsic value, with market participants forming private beliefs and trading accordingly. Unpredictable information releases continuously update such beliefs, eventually altering intrinsic value, though trading *per se* should not affect prices.

This orthodox view of financial markets has faced mounting challenges. Recent advances—largely ‘fueled’ by econophysicists –, position order flow (i.e., the granular mechanics of buy and sell order execution physically occurring in the market) as a statistically dominant driver of price movements [Bouchaud et al., 2018]. Emphasis has therefore shifted from (exogenous) information shocks to (endogenous) market impact, with refined distinctions between transient (reversible) and permanent (persistent) effects [Neuman et al., 2023b]. An influential and novel *order-driven* view inverts traditional causality supported by the previous tenets: rather than fundamental value determining prices, order flow poses itself as the main driver of price changes. As [Gabaix and Koijen, 2021b] put it, “*the mystery of apparently random movements of the stock market, hard to link to fundamentals, is replaced by the more manageable problem of understanding the determinants of flows in inelastic markets.*” Empirical support for this perspective abounds in high-frequency data, where metaorders and liquidity provision reveal predictable, concave market impact functions [Almgren et al., 2005; Cont, 2001; Bouchaud, 2022a].

Yet this order-driven paradigm raises a pivotal question: where does information fit in price formation? Possibly, the state of affairs is best summarized in [Bouchaud, 2022a]: *If order flow is the dominant cause of price changes, ‘information’ is chiefly about correctly anticipating the behaviour of others, as Keynes envisioned long ago, and not about fundamental value.* In other words, information potentially acts as a trigger for order flow which, in most situations driven by inelasticity and herding, is the actual driver of price changes.

To shed light on the ongoing debate over whether price changes arise primarily from

endogenous patterns within financial markets or exogenous informative shocks, this paper employs a rigorous empirical approach based on public market data, advancing the understanding of the interplay between information and order flow in driving price dynamics. The analysis focuses on public releases of macroeconomic information (specifically, Federal Open Market Committee (FOMC) announcements) which plausibly rule out the presence of insiders in the sense of [Kyle, 1985]. Furthermore, FOMC announcements are scheduled, public, and precisely timestamped, thus eliminating (or significantly reducing) possible arising concerns about information leakage or any anticipation effect. These macroeconomic announcements are known to generate sharp, well-identified information shocks that allow me to cleanly separate the announcement effect from pre-existing trends. Moreover, the high-frequency nature of limit order book data allows me to observe the tick-by-tick (later aggregated to a minute-by-minute in the quantitative analysis) response of both prices and order flow, providing for the required granularity of data needed to distinguish causal from merely correlational channels.

Despite the effects of macroeconomic announcements on financial markets have been extensively studied [Andersen and Bollerslev, 1998; Bollerslev et al., 2018], to the best of my knowledge few if any have examined the distinct causal contributions of exogenous information and endogenous order flow to high-frequency price dynamics of individual securities. The research design comprises two complementary steps. First, I conduct an event study around FOMC announcements, analysing the evolution of cumulative average abnormal order flow imbalance (CAAOFI) in a ± 15 -minute window around each announcement. Second, I estimate a multiple panel regression decomposing excess returns into an adverse selection component, a proxy for the information channel, IC_{lag} , and an order flow imbalance component (OFI), with firm and event fixed effects and instrumental variables to address potential endogeneity concerns.

The results support a microstructural account in which lagged informed trading drives price reversals. The adverse selection coefficient is negative and highly significant across all specifications ($\hat{\beta} = -0.0014$, $t = -20.7$ under Newey-West HAC; $t = -10.0$ under two-way clustering), consistent with temporary price overshooting following informed order flow [Kyle, 1985; Glosten and Harris, 1988]. Informed trading intensity spikes sharply at the moment of announcement and remains elevated for approximately 30 minutes, indicating a sustained episode of adverse selection rather than instantaneous price adjustment. Instrumental variable estimation confirms the causal nature of this channel ($\hat{\beta} = -0.0016$, $t = -16.3$), while order flow imbalance loses significance under IV correction, revealing simultaneous price discovery. Cross-sectional analysis reveals that the adverse selection effect is significantly stronger for large-tick stocks ($\hat{\beta}_{\text{interaction}} = -0.000111$, $t = -5.20$, $p < 0.001$), where market-makers face more constrained quote adjustment and greater exposure to informed order flow [Parlour, 1998; Foucault et al., 2005]. The adverse selection effect is negative in all sixteen FOMC events and statistically significant in fourteen, confirming it as a systematic feature of informed trading around monetary policy announcements rather than an 'artefact' of any single unusual event. Collectively, these findings contribute to theoretical models of price formation and carry direct implications for the design of algorithmic trading strategies.

Notably, extensive robustness checks further reveal an interesting tension between theoretical correctness and empirical fit: while the linear OLS specification satisfies the Ramsey RESET test for functional form at conventional significance levels, a square-root transformation of order flow imbalance, motivated by the well-documented concavity of the price impact function [Bouchaud et al., 2018], yields a remarkable marginally superior fit. This trade-off between the theoretical appeal of the square-root specification and the statistical

parsimony of the linear model is left as an open question for future work.

The rest of the paper is organized as follows. Section 3.2 synthesizes the literature and derives testable hypotheses. Section 3.3 describes the data and methods. Section 3.4 reports main results, including baseline estimates, robustness checks, endogeneity tests, and heterogeneity analyses. Section 3.6 concludes with implications for microstructure theory and future research.

3.2 Literature review

Understanding what moves asset prices is one of the central questions in financial economics. In this regard, the EMH has long provided the foundational framework for understanding price formation in financial markets, with the canonical answer that “*prices move only when new information arrives*”. Yet, half a century of empirical research has documented systematic anomalies contradicting this canonical view [Shiller, 1981; Cutler et al., 1989a; Lempérière et al., 2014]. Concurrently, a growing body of literature from the strand of econophysics has lately proposed that order flow (i.e., the net buying and selling pressure of market participants in the limit order book) plays an independent role in driving price dynamics [Bouchaud et al., 2018; Gabaix and Koijen, 2021b].

On the one hand, the EMH posits that prices reflect all available information, and prices result from the consensus between fully rational agents who would otherwise immediately arbitrage away any deviation from the fundamental value [Fama, 1970a; Samuelson, 2016]. Consequently, price changes can only be the result of unanticipated news and are totally unpredictable. In this view, prices behave as martingales which, by definition, satisfy the following property:

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s, \quad \text{for all } 0 \leq s \leq t.$$

and trading itself has no lasting impact on price dynamics.

In this vein, a large body of research has demonstrated the importance of exogenous informational shocks (i.e., public information) in explaining price deviations. For instance, [French and Roll, 1986] provide early evidence that stock return variances are significantly higher on trading days than on non-trading days, suggesting that the process of trading itself contributes to price discovery. This finding established that public information arrival and trading activity are jointly responsible for price deviations. Public news, earnings announcements, and unanticipated news releases (i.e., ESG-related disclosures) shape investor expectations and can generate abnormal returns and changes in liquidity. For example, [Salarin et al., 2024] apply an event-study framework to examine intraday market reactions to ESG news, showing that trading volume and prices adjust within minutes of disclosure. Related evidence is also provided by [Ranaldo, 2006], who studies intraday market dynamics around scheduled public information arrivals, and by [Strauß, 2017], who shows that economic news disseminated by wire services quickly translates into abnormal returns and higher trading intensity. In terms of high-frequency settings, [Andersen and Bollerslev, 1998] document systematic intraday volatility spikes at scheduled announcement times, establishing the event-study framework for high-frequency identification, and [Bollerslev et al., 2018] analyse volume, volatility, and returns around FOMC announcements specifically, highlighting systematic price adjustments in the minutes following disclosure. More recently, [Kannianen and Yue, 2019] employ high-frequency data to show that certain announcements induce sharp return jumps immediately after release, and occasionally reveal signs of information leakage.

On the other hand, in this “*platonic view*”, neither microstructural effects nor the process of trading itself can affect prices, except perhaps very briefly [Bouchaud, 2022a]. In fact, researchers initially did not regard order flow as a particularly informative perimeter that market prices would be assumed to reflect. More recently, based on empirical studies of granular data on market interaction and the mechanics of trading activity, order flow has emerged as a statistically consistent driver of price dynamics [Bouchaud et al., 2018]. Early critiques document anomalies inconsistent with the EMH, such as [Shiller, 1981], who highlight the excess-volatility puzzle, arguing that asset prices fluctuate far more than can be justified by fundamentals. [Jensen, 1978b] and [Barber and Odean, 1999b] emphasize excess trading relative to rational benchmarks, while [Cutler et al., 1989b] shows that large price moves often occur in the absence of identifiable news. Similarly, the trend-following puzzle reveals persistent, predictable return patterns over centuries and across asset classes [Lempérière et al., 2014]. These anomalies suggest that order flow, rather than solely information, plays a central role in driving price movements.

On top of that, noise trading provides a theoretical mechanism linking trading activity to liquidity and volatility. [Black, 1986] argued that noise is essential for market functioning, while models such as [Kyle, 1985] and [Glosten and Milgrom, 1985] formalized the interplay between informed traders, noise traders, and market-makers. In Kyle’s framework, market-makers infer information from order flow corrupted by noise, producing efficient prices on average but failing to capture empirically observed persistence in order flow [Lillo, 2022]. Glosten and Milgrom demonstrated that adverse selection drives bid-ask spreads, with noise traders ensuring liquidity provision. Finally, [Glosten and Harris, 1988] further decompose the bid-ask spread into adverse selection and order processing components, providing the theoretical foundation for the information component measure IC_{lag} employed in this paper.

Hence, the *order-driven view* of markets emphasizes the endogenous dynamics of order flow as a primary driver of short-term price changes [Bouchaud et al., 2004b, 2018; Bouchaud, 2022a]. Within this perspective, trades—whether informed or not—exert a persistent price impact, that is the mechanical effect of trades on prices. Empirical studies from market microstructure literature consistently show that aggressive buy (sell) orders push prices upward (downward), even absent fundamental information [Bouchaud, 2022b]. Among the theories of price-impact, [Almgren and Chriss, 2000] distinguished between temporary, transient, and permanent impact, while [Gatheral, 2010] formalized impact dynamics with the “*no-dynamic-arbitrage*” principle, showing that permanent impact must be linear to avoid price manipulation. More recently, [Neuman et al., 2023a] extended propagator models with learning approaches.

In contrast to the EMH, the order-driven view of market microstructure highlights that this persistence is not only due to private information but also to the mechanical and strategic interaction of orders in the limit order book. This stream of literature positions order flow as a key explanatory factor for excess returns at high-frequency domains, thereby shifting attention from purely informational interpretations of price changes toward a microstructural one.

Bridging these two perspectives, [Gabaix and Koijen, 2021b] propose the Inelastic Market Hypothesis (IMH), which provides an economic rationale for why order flow has lasting effects on prices even in the absence of information. Their model shows that institutional mandate constraints render aggregate demand for stocks relatively inelastic. Consequently, a 1% purchase of shares raises prices by about 1% (i.e., an inflow of one dollar into the equity market translates into a five-dollar increase in market capitalization). This interpretation implies order flow itself, rather than fundamentals, primarily determines prices, helping to explain excess volatility, feedback loops, and trend-following dynamics. The IMH thus

offers a unifying framework in which both information events and liquidity-driven flows shape excess returns, complementing earlier empirical observations of “price moves without news” [Cutler et al., 1988; Cornell, 2013].

Taken together, this body of evidence reveals a tension between the EMH and the empirical reality of market microstructure. Whereas the EMH assumes that only new information moves prices, modern evidence suggests that liquidity provision, order flow correlations, and inelastic supply-demand dynamics are central to understanding price formation. Hence, the disentanglement of how much of the observed price adjustment is due to direct news incorporation versus the subsequent cascade of order flow remains an open empirical question.

The aforementioned literature motivates two testable hypotheses. On one side, the order-driven view of markets [Bouchaud et al., 2018; Gabaix and Koijen, 2021b] suggests that contemporaneous liquidity pressures are the main causal drivers of short-term excess returns [Cont et al., 2014]. On the other side, adverse selection models posit that informed trading around information events generates temporary price overshooting with subsequent correction [Glosten and Milgrom, 1985; Glosten and Harris, 1988; Kyle, 1985]. Both hypotheses are directly testable using recently available L2¹ tick-by-tick limit order book data around FOMC announcements.

Thus, inspired by [Jeon et al., 2023], who examines how information and order flow influence return jumps, I propose to invert the causal lens and analyze to what extent can excess returns be decomposed into an information and an order flow component. In light of this, I aim to test the following set of hypotheses :

- H1.** *Contemporaneous order flow imbalance is the main causal driver of excess returns in a high-frequency setting*
- H2.** *Informed trading—in the form of the adverse selection component of order flow—negatively predicts subsequent excess returns*

3.3 Data and Methodology

3.3.1 The Limit Order Book

A Limit Order Book (LOB) the set of all active orders(i.e. all limit orders). It is an order-driven market ² that is responsible for matching sellers with the buyers firstly according to price priority, and secondly by queue priority. Figure 2.1 is an illustrative example of how the matching of orders occurs in the LOB. An order $z = (p_z, q_z, t_z)$ submitted at time t_z with limit price p_z and size q_z (with $q_z > 0$ for buys and $q_z < 0$ for sells) is a commitment to *buy* (resp. *sell*) up to $|q_z|$ units of the asset at a price no greater than (respectively no less than) p_z [Bouchaud et al., 2018]. LOBs thus display the available depth at different levels of the order book. Let $V(p)$ denote the total volume of limit orders submitted at price p . Formally,

$$V(p) = \sum_{z \in L : p_z = p} q_z, \quad (3.1)$$

where L is the set of active limit orders, and q_z is the signed size of order z .

Among the worth-mentioning aspects of a LOB, I find :

¹L2 market data refers to full limit order book data showing all active bid and ask quotes at multiple price levels, as opposed to L1 data which provides only the best bid and ask.

²Traders can choose whether to provide or consume liquidity by submitting different types of orders to a LOB. Liquidity provision is decentralized and becomes an endogenous process. Limit order books are therefore more flexible than quote driven markets

- The *bid price* at time t is the highest stated price among active buy orders at that time:

$$b(t) := \max_{z \in L: q_z > 0} p_z. \quad (3.2)$$

- The *ask price* at time t is the lowest stated price among active sell orders at that time:

$$a(t) := \min_{z \in L: q_z < 0} p_z. \quad (3.3)$$

To distinguish these from other limit prices, one often refers to them as the *best bid* and *best ask*, respectively.

- The *bid-ask spread* at time t is defined as

$$s(t) := a(t) - b(t). \quad (3.4)$$

- The *mid-price* at time t is

$$x(t) := \frac{1}{2} [a(t) + b(t)]. \quad (3.5)$$

- The *tick-size* δ of a LOB is the smallest permissible price increment between different orders. All orders must arrive with a price specified to the accuracy of δ ; that is,

$$|p_i - p_j| \geq \delta \quad \forall i, j. \quad (3.6)$$

As a final note, limit order books (LOBs) operate with finite resolution parameters: there is typically a minimum tick-size (price increment) and a minimum lot size (trading quantity).

I obtain the “*message file*” and “*order book file*” from the LOBSTER³ database [Huang and Polak, 2011]. Notice that, since my sample is composed of all US equities with large capitalization, they all have a constant tick-size = 1 cent. The message file contains information on all the orders sent to the NASDAQ exchange: the timestamp in nanosecond resolution, the type of order (e.g. limit-order, cancellation, market-order), the size and the price. From now on, I may write market instead of NASDAQ. The order book is reconstructed by LOBSTER and presents a snapshot of the first ten levels of the order book as a result of each message. The book levels refer to the number of shares available at different bid and ask prices. The first level gives the number of shares at the highest bid price and the lowest ask price. The next best bid and ask prices and sizes refer to the second level and so on. Figure

For the sake of clarity, let me propose a toy-example of how an order book works: Table Table 3.2 presents an illustrative example of a message book. Each row of the message book corresponds to a different trading event. Specifically, for each event I have the following information presented in a separate columns :

1. **Time:** measures the arrival of each event in seconds after midnight. Specifically, the LOBSTER dataset uses nanoseconds time resolution as the exchange. For example, a time equal to 34200.4637017390 stands for clock time 09:30:00.463702am EST.
2. **Type:** represents the different type events categorized as:
Type ‘1’ limit orders (passive orders) are buy or sell orders sent to the exchange platform at a specified price and whose execution is not guaranteed. *Type ‘2’* and

³LOBSTER: Limit Order Book System- The Efficient Reconstructor n.d.

Table 3.1: Message Types in the LOBSTER Dataset

Type	Description
1	Submission of a limit order
2	Cancellation
3	Deletion
4	Execution of a visible limit order
5	Execution of a hidden limit order

Type ‘3’ orders represent partial or complete cancellations: the agents remove orders that previously placed in the market. *Type ‘4’* orders are buy or sell orders that are executed, and they can be either limit or market order. Lastly, *Type ‘5’* orders represent executions against an hidden order, i.e. orders that are not visible in the reconstructed order book.

Thus, I can summarize the event set is as follows :

$$\mathcal{E} := \left\{ LO_{ask_D}, CO_{ask_D}, LO_{ask_T}, CO_{ask_T}, MO_{ask_T}, LO_{ask_{IS}}, LO_{bid_{IS}}, LO_{bid_T}, CO_{bid_T}, MO_{bid_T}, LO_{bid_D}, CO_{bid_D} \right\}. \quad (3.7)$$

3. **Size:** is the number of shares available at a given price.
4. **Price:** corresponds to the order price and is expressed in dollars times 10000 (e.g., a price of \$165.6 is given by 1656000).
5. **Direction:** Bid/Ask simply tells me if the order is a bid or an ask. Bid orders are sent by traders who would like to buy the asset, while ask orders are sent by agents who would like to sell.

Moreover, in Table 3.3, I present the order book corresponding to exactly the same trade events depicted in Table 3.2. The order book is reconstructed recursively from the message updates. For ease of illustration, my example includes only the first two levels. As depicted in Table 3.3, for each trade event, I show the resulting ask/bid prices and sizes for the different levels. These illustrate the sell/buy orders that agents are willing to transact, sorted according to price. Regarding the order book dynamics, one can see that the best ask and bid prices change constantly depending on the orders arriving from the agents. For instance, the second order in the message book is an execution on the ask side as a result of a high bid. As it can be seen from the second row in Table 3.2, this order corresponds to a bid of 16544700 for 50 shares. The first row in the order book snapshot in Table 3.3 shows that there was an Ask 1 at the same price for the same number of shares. Hence, there is a match and the order is transacted. Given that there were exactly 50 shares available on the Ask 1, the new Ask 1 becomes the offer that used to be at level 2. The new state of the order book as a result of the order is given in the second row of Table 3.3. As a final example, consider the third row in Table 3.2. This is an ask order at 16560000 for 100 shares. This order joins the existing Ask 1 in the second row of Table 3.3 and results in a total size of 101 shares available at that price, as shown in the third row of Table 3.3. These examples show how orders change the order book. From the order book one can obtain valuable information such as the mid price, the spread and order flow information.

Table Table 3.3 shows the corresponding *order book snapshot* for the same sequence of events. The book is reconstructed recursively from message updates, and each row reports the state of the book after the corresponding message. For simplicity, only the first two levels are shown.

Table 3.2: Illustrative example of a LOBSTER message file.

Time (sec)	Type	Order ID	Size	Price	Direction
34200.46	1	101	100	1656000	Ask
34200.47	1	102	50	1654470	Bid
34200.48	4	102	50	1654470	Bid
34200.49	1	103	100	1656000	Ask

Table 3.3: Illustrative example of a reconstructed LOBSTER order book (two levels).

Ask Price 1	Ask Size 1	Bid Price 1	Bid Size 1
165.60	100	165.45	50
165.60	100	165.44	0
165.60	200	165.44	0

3.3.2 Small vs. Large tick stocks

[Bouchaud et al., 2018] define the 'tick-size' as the *cost of price priority*⁴. Considering that, across US equity exchanges, tick-sizes are expressed in decimal points and uniformly standardised at \$0.01 for all stocks, an LOB displaying a large spread gives traders a full spectrum of choice on where to place their limit orders : at best, inside the spread or deeper in the book. Focusing on the first two, a limit order placed '*at best*' benefits of a better execution price than an order inside the spread, but other limit orders at the best might have a higher priority (e.g., the ones submitted earlier and queuing). Vice-versa, a limit order submitted inside the spread marks a new best price and, therefore, benefits from price priority over all other limit orders (of the same side of the book), though executed at a worse price. The *cost of price priority* lies in this idea : in order to gain price priority, a trader has therefore to '*pay one tick*'. In this regard, it is worth mentioning that the market microstructure literature has recently devoted considerable attention to the features and dynamics surrounding tick-size regulation [Jain et al., 2024b,a].

In general, the LOBs of small tick stocks shows an average spread which is a large multiple of the tick-size, typically $\langle s(t) \rangle \geq 3\delta$. Sometimes, for very small tick stocks, the spread is as large as 30 times the tick. In addition, the spread tends to fluctuate significantly throughout time, as the LOB is usually sparse, and therefore several potential price levels are left unoccupied by limit orders. Furthermore, the size of the first gap $d1$, (i.e. the spread between the best and the second best price), is on average proportional to the spread. In terms of liquidity, small-tick stocks display lower liquidity levels, i.e., the available depth at the best is a small multiple of the average market order size. Moreover, empirical evidence reveals that traders make heavy use of hidden orders.

In contrast, the LOBs of large tick stocks presents somehow the opposite properties : large-tick stocks tend to have an average spread that is close to one tick, $\langle s(t) \rangle \leq 1.5\delta$, as the spread does not fluctuate much throughout time. Consequently, the first levels in the LOB are, in average, densely populated (i.e. $d1 = \tau$), liquidity is comparatively large, and the available depth at the best is a large multiple of the average market order size. Contrarily to what happens in small-tick stocks, here traders make relatively little use of hidden orders, and price priority is preferred to time priority, as the locked spread make it costly to place orders at a '*better*' price than other traders. Figure 2.4 displays an instant

⁴Quoting the authors : 'The tick-size ϑ dictates how much more expensive it is for a trader to gain the priority associated with choosing a higher (respectively, lower) price for a buy (respectively, sell) order.'

of the Limit Order Book for two stocks (TSLA and BAC) respectively denoted as small-tick and large-tick stocks. It perfectly depicts the microstructural properties of these two classes of stocks outlined above.

Recent developments in market microstructure literature have highlighted the fundamental role of tick-class specification, for it being an important feature providing a natural taxonomy for comparing liquidity conditions, trading strategies, and price dynamics for equities. Hence, categorizing stocks by tick class has become a standard approach in both empirical and theoretical studies of financial markets. For my research purposes, a tick-categorization allows me to test whether the relative contributions of public information and order flow to excess returns vary systematically with market granularity. Relying on this, I consider it as a natural dimension along which to study the decomposition of excess returns into information and order flow components in order to deepen my assess whether price dynamics are predominantly liquidity or information -driven.

3.3.3 Data

Our sample consists of high-frequency limit order book data for five constituents from the Nasdaq-100 Index (i.e., MSFT, IBM, BAC, GS, and CSCO, see Table 3.4) obtained from the LOBSTER database, which provides an easy-to-use interface for reconstructing full limit order book dynamics from NASDAQ ITCH⁵ data with a resolution timestamped to the nanosecond. Each trading day includes a detailed *message file*, recording every event (i.e., order submission, cancellation, deletion, and execution), as well as an *order book file*, which provides a 1:1 mapping of the current state of the book—immediately matching the event from the *message file* - up to 40 levels, hence capturing the whole depth after each event occurrence. This allows researchers to work with the most recent and granular representation of the market, making it particularly suited for high-frequency empirical studies. In addition to this, a dataset containing FOMC announcements (hereafter FOMC, see Table 3.5) timestamped to the millisecond and covering regular trading hours on each day related to an FOMC from January 2021 - December 2024 completes my data collection. These announcements happen 8 times per year at 02:00 p.m. sharp, so I have a total of 16 events in total.

Table 3.4: Our sample : for each stock, I report the ticker, the extended name, and the sector according to the GICS⁶ taxonomy.

Ticker	Stock Name	Sector	Tick Class
MSFT	Microsoft Corporation	Information Technology	Small-tick
IBM	International Business Machines Corporation	Information Technology	Small-tick
BAC	Bank of America Corporation	Financials	Large-tick
GS	The Goldman Sachs Group, Inc.	Financials	Small-tick
CSCO	Cisco Systems, Inc.	Information Technology	Large-tick

Following the market microstructure literature on tick-size classification [Bouchaud et al., 2018], Table 3.4 partitions the sample into small-tick and large-tick stocks according to their relative tick-size. The classification is based on the average $\langle \sigma \rangle$ spread measured in units of the relative exchange tick-size θ , denoted by $\langle \sigma \rangle / \theta$ and defined as follows :

- **Small-tick:** $\langle \sigma \rangle \geq 3\theta$
- **Medium-tick:** $1.5\theta \lesssim \langle \sigma \rangle \lesssim 3\theta$

⁵NASDAQ's native data feed protocol

3.3. Data and Methodology

- **Large-tick:** $\langle \sigma \rangle \lesssim 1.5\theta$

Specifically, CSCO and BAC are large-tick stocks, with spreads typically close to one tick; MSFT⁷ This distinction is central to my analysis: large-tick stocks are expected to exhibit stronger mechanical price impact due to their dense order books, whereas small-tick stocks display greater heterogeneity in spread and depth dynamics.

Table 3.5: FOMC announcement dates used in the sample, February 2023–December 2024

Date	Time (ET)
Feb 01, 2023	02:00 p.m. ET
Mar 22, 2023	02:00 p.m. ET
May 03, 2023	02:00 p.m. ET
Jun 14, 2023	02:00 p.m. ET
Jul 26, 2023	02:00 p.m. ET
Sep 20, 2023	02:00 p.m. ET
Nov 01, 2023	02:00 p.m. ET
Dec 13, 2023	02:00 p.m. ET
Jan 31, 2024	02:00 p.m. ET
Mar 20, 2024	02:00 p.m. ET
May 01, 2024	02:00 p.m. ET
Jun 12, 2024	02:00 p.m. ET
Jul 31, 2024	02:00 p.m. ET
Sep 18, 2024	02:00 p.m. ET
Nov 07, 2024	02:00 p.m. ET
Dec 18, 2024	02:00 p.m. ET

All announcements at 02:00 p.m. Eastern Time. Sample covers 16 FOMC events over two calendar years.

3.3.4 Methodology

This study aims at disentangling how much of high-frequency excess returns around major public news events (e.g., FOMC announcements) can be attributed to the information component versus order flow dynamics. In other terms, my objective is to test whether price adjustments are mainly driven by the informational content of an announcement (i.e., news, tone, monetary policy surprises), or by mechanical order flow imbalances (i.e., trading pressure, liquidity frictions, inelastic demand).

To do so, I begin by constructing the dataset from NASDAQ LOBSTER message and order book files. Table Table 3.2 and Table 3.3 show an illustrative example of the primary data files for the reader. To ensure comparability across securities and to avoid distortions, I apply the following filters: (i) I exclude auctions and remove the first and last ten minutes of trading each day, as opening and closing periods are affected by institutional frictions

⁷Notice that, in the previous chapter, I adopt a three-way tick-size classification (small, medium, large) following Briola et al. [2024], under which MSFT is classified as medium-tick. The present paper relies on the standard binary classification (small and large tick stocks) which collapses the medium category, resulting in MSFT being grouped together with the other small-tick stocks.

and batch mechanisms; (ii) I drop days subject to trading halts; (iii) I require continuous book coverage to guarantee that the order book snapshots can be accurately reconstructed; and (iv) I classify stocks into tick-size categories-small- or large-tick-based on the fraction of time that the quoted spread equals one tick. This classification ensures that my analysis accounts for heterogeneity in market microstructure across different liquidity regimes.

For each stock, the mid-price is defined as

$$m_t = \frac{1}{2}(a_t + b_t), \quad (3.8)$$

where a_t and b_t denote the prevailing ask and bid quotes, respectively. The spread is given by $s_t = a_t - b_t$, while q_t^a and q_t^b denote the standing depth at the best ask and best bid. Each trade is also associated with a volume v_τ .

Returns are then measured as log changes in the mid-price,

$$r_{t,\Delta} = \log m_{t+\Delta} - \log m_t, \quad (3.9)$$

and Excess Returns $ER_{t,\Delta}$ are obtained using the standard market model estimated over the trading day until 60 minutes prior to the FOMC announcement:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it} \quad (3.10)$$

where R_{it} is the log return of stock i at minute t , R_{mt} is the contemporaneous return on the NASDAQ Index Futures, and the parameters α_i and β_i are estimated using OLS over the pre-event estimation window $[-360, -60]$ minutes relative to the announcement.

Excess returns are then defined as the residuals from this model:

$$ER_{it} = R_{it} - \hat{\alpha}_i - \hat{\beta}_i R_{mt} \quad (3.11)$$

The information component is constructed following [Glosten and Harris, 1988], who decompose the bid-ask spread into an adverse selection component and an order processing component. For each stock i at minute t , I define the information component as:

$$IC_{it} = \text{sign}(OF_{it}) \times (m_{i,t+1} - m_{it}) \quad (3.12)$$

where $\text{sign}(OF_{it})$ is the direction of net signed order flow at minute t (positive for net buying pressure and negative for net selling pressure) and $(m_{i,t+1} - m_{it})$ is the one-minute-ahead change in the mid-price. Intuitively, IC_{it} is positive when the market moves in the direction of order flow (consistent with informed trading) and negative when prices reverse, indicating that the initial price impact was transitory. Following [Kyle, 1985], a large positive IC_{it} signals that informed traders successfully pushed prices toward their private valuation, while a subsequent reversal reflects the market-maker's correction of the temporary overshoot. I then proceed and lag the regressor IC_{it} by one period to obtain $IC_{lag,it} = IC_{i,t-1}$, ensuring that the regressor is predetermined with respect to the dependent variable ER_{it} and avoiding simultaneity bias.

While IC_{lag} captures the information content of executed trades, the second regressor measures the liquidity pressure of the limit order book at timestamp t . More specifically, I construct the Order Flow Imbalance measure (OFI) following [Cont et al., 2014] where, for each message event n , the OFI contribution e_n captures the net change in depth at the best bid and ask relative to the previous event as:

$$e_n = \underbrace{\mathbf{1}_{P_n^B \geq P_{n-1}^B} q_n^B - \mathbf{1}_{P_n^B \leq P_{n-1}^B} q_{n-1}^B}_{\text{bid-side contribution}} - \underbrace{\left(\mathbf{1}_{P_n^A \leq P_{n-1}^A} q_n^A - \mathbf{1}_{P_n^A \geq P_{n-1}^A} q_{n-1}^A \right)}_{\text{ask-side contribution}}, \quad (3.13)$$

where P_n^B (P_n^A) denotes the best bid (ask) price at event n , and q_n^B (q_n^A) denotes the corresponding depth. The bid-side contribution increases either when the best bid price rises or additional depth arrives at the current best bid, and decreases when the best bid falls or depth is removed. The ask-side contribution exhibits symmetric, though opposite, behavior. Consequently, $e_n > 0$ indicates net buying pressure, while $e_n < 0$ indicates net selling pressure.

The cumulative OFI over a time window $[t - h, t]$ aggregates these event-level contributions:

$$\text{OFI}(t, h) = \sum_{n: t-h < t_n \leq t} e_n. \quad (3.14)$$

I then aggregate all of the constructed variables over non-overlapping 60-second intervals for examining FOMC event windows. The reason for this being that, at tick-by-tick frequency, OFI and order flow are generally noisy (e.g., individual trades are small and often driven by pure liquidity motives with no information content). Aggregating to 1 minute averages out this microstructure noise and makes the signal-to-noise from informed trading ratio stronger and more detectable. Also, this seems to be a standard approach in the market microstructure literature, as [Cont et al., 2014] use 10-second to 1-minute bins while [Andersen and Bollerslev, 1998] use 5-minute bins.

3.3.5 Cumulative Average Abnormal Order Flow

Before the regression analysis, I deem it interesting to examine whether signed order flow responds abnormally around such FOMC announcements, therefore providing a preliminary evidence of liquidity pressure around scheduled macroeconomic events. Hence, this first empirical section follows an event-study design adapted to intraday frequency in line with the renowned methodology proposed by [MacKinlay, 1997]. As for the main parameters of the model, I set 30-minute window $[-15, +15]$ around each FOMC announcement at $t = 0$ and compute the Cumulative Average Abnormal Order Flow Imbalance (CAAOFI) as:

$$\text{CAAOFI}(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} \text{AAOF}_{jt} \quad (3.15)$$

where $\text{AAOF}_{jt} = \frac{1}{N} \sum_{i=1}^N (OF_{ijt} - \overline{OF}_{ij})$ is the cross-sectional average of abnormal order flow at event-time t , and $\overline{OF}_{ij}$ is the firm-specific rolling mean computed over the 60-minute pre-event estimation window. Given the well-documented non-normality of high-frequency returns and order flow distributions, I assess statistical significance using two non-parametric methods: a bootstrap procedure with 100,000 replications and a Wilcoxon signed-rank test, both of which do not require any *a priori* assumption about the underlying data distribution. The bootstrap serves as my primary inference method; the Wilcoxon results are presented as a robustness check. Results for this preliminary analysis are found in Section 3.4.2.

Finally, to decompose the drivers of excess returns, I estimate the following panel regression:

$$ER_{it} = \beta_1 IC_{lag,it} + \beta_2 \text{OFI}_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \quad (3.16)$$

where $IC_{lag,it}$ is the lagged adverse selection component of order flow, OFI_{it} is order flow imbalance, α_i are firm fixed effects, and γ_t are FOMC event fixed effects. Standard errors are Newey-West HAC with five lags to correct for residuals' heteroskedasticity serial correlation.

3.4 Empirical Results

3.4.1 Exploratory Data Analysis

Table 3.6 reports summary statistics for the main variables across all five firms and 16 FOMC event windows, while Figure 3.1 illustrates their distributions.

ExcessReturn displays the fat-tailed, leptokurtic distribution (excess kurtosis = 1.21) typical of high-frequency returns [Cont, 2001], with a slightly negative mean (-0.0002), consistent with the net selling pressure documented above (see Section 3.4.2). The median is -0.0002, confirming that the distribution is approximately symmetric around a mildly negative centre. The information component IC_{lag} exhibits extreme concentration at zero (e.g., the median is precisely 0) with occasional large values in both tails (excess kurtosis = 4.49), implying that most minutes involve routine liquidity trading with infrequent episodes of informed activity. Order flow imbalance OFI_m displays a similar distributional pattern to IC_{lag} , with median exactly zero and moderate excess kurtosis (5.21), consistent with the concentration of limit order book activity near zero with occasional large imbalances. Lastly, Signed Order Flow OF is strongly left-skewed (skewness = -2.67, excess kurtosis = 21.28), reflecting the dominance of net selling episodes in the sample. This phenomenon is consistent with institutional investors reducing equity exposure around monetary policy announcements.

The correlation plot from Figure 3.2 provides a first-hand visual representation of the—possible—linear relationship between the main variables. Overall, it confirms that IC_{lag} and OFI_m are nearly orthogonal ($r = -0.11$), thus ensuring that the two predictors capture genuinely distinct dimensions of market activity in the joint regression specification. OFI and *ExcessReturn* display the strongest pairwise correlation ($r = 0.39$), in line with the well-established connection between order-flow imbalances and contemporaneous price changes. Order-flow imbalance from the limit order book seems to be the best independent predictor of excess returns. In economic terms this makes sense, as OFI captures the aggressive repositioning of limit orders around information events, which is a leading indicator of price movement. Interestingly, the proxy for the information component (i.e., IC_lag is negatively associated to Excess Returns ($r = -0.31$), suggesting that high adverse selection in the previous minute predicts lower excess returns in the current minute. This is consistent with price reversal after informed trading : prices overshoot when informed traders act (by front-running the announcement), then partially correct. Put differently, informed traders shift price discovery into the pre-event window, therefore attenuating the post-announcement return.

Table 3.6: Summary statistics for the main variables across all five firms and 16 FOMC event windows. The sample consists of 15,594 firm-minute observations after winsorisation at the 1% level. *ExcessReturn* is the market-model residual return. IC_{lag} is the lagged adverse selection component of order flow. OF is signed executed order flow in shares. OFI_m is order flow imbalance scaled to millions. All variables are resampled from tick-by-tick to minute-frequency.

Variable	N	Mean	SD	Min	P25	Median	P75	Max	Skew	Kurt
<i>ExcessReturn</i>	15,594	-0.0002	0.0010	-0.0032	-0.0007	-0.0002	0.0004	0.0028	-0.023	1.21
IC_{lag}	15,594	0.0031	0.1817	-0.6354	-0.0450	0.0000	0.0450	0.7000	0.277	4.49
OF	15,594	-21,151	62,473	-480,458	-21,725	-2,713	-49	374,554	-2.668	21.28
OFI_m	15,594	-0.0004	0.0120	-0.0478	-0.0028	0.0000	0.0022	0.0443	-0.220	5.21

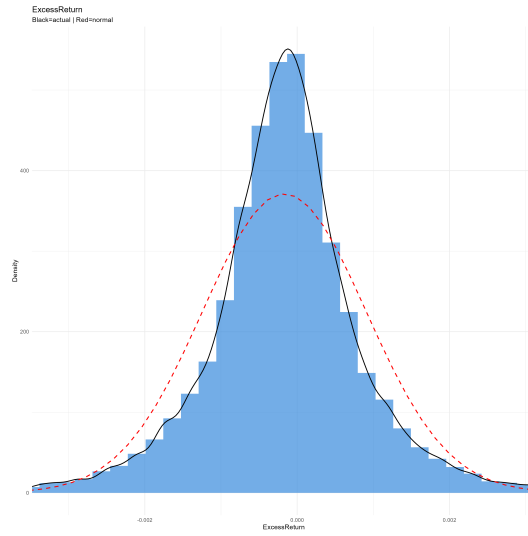
All variables resampled from tick-by-tick to 1-minute frequency using non-overlapping intervals and winsorised at the 1% level. Kurtosis reported as excess kurtosis.

$IC_{lag} = \text{sign}(OF_t) \times (\text{MidPrice}_{t+1} - \text{MidPrice}_t)$, lagged one period.

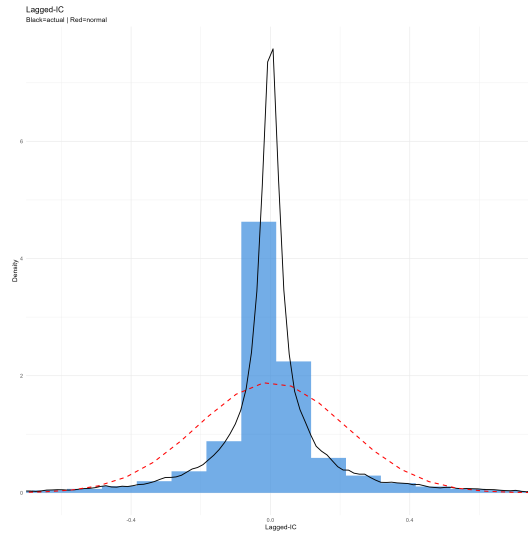
Sample: 6 firms $\times$ 16 FOMC events, 1-minute frequency, ± 60 minutes around each announcement.

Figure 3.3 presents the bivariate relationships between the main variables of interest.

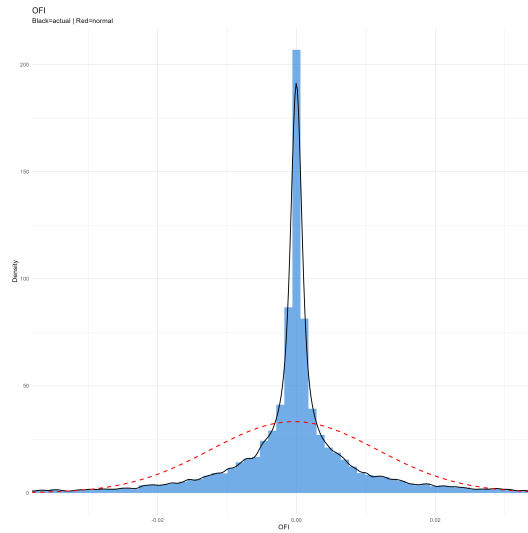
3.4. Empirical Results



(a)



(b)



(c)

Figure 3.1: Distribution of—respectively -Excess Returns, Information Component, and Order Flow Imbalance across all FOMC announcement windows. The black curve is the kernel density estimate; the red dashed curve is the normal distribution fitted to the same mean and variance. Top pane: *ExcessReturn*; *IC_{lag}*. Middle pane : *IC_{lag}*; Bottom pane: *OFI*

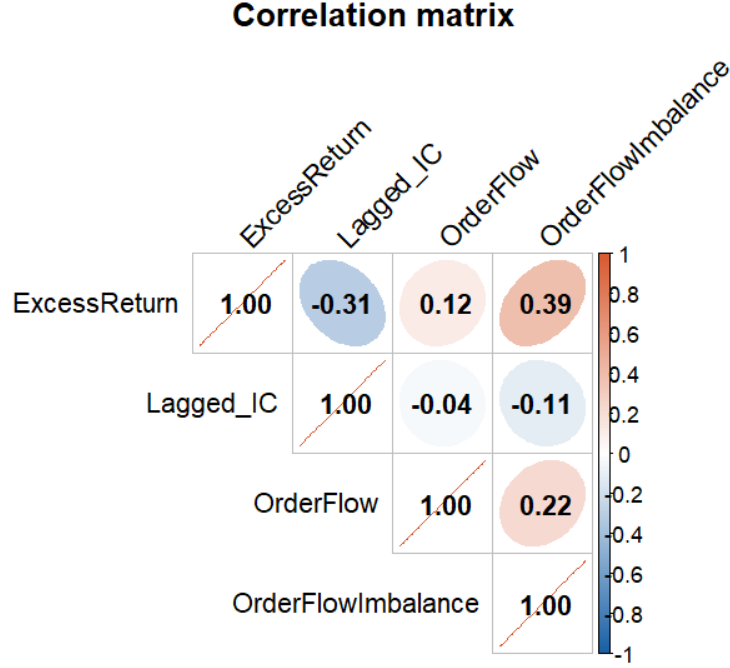


Figure 3.2: Pairwise correlation matrix between *ExcessReturn*, IC_{lag} , and *OFI*. Pearson correlation coefficients are reported for all variable pairs.

Panel (a) - *ExcessReturn* vs. IC_{lag} - shows a clear negative slope, with the loess⁸ and linear fit broadly coinciding in the tails with a slight kink near zero. Panel (b) - IC_{lag} vs. *OFI* - supports the near-orthogonality of the selected predictors. The loess is essentially flat across the entire range of *OFI*, despite a slight negative drift at extreme *OFI* values, but negligible. Hence, from a visual inspection, the regression design seems exempt from multicollinearity concerns. Finally, in Panel (c) - *ExcessReturn* vs. *OFI* - the loess exhibits a clear S-shaped relationship: approximately flat for extreme negative *OFI* values, with a steep positive slope near zero, then flattening again for large positive *OFI*. This indicates that a linear fit would most likely underestimate the true relationship in the centre, while overestimating it at the extremes, thus suggesting inadequacy for modelling *OFI*. Moreover, the sharp transition at zero and the flattening at the extremes are consistent with the theory of diminishing marginal price impact of market orders in the LOB: larger one-sided book depth produces progressively smaller incremental price effects, analogous to the concavity documented for executed order flow by Almgren et al. [2005]. This is also known as the square-root law of market impact, which predicts that price impact scales as a concave function of order flow rather than linearly [Bouchaud et al., 2018]. This motivates the investigation of power transformations of *OFI*. In particular, the square-root transformation:

$$OFI_sqrt = \text{sign}(OFI) \times \sqrt{|OFI|} \quad (3.17)$$

preserves the sign of order flow imbalance while compressing its magnitude, effectively down-

⁸Locally Estimated Scatterplot Smoothing (*LOESS*) is a non-parametric regression technique employed in scatterplots to effectively model complex relationships and identify trends in noisy data by creating smoothed lines fitting multiple locally weighted polynomial regressions.

weighting extreme values and allowing the model to capture the concave relationship between order flow and returns suggested by the scatter plot.

As reported in the robustness of Section 3.5, this transformation yields a marginally superior fit relative to the linear specification, though at the cost of a less direct theoretical interpretation.

3.4.2 Intraday Event-Study on Order Flow

Figures 3.4a through 3.4f present the CAAOFI plots for all five stocks. The results reveal a clear cross-sectional pattern aligned with the tick-size classification of the sample. The two small-tick stocks (GS and IBM) exhibit the strongest and most statistically significant abnormal order flow dynamics: for both, the CAAOFI trends monotonically negative from the pre-announcement window, accelerates sharply at $t = 0$, and continues declining in the post-announcement period. This pattern aligns with the net selling pressure triggered by the FOMC announcement and the theoretical prediction that order flow is more informative in small-tick stocks, where price competition dominates queue competition [Parlour, 1998].⁹

In contrast, the pattern does not hold uniformly across all stocks. For the large-tick stocks (CSCO and BAC), the CAAOFI displays a similar directional trend to the sample average, but the Wilcoxon test fails to confirm statistical significance relative to the null of a zero median for either stock. AMZN and MSFT present more ambiguous results: AMZN shows a mild negative post-announcement drift, albeit with extremely wide confidence bands under both non-parametric tests; MSFT exhibits a sharp reversal at $t = 0$, shifting from positive pre-announcement levels to negative post-announcement values—a directionally intriguing but interpretively challenging pattern.

Taken together, the CAAOFI analysis provides preliminary evidence that FOMC announcements generate abnormal selling pressure on average, concentrated in small-tick stocks where order flow is most informative. These findings motivate the regression analysis in Section 3.4.3, where I formally decompose excess returns into information and order flow components.

Figures 3.5 to 3.7 present the time evolution of the mean absolute values for the time-series in the model. Looking at the Information component, every stock shows a sharp spike at $t = 0$ with clear pre-announcement flatness independently from the tick-size they belong to. Notably, the spike is largest for small-tick stocks (GS, IBM, MSFT reach 0.4-1.5) and smallest for large-tick stocks (BAC, CSCO reach 0.05-0.15). For what concerns the Order Flow, it shows to be noisy but directional. However, no clear systematic spike at $t = 0$ across stocks is detected.

Lastly, the faceted plots of Excess Returns by tick-size facet shows another clear picture: almost near-zero pre-announcement mean, with increased volatility post-announcement.

3.4.3 Estimation Results

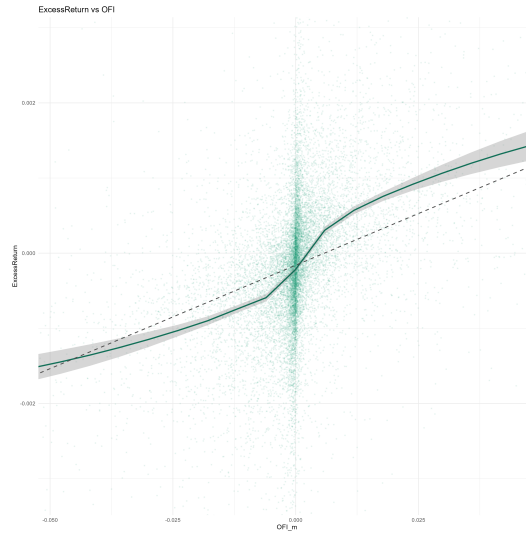
At this stage, I proceed to estimate a multiple panel regression model with firm and event fixed effects, using Newey-West HAC standard errors in order to correct for heteroscedasticity and serial correlation in the residuals.

Table 3.7 reports OLS estimates of excess returns on the information component and order flow imbalance, with firm and FOMC event fixed effects throughout. Standard errors are Newey-West HAC with five lags, correcting for the heteroscedasticity and serial correlation confirmed by the Breusch-Pagan test ($BP = 443.7$, $p < 0.001$) and Durbin-Watson statistic

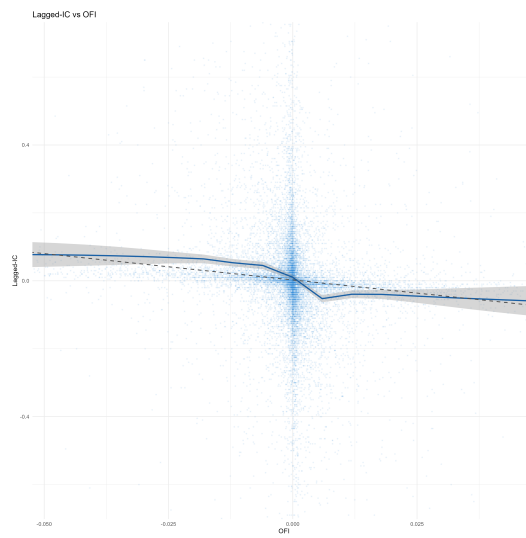
⁹Robustness checks using the Wilcoxon non-parametric test appear in Appendix C



(a) AMZN



(b) BAC



(c) BAC

Figure 3.3: Bivariate scatterplots superimposed with loess (solid) and linear (dashed) fits. Top pane: *ExcessReturn* versus IC_{lag} . Middle pane: IC_{lag} versus *OFI*. Bottom pane: *ExcessReturn* versus *OFI*

3.4. Empirical Results

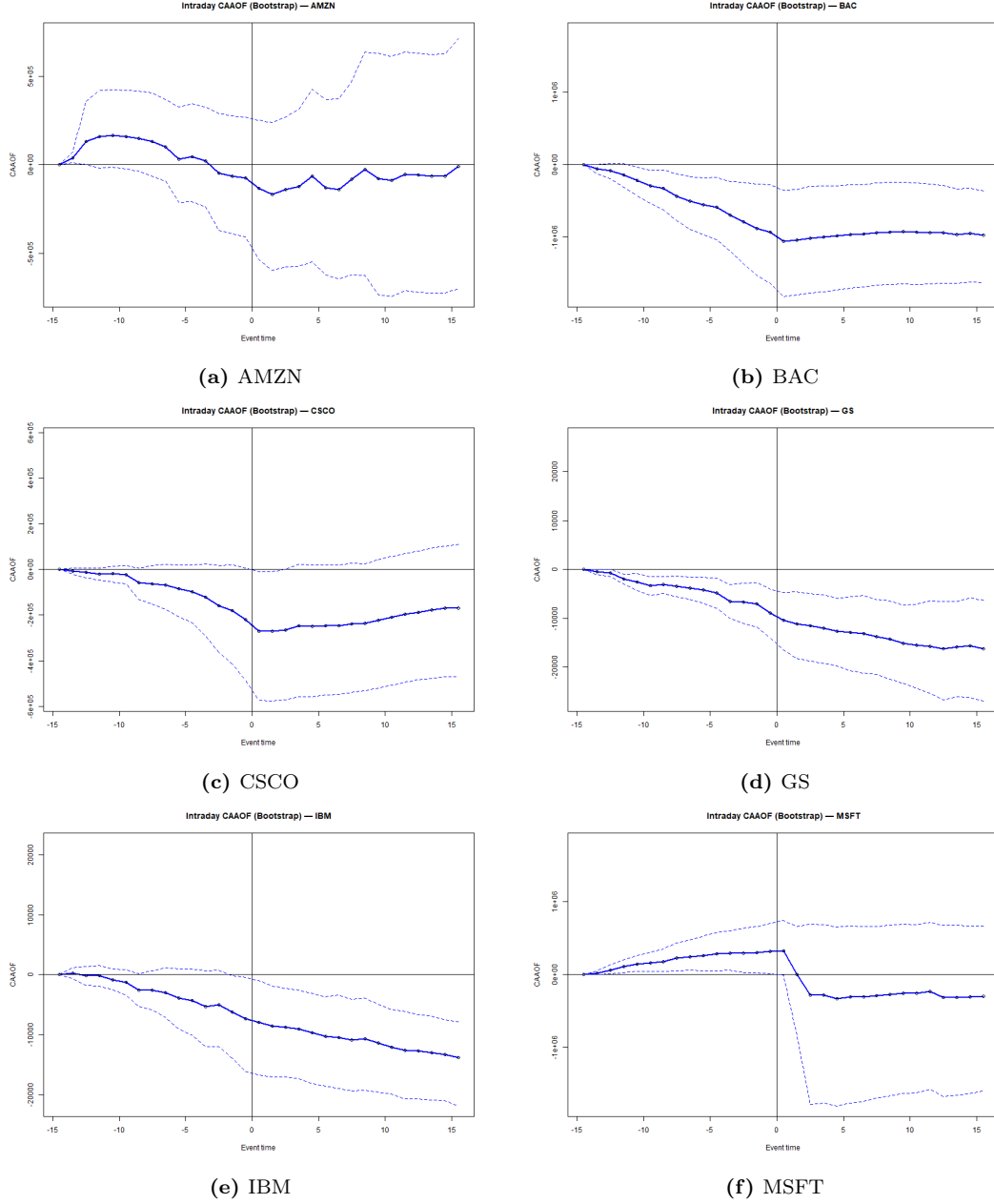


Figure 3.4: Bootstrap confidence intervals for CAAOFI across five stocks: Amazon (AMZN), Bank of America (BAC), Cisco (CSCO), Goldman Sachs (GS), IBM, and Microsoft (MSFT). Each panel displays the bootstrap distributions of CAAOFI coefficient estimates.

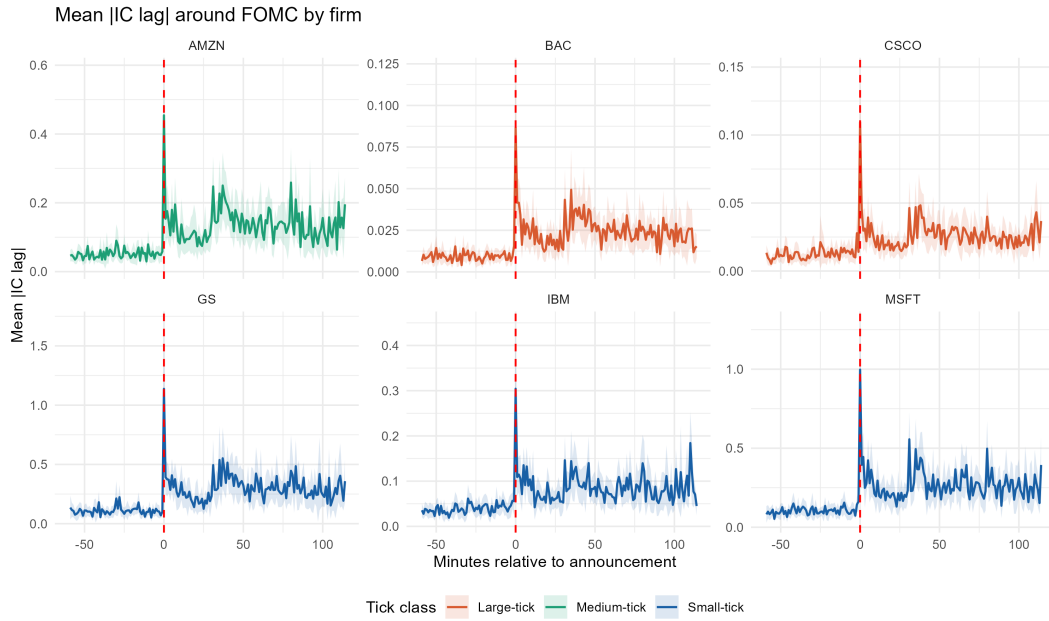


Figure 3.5: Mean absolute adverse selection component $|IC_{lag}|$ around FOMC announcements by firm. The red dashed vertical line marks $t = 0$, the moment of the announcement. Shaded bands represent 95% confidence intervals. Large-tick stocks (BAC, CSCO) are shown in red, medium-tick (AMZN) in green, and small-tick (GS, IBM, MSFT) in blue. All stocks display a sharp spike in informed trading intensity at $t = 0$, with the spike magnitude notably larger for small-tick stocks, consistent with greater adverse selection dynamics in markets where price competition dominates queue competition.

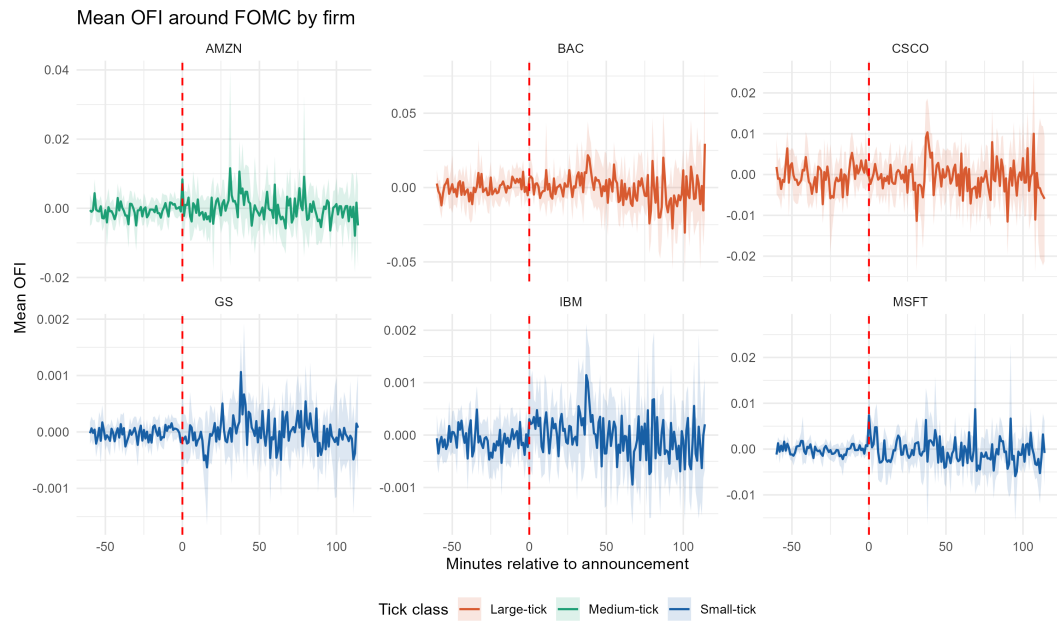


Figure 3.6: Mean Order Flow Imbalance (OFI) around FOMC announcements by firm. The red dashed vertical line marks $t = 0$. Shaded bands represent 95% confidence intervals. Unlike $|IC_{lag}|$, OFI shows no systematic spike at the announcement time across stocks, consistent with its endogenous nature, OFI and returns respond simultaneously to the same information shock rather than OFI leading returns.

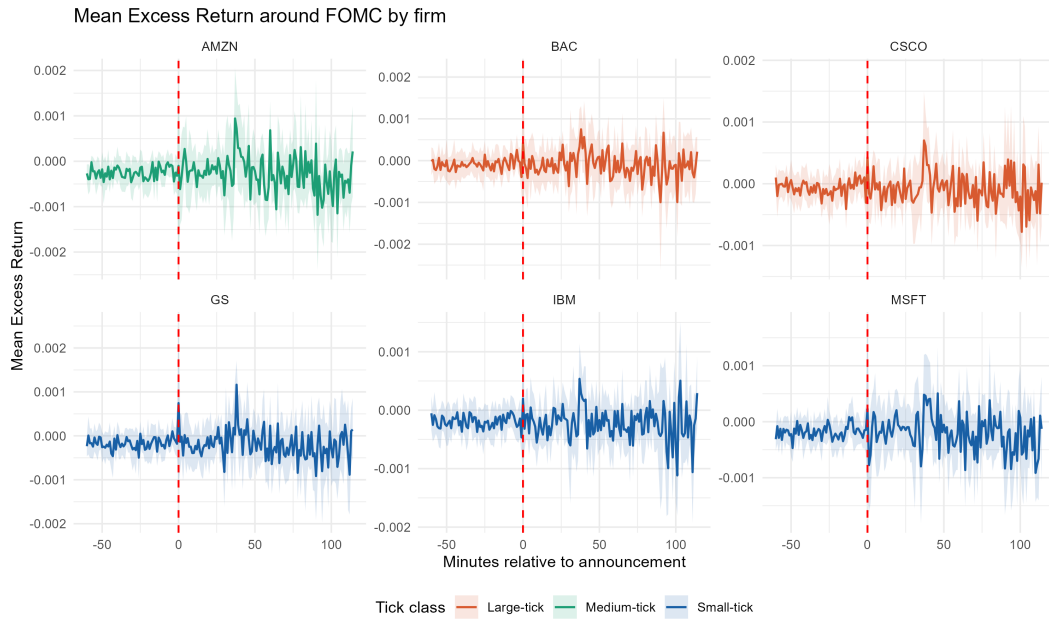


Figure 3.7: Mean Excess Return around FOMC announcements by firm. The red dashed vertical line marks $t = 0$. Shaded bands represent 95% confidence intervals. Excess returns are approximately zero pre-announcement with increased volatility post-announcement, reflecting heterogeneous market reactions across the 16 FOMC events. The absence of a systematic directional pattern is consistent with the event fixed effects absorbing the common macro shock.

(DW = 1.808, $p < 0.001$) respectively. Variance inflation factors are below 1.15 for all regressors, confirming the absence of multicollinearity. Column (1) presents the univariate IC_{lag} specification. The adverse selection coefficient is negative and highly significant ($\beta = -0.0017$, $t = -23.4$, $R^2 = 14.6\%$), establishing a clean univariate information channel: high informed trading intensity at $t-1$ predicts lower excess returns at t , consistent with temporary price overshooting following informed order flow as documented by Glosten and Harris (1988). Column (2) presents the univariate OFI specification. OFI is positive and highly significant ($\beta = 0.038$, $t = 38.9$, $R^2 = 24.7\%$) - the highest single-predictor R^2 in the table, reflecting that order book imbalance is the strongest marginal predictor of short-horizon returns, consistent with [Cont et al., 2014]. Column (3) presents the preferred joint specification with IC_{lag} and OFI. Both coefficients remain highly significant - IC_{lag} at $\beta = -0.0014$ ($t = -20.7$) and OFI at $\beta = 0.035$ ($t = 36.9$) - with $R^2 = 31\%$. The Ramsey RESET test nearly passes at conventional significance levels ($F = 4.09$, $p = 0.017$), indicating adequate functional form specification. The scatter plot of OFI against excess returns reveals a nonlinear relationship with a kink at zero (Figure 3.3). A systematic search over power transformations confirms that the square root transformation improves single-predictor fit (AIC improvement of 710 points) and is consistent with the square root law of market impact [Almgren et al., 2005]. However, when combined with IC_{lag} , the square-root transformation introduces residual cross-product nonlinearity confirmed by a GAM tensor interaction ($F = 133.1$, $p < 0.001$). I therefore retain the linear OFI specification in the main results and report square-rooted OFI results in the robustness table. Including signed order flow OF as an additional regressor leaves both IC_{lag} and OFI coefficients unchanged while adding negligible explanatory power ($\Delta R^2 < 0.001$) at the cost of residual nonlinearity. I, therefore, retain the parsimonious two-predictor specification. Column (4) reports the preferred specification under two-way clustering by firm and FOMC event—the most conservative standard error correction available for this panel structure—yielding IC_{lag} $t = -10.03$ ($p = 0.0002$) and OFI $t = 4.94$ ($p = 0.004$), confirming that both findings survive the strictest possible error correction despite the limited number of clusters (96 firm-event pairs). The IC_{lag} coefficient of -0.0014 implies that a one-unit increase in lagged adverse selection predicts a 0.14 bp decrease in excess return in the following minute. Under [Kyle, 1985]’s framework, this reflects partial correction of the temporary price overshoot generated by informed traders. Concurrently, market-makers are unable to perfectly distinguish informed from uninformed trades, and ultimately allow prices to temporarily exceed the new efficient level, with the correction occurring in the subsequent minute.

On one side, results provide mixed but informative evidence with respect to **H1**. While contemporaneous order flow imbalance (*OFI*) is a strong and significant predictor of excess returns under OLS ($\hat{\beta} = 0.038$, $t = 38.9$), instrumental variable estimation reveals that this relationship is driven by simultaneity rather than causality, meaning that once the endogeneity of *OFI* is corrected via IV, order flow imbalance loses statistical significance entirely. Therefore, **H1** is not supported in its causal interpretation: contemporaneous order flow reflects simultaneous price discovery rather than constituting an independent causal driver of excess returns. On the other hand, the findings provide strong and unambiguous support for **H2**. The adverse selection component IC_{lag} is negative and highly significant across all specifications ($\hat{\beta} = -0.0014$, $t = -20.7$ under Newey-West HAC; $t = -10.0$ under two-way clustering), confirming that high informed trading intensity at $t - 1$ predicts return reversal at t , consistent with temporary price overshooting following informed order flow. Instrumental variable estimation strengthens this conclusion: the IC_{lag} coefficient grows in magnitude under IV correction ($\hat{\beta} = -0.0016$, $t = -16.3$), confirming that OLS provides a conservative estimate of the true causal effect of informed trading on subsequent price

dynamics. **H2** is therefore confirmed and its causal interpretation is validated.

Table 3.7: Baseline regression results. The dependent variable is *ExcessReturn*. M1 and M2 are univariate specifications. M3 is the preferred joint specification with Newey-West HAC standard errors (lag = 5). M4 reports M3 with two-way clustering by firm and FOMC event. All specifications include firm and FOMC event fixed effects. *t*-statistics in parentheses.

	(M1) IC only	(M2) OFI only	(M3) IC + OFI	(M4) IC + OFI
IC_{lag}	-0.0017*** (-23.4)		-0.0014*** (-20.7)	-0.0014*** (-10.0)
OFI		0.0375*** (38.9)	0.0345*** (36.9)	0.0345** (4.9)
Firm FE	Yes	Yes	Yes	Yes
Event FE	Yes	Yes	Yes	Yes
SE	NW	NW	NW	Clustered
R^2	0.146	0.247	0.309	0.308
RESET p	0.001	0.000	0.017	—
N	15,594	15,594	15,594	15,594

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. NW = Newey-West HAC, lag = 5. Clustered = two-way by Firm $\times$ FOMC event.

3.4.4 Endogeneity and Causal Identification

Order flow imbalance and excess returns may be jointly determined within the minute: a single informational shock arriving at t moves prices and simultaneously provokes the quote revisions and depth changes that constitute OFI_t . OLS therefore cannot separate order flow *causing* price movement from order flow and prices *responding together* to a common cause. In fact, the Wu-Hausman test firmly rejects the null of exogeneity ($F = 14.67$, $p < 0.001$), thus implying that the OLS estimate of β_{OFI} is not merely imprecise but inconsistent. Therefore, I address this by instrumenting OFI_t with lagged OFI (OFI_{t-1}) and lagged signed order flow (OF_{t-1}), both predetermined with respect to the innovation arriving at t . The first-stage F -statistics confirm the *relevance* criterion ($F = 56.9$ and $F = 1,461$ respectively), well above the conventional threshold of ten [Staiger and Stock, 1994].

With two instruments and one endogenous regressor the system is overidentified by one degree of freedom, and the Sargan fails to reject the null ($p = 0.936$). However, for the sake of clarity, let me state the following: *i*) the test evaluates only the *overidentifying* restrictions (i.e., whether the two instruments deliver mutually consistent estimates, conditional on at least one already being valid); *ii*) a failure to reject is not evidence of validity, and a p -value of 0.936 is as consistent with low power as with strong support; *iii*) both instruments are lagged order-flow measures, so if the exclusion restriction fails for one it will likely fail for the other as well, and in the same direction. The result of this test is, therefore, the absence of evidence against the strategy, not its confirmation.

The *exclusion* criterion requires that lagged order flow affect current excess returns *only* through contemporaneous OFI . Unlike relevance, it cannot be tested directly and must be defended on theoretical grounds. The most credible violation follows from a stylised fact documented in Section 2.2.6 of this dissertation, namely the long memory of order flow [Lillo and Farmer, 2003; Bouchaud et al., 2018]. Since OF_{t-1} is strongly autocorrelated, any channel through which past flow forecasts current returns directly (i.e., short-horizon momentum, the gradual diffusion of an information shock across several minutes, or a large parent order worked through a multi-minute execution schedule), would place the instrument in the structural equation. Formally, this may arise either through a direct path from OF_{t-1}

to r_t , or through a common component leaving $\text{corr}(OF_{t-1}, \varepsilon_t) \neq 0$. Four features of the research design limit, without eliminating, this concern. First, identification comes from tightly bounded windows around discrete, precisely timestamped FOMC announcements, which leaves comparatively less room for slow information diffusion than would a design based on unscheduled news. Second, to a certain extent, firm and event fixed effects absorb the persistent components, so the identifying variation is within firm-event. Third, IC_{lag} enters the regression directly and is constructed precisely to capture the signed price reversal that follows informed trading; in this view, the most obvious direct channel of returns predictability is therefore controlled for rather than left in the error term. Fourth, and in my opinion most importantly, the likely direction of the bias works *against* the finding rather than for it: a violation of this kind would make lagged order flow appear to predict returns directly and would inflate $\hat{\beta}_{OFI}$. Instead, the data show the exact opposite: the coefficient collapses once the instrument is applied. Table 3.8 presents OLS and IV estimates side by side. To gauge economic magnitude, throughout this section I report the effect of a one standard deviation change in each regressor, given by $\hat{\beta} \times \sigma_X$ and expressed in bps. Under IV estimation the adverse selection coefficient IC_{lag} grows in magnitude ($\hat{\beta} = -0.0016$, $t = -16.3$), indicating that OLS was conservative. By contrast, the OFI coefficient falls from 0.0345 to 0.0019 and retains only marginal significance ($p = 0.065$), indicating that the strong positive OLS relationship between order flow imbalance and returns reflects simultaneous price discovery (i.e., just price moves and order flow happening at the same time), rather than a causal price impact channel. In economic terms, using the standard deviations reported in Table 3.6, a one standard deviation increase in the lagged information component ($\sigma = 0.1817$) corresponds to roughly -2.5 bps of excess return per minute (about a quarter of the 10 bp standard deviation of one-minute excess returns), under IV to approximately -2.9 bp. Applying the same exercise to order flow imbalance ($\sigma = 0.0120$) is instructive precisely because the estimators disagree: under OLS a one standard deviation move maps to approximately 4.1 bps, under IV to approximately 0.228bp. Employing an instrumental variable, therefore, shrinks the order flow channel roughly eighteen-fold, to a magnitude that is economically negligible at the one-minute horizon, while leaving the information channel intact. So, under the exclusion restriction defended above, this distinction is the paper’s main takeaway: the adverse selection channel survives correction for simultaneity, whereas the order flow imbalance channel does not. A final comparison puts these magnitudes in perspective. One tick corresponds to approximately 2.9 bps for a large-tick stock such as BAC and 0.3 bps for a small-tick stock such as GS. The pooled adverse selection effect of roughly 2.5 bps per minute is therefore smaller than a single price increment in the former case and close to an order of magnitude larger in the latter. The channel identified here is economically meaningful, but it is not straightforwardly exploitable at the one-minute horizon in the large-tick regime, where the predicted reversal is smaller than the minimum price movement required to capture it. Notably, this runs counter to intuition: Section 3.5.1 shows that the adverse selection effect is *strongest* in large-tick stocks, that is, precisely in the regime where the coarse price grid makes it least straightforward to act upon. Both facts follow from the same institutional feature—where the spread is pinned at one tick, market makers cannot adjust quotes gradually, so the price adjustment is concentrated into a single minute rather than smeared across several.

Nevertheless, I am aware that this identification strategy relies on untestable assumptions and a specific sample (i.e., six NASDAQ-listed stocks across sixteen FOMC announcements in 2023–2024), meaning that these findings apply specifically to high-impact monetary policy periods, rather than proving that order flow is broadly irrelevant to price formation.

Table 3.8: OLS versus IV estimates. The dependent variable is *ExcessReturn*. The endogenous variable is *OFI*, instrumented by its own lag and the lag of signed order flow (OFI_{t-1}). All specifications include firm and FOMC event fixed effects. Diagnostic tests reported at the bottom. *t*-statistics in parentheses.

	(1) OLS	(2) IV
IC_{lag}	-0.0014*** (-20.7)	-0.0016*** (-16.3)
OFI	0.0345*** (36.9)	0.0019 (1.8)
Firm FE	Yes	Yes
Event FE	Yes	Yes
SE	NW	Conventional
R^2	0.309	0.255
N	15,594	15,488
<i>IV diagnostic tests</i>		
Weak instruments F	—	56.9***
Wu-Hausman F	—	14.67***
Sargan p	—	0.936

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, · $p < 0.10$.

Instruments: OFI_{t-1} , OFI_{t-1} . Sargan test of overidentifying restrictions.

3.5 Robustness

Table 3.9 collects the IC_{lag} coefficient across all specifications estimated in this paper. The adverse selection effect is negative and statistically significant in every single specification without exception. The coefficient ranges from -0.00123 to -0.00156 across eight OLS specifications—a factor of 1.27x across the full range—indicating exceptional stability. The finding is robust to the choice of OFI functional form (linear, square root, natural spline with 7 degrees of freedom), the inclusion of additional controls (signed order flow, lagged returns), the standard error correction method (Newey-West HAC, two-way clustering by firm and event), the estimation method (OLS, IV), and sample splits by tick-size regime and monetary policy surprise magnitude. The *t*-statistic never falls below 8.89 in any specification—more than four times the conventional threshold of 2 under any error correction method. Column (2) replaces linear OFI with the square root transformation motivated by the square root law of market impact [Almgren et al., 2005]; the IC_{lag} coefficient is unchanged at $\beta = -0.0012$ ($t = -19.1$), confirming robustness to the functional form of OFI.

3.5.1 Cross-Sectional Heterogeneity

To facilitate cross-sectional comparison, all variables entering the interaction specification are standardised within each firm-event cell prior to estimation. Table 3.10 reports the IC_{lag} coefficient estimated separately for large-tick and small-tick stocks, using within-firm standardised variables to ensure comparability across regimes. The adverse selection effect is significantly stronger for large-tick stocks ($\beta = -0.000350$, $t = -14.75$) than for small-tick stocks ($\beta = -0.000232$, $t = -14.76$), with the difference formally confirmed by an interaction test ($\beta_{\text{interaction}} = -0.000111$, $t = -5.20$, $p < 0.001$). This heterogeneity is consistent with the theoretical predictions of [Parlour, 1998] on how tick-size affects market-maker behaviour and adverse selection dynamics. In large-tick stocks—where the spread is fixed at approximately one tick—market-makers cannot adjust quotes gradually in response to informed order flow.

3.6. Conclusions

They face a binary choice: remain in the queue and risk adverse selection, or cancel and lose queue position. When informed trading occurs, the price adjustment is therefore abrupt and concentrated within a single minute, producing a larger per-minute reversal. In small-tick stocks, market-makers continuously adjust quotes in small increments, spreading the price impact over multiple minutes and reducing the per-minute correction. The adverse selection effect is significantly stronger for large-tick stocks ($t = -5.20$, $p < 0.001$), consistent with the theoretical prediction that market-makers in queue-driven markets face more constrained adjustment to informed order flow [Parlour, 1998; Foucault et al., 2005].

Table 3.9: Robustness of the adverse selection coefficient IC_{lag} across all specifications. The dependent variable is *ExcessReturn* throughout. Each row reports the IC_{lag} coefficient, t -statistic, and R^2 for a different specification or subsample. Firm and FOMC event fixed effects included in all specifications. NW = Newey-West HAC standard errors (lag = 5).

Specification	$\hat{\beta}_{IC}$	t -stat	p -value	R^2
<i>Main specifications</i>				
(1) Baseline: IC + OFI_m (NW)	−0.001404	−20.72	< 0.001	0.309
(2) OFI_sqrt specification	−0.001248	−19.05	< 0.001	0.340
(3) Adding OF_m	−0.001399	−20.66	< 0.001	0.310
(4) SplineS OFI (df = 7)	−0.001228	−18.76	< 0.001	0.343
<i>Standard error corrections</i>				
(5) Two-way clustered (Firm × Event)	−0.001404	−10.03	< 0.001	—
(6) IV estimation	−0.001561	−16.30	< 0.001	—
<i>Subsamples</i>				
(7) Low surprise events	−0.001274	−9.49	< 0.001	0.381
(8) High surprise events	−0.001430	−8.89	< 0.001	0.265
(9) Large-tick stocks (std)	−0.000350	−14.75	< 0.001	0.510
(10) Small-tick stocks (std)	−0.000232	−14.76	< 0.001	0.318

*** $p < 0.001$. All IC_{lag} coefficients negative and significant without exception.

Rows (9)–(10) use within-firm standardised IC_{lag} — not directly comparable to rows (1)–(8).

Low/high surprise split at median $|MPS_{orth}|$ following [Bauer and Swanson, 2023].

3.6 Conclusions

This paper investigates the causal relationship between informed trading and price dynamics around Federal Open Market Committee announcements, contributing to the ongoing debate on whether prices are driven by information or by order flow. Using high-frequency limit order book data for five NASDAQ-listed stocks across 16 FOMC events from 2023 to 2024, I decompose excess returns into an information component and an order flow component and test their respective contributions to intraday price dynamics.

The results jointly support a microstructural account in which informed trading, rather than mechanical order flow, is the proximate causal driver of high-frequency price reversals. The adverse selection component IC_{lag} is negative and highly significant across all specifications, confirming that high informed trading intensity at $t - 1$ predicts return reversal at t consistent with temporary price overshooting following informed order flow as predicted by [Kyle, 1985] and [Glosten and Harris, 1988]. At the moment of the announce-

Table 3.10: Cross-sectional heterogeneity by tick-size regime. The dependent variable is *ExcessReturn*. Columns (1) and (2) report estimates for large-tick (BAC, CSCO) and small-tick (GS, IBM, MSFT) stocks respectively, using within-firm standardised regressors to ensure comparability across regimes. Column (3) reports the formal interaction test on the pooled sample excluding AMZN. All specifications include firm and FOMC event fixed effects with Newey-West HAC standard errors (lag = 5). *t*-statistics in parentheses.

	(1) Large-tick	(2) Small-tick	(3) Interaction
IC_{lag_std}	−0.000350*** (−14.75)	−0.000232*** (−14.76)	−0.000232*** (−15.30)
$IC_{lag_std} \times \mathbb{I}[\text{Large}]$			−0.000111*** (−5.20)
OFI_sqrt_std	0.000383*** (33.5)	0.000390*** (33.5)	0.000383*** (33.5)
Firm FE	Yes	Yes	Yes
Event FE	Yes	Yes	Yes
SE	NW	NW	NW
R^2	0.510	0.318	—
N	5,011	7,932	12,943

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Large-tick: BAC, CSCO. Small-tick: GS, IBM, MSFT. AMZN (medium-tick) excluded from Column (3).

Variables standardised within firm by dividing by firm-specific standard deviation.

ment, informed trading intensity spikes sharply and remains elevated for approximately 30 minutes, suggesting that FOMC announcements trigger a sustained episode of adverse selection rather than a single instantaneous price adjustment. The 30-minute decay is consistent with the semi-strong form of market efficiency, as the public information content of the Fed statement is incorporated quickly, and the temporary price impact of informed trading is corrected through subsequent liquidity trading as market-makers update their beliefs. Cross-sectional analysis reveals that the adverse selection effect is significantly stronger for large-tick stocks ($\beta_{\text{interaction}} = -0.000111$, $t = -5.20$, $p < 0.001$), where market makers face more constrained quote adjustment and greater exposure to informed order flow [Parlour, 1998; Foucault et al., 2005]. This heterogeneity validates the mechanism and rules out the possibility that the IC_{lag} finding is a statistical artifact of the panel structure.

However, this study is not exempt from limitations. Alongside the identification concerns discussed in Section 3.4.4, two additional restrictions deserve emphasis. First, while counting a total of 15,594 firm-minute observations, the effective number of independent information events is 16, and the firm dimension is far too small for cluster asymptotics to apply. Hence, the two-way clustered standard errors reported in Table 3.7 should be read as indicative rather than exact. Second, the sample covers 2023-2024, a period of unusually elevated policy rate uncertainty in which FOMC communication was closely scrutinised. The mechanism identified here concerns high-attention monetary policy events, and its magnitude in quieter policy regimes remains an open question.

Several directions remain open for future research. First, extending the sample to a longer time series covering multiple monetary policy regimes would provide greater statistical power for the surprise-magnitude heterogeneity test. Secondly, varying the aggregation frequency (from the current one-minute intervals to sub-second horizons) would sharpen the identification of the information channel by reducing the simultaneity between order flow

3.6. Conclusions

and returns. Finally, incorporating alternative OFI measures that account for quote updates at multiple levels of the limit order book [Cont et al., 2014] would provide a more complete picture of how liquidity provision responds to informed trading around macroeconomic announcements.

Conclusive Remarks

In this dissertation, I have explored the role of public information in shaping short-term price dynamics in a rapidly evolving financial ecosystem. Exploiting the unprecedented availability of granular, high-frequency data, my aim was to revisit a timeless and contested question through the lenses of modern market microstructure theory. Likewise, the three papers that compose this thesis tentatively approach this broad question from complementary empirical angles, each building on the residual questions left open by its predecessor and contributing with a distinct methodological rigor to the overarching argument. Taken together, they paint a coherent and nuanced picture of whether and how the information-processing mechanism unfolds in modern financial markets. The overarching theoretical framework of the present work is the striking contrast between two competing frameworks: the *Efficient Market Hypothesis*, which attributes information a direct causal role in price formation; and the recently proposed *Inelastic Market Hypothesis*, which argues that order flow, rather than information *per sé*, is the proximate driver of price dynamics. While adjudicating definitively between these two views—or, as [Bouchaud, 2022b] provocatively frames it ‘[...] [to] possibly hammer a final nail into the coffin of the *Efficient Market Hypothesis*’ - lies beyond the scope of this dissertation, the evidence that emerges from the data is not easily reconciled with the orthodox neoclassical view of efficient markets. The balance of findings leans towards an *adaptive complex systems* perspective of financial markets, in which heterogeneous agents, strategic order flow, and market in-elasticity jointly determine how prices respond to the arrival of public information. A perspective that, as this dissertation argues, deserves a more central place in the empirical finance literature than it has so far received.

Nonetheless, it is worth being explicit about one apparent tension in these concluding remarks. The third paper finds that the order flow imbalance coefficient collapses from 0.0345 under OLS to 0.0019 under instrumental variable estimation—an eighteen-fold attenuation, to roughly two tenths of a basis point per standard deviation—while the adverse selection coefficient moves in the opposite direction, growing in magnitude to -0.0016 ($t = -16.3$) and remaining negative in all sixteen FOMC events, statistically distinguishable from zero in fourteen of them. Taken at face value, this may seem to sit awkwardly beside a conclusion inclined toward the order-driven view. The tension is less sharp than it appears. Bouchaud [2022b] formulates the order-driven claim as the proposition that prices move with order flow *whether informed or non-informed*, and argues that “the lion’s share of the volatility is induced by trades”. What the third paper establishes is a decomposition of that flow rather than a rejection of it: at the one-minute horizon around scheduled macroeconomic announcements, it is the informed component that survives correction for simultaneity. Informed order flow remains order flow, and the result therefore refines the order-driven proposition rather than contradicting it. A measurement distinction sharpens the point further. The quantity instrumented here is the imbalance of depth and quotes at the best limits [Cont et al., 2014]—a book-level state variable that includes orders never executed. The object of

the latent liquidity theory is the metaorder, executed incrementally over minutes to days, whose impact is governed by the square-root law. These are different quantities measured at different scales, and a null result on the former is not evidence against the latter. What can be claimed is accordingly narrower, and more conservative, than a verdict on either hypothesis: around these events and at this frequency, it is the informational content of order flow, rather than its mechanical imbalance, that moves prices.

The first paper established that financial markets do not process firm-specific ESG news uniformly. Using minute-level data for Dow Jones Industrial Average constituents combined with precisely timestamped ESG news releases classified by a Large Language Model (LLM) across both pillar (E, S, G) and sentiment dimensions, the paper documented significant heterogeneity in abnormal returns and trading volumes following ESG announcements. Reactions varied systematically across ESG pillars, with Governance news tending to generate more pronounced and faster price responses than Environmental or Social disclosures. In several cases intraday reactions were muted or statistically insignificant. To a certain extent, one could argue that such patterns might be consistent with a rapid pre-announcement absorption or anticipatory effect. However, the minute-level design cannot distinguish these from mere limited statistical power in pillar-sentiment subsamples with few events, nor from the 'attenuation' induced by classification error discussed in Section 1.8.1. I therefore treat them as inconclusive results, rather than fragmented evidence against the semi-strong form of the EMH. The paper's key methodological contribution was the LLM classification framework, which addressed the semantic heterogeneity inherent in ESG disclosures more rigorously than prior approaches based on keyword matching or simple sentiment dictionaries.

The second paper took the analysis inside the limit order book, motivated by the recognition that minute-level event study methodology might capture the outcome of the information absorption process, not its detailed mechanism. Using high-frequency LOBSTER L2 data for six NASDAQ-listed stocks classified by relative tick-size, the paper first documented the main stylized facts of intraday LOB behaviour-including the U-shaped intraday pattern of trading activity, the widening of bid-ask spreads around news events, and the clustering of order flow imbalances in the minutes immediately following information releases. Crucially, the paper showed that the intensity and timing of these patterns vary systematically across news categories: scheduled macroeconomic announcements generate sharp, symmetric, and short-lived LOB responses consistent with rapid efficient price discovery, while unscheduled political and controversial corporate disclosures generate more prolonged, asymmetric, and heterogeneous responses consistent with gradual and contested belief updating. Following the methodology advanced by [Campigli et al., 2022], the paper's primary theoretical backbone was the *Score-Driven Aggregated Modified Hasbrouck* model (SDAMH), which derives from [Hasbrouck, 1991], and allows instantaneous price impact to evolve as a time-varying parameter driven by the score of the predictive likelihood. This specification captures the rapid belief updating that characterises high-frequency markets around information events without imposing parametric assumptions on the timing or magnitude of structural breaks-a flexibility that proved essential for modelling the heterogeneous dynamics observed across news categories.

The third paper completed the arc of the dissertation by asking not merely whether price impact varies around information events, but *why* it varies and through which causal channel. Focusing on Federal Open Market Committee announcements, as the utmost exemplification of scheduled macroeconomic releases in financial markets, the paper constructed an adverse selection measure, the information component of order flow IC_{lag} , and tested whether it causally predicts excess returns at the one-minute frequency. The adverse selec-

tion coefficient, constructed following [Glosten and Harris \[1988\]](#) and rooted in the informed trading framework of [Kyle \[1985\]](#) and [Glosten and Milgrom \[1985\]](#), was negative and highly significant across all specifications and error correction methods, consistent with temporary price overshooting following informed trading. Instrumental variable estimation confirmed the causal nature of this channel, while order flow imbalance lost significance once endogeneity was corrected, a finding that reflects simultaneous price discovery rather than a causal price impact channel. The adverse selection effect proved stable across all FOMC events in the sample and significantly stronger for large-tick stocks, consistent with the more abrupt price adjustment dynamics characteristic of queue-driven markets [[Parlour, 1998](#); [Foucault et al., 2005](#)].

Collectively, these papers present an interesting picture: the overall evidence consistently points in the same direction: information does not, solely, drive prices uniformly, instantaneously, or mechanically. Rather, it operates heterogeneously and progressively, mediated by order flow dynamics that the data suggest are only partially attributable to informational content. These findings are broadly consistent with the Inelastic Market Hypothesis [[Gabaix and Koijen, 2024](#)] and the order-driven perspective [[Bouchaud et al., 2018](#)], without fully endorsing either. More robustly, as mentioned in the opening of this dissertation, evidence supports viewing financial markets as *adaptive complex systems*, where the same piece of information triggers multiple responses depending on who is trading, when, and under what microstructural conditions.

No empirical research comes without limitations, and this dissertation is no exception. The first paper is limited by the very *radius of action* of its data. I mean that the sample is restricted to a short time frame, only Dow Jones Industrial Average constituents, widely known to be large, liquid, and heavily filled with tech firms. Though this was merely due to the availability of data and it was not intentional 'cherry-picking', I understand that it may not be representative of the broader universe of stocks (as, for instance, the S&P500 is) for which ESG news might potentially be more informative, as smaller, less liquid stocks may exhibit very different intraday responses to ESG disclosures. Moreover, though the LLM classification framework has been validated against a manually annotated ground truth, questions as to whether other LLMs would provide better performance in the classification task is genuinely reasonable. The second paper faces a somehow similar but more practical 'challenge' inherent to high-frequency data : the 'relatively' small universe of stocks. More than a challenge it looks like a constraint, as the availability of high-frequency limit order book data that I have celebrated throughout this dissertation comes, somewhat paradoxically, with a prohibitive financial cost that, in my honest opinion at least, severely restricts the cross-sectional scope of any of my studies relying on it. While the six stocks were carefully chosen to span the full taxonomy of relative tick-size, I truthfully recognize that a universe of six stocks may not be sufficient to produce results that *generalise* across the full cross-section of NASDAQ-listed equities, even though, to be clear, that had never the objective of this paper. What I do claim, and what I believe the evidence supports, is that the findings are robust across multiple specifications, subsamples, and estimation approaches and robustness which is, in my view, a genuine and meaningful quality in empirical research, irrespective of sample size. Finally, the last paper is the most controversial one : as much as the quantitative methodology reflects all the statistical steps that are usually involved, the limitation might lie in the construction of the variables. Once again, the same problem related to the market-model choice for the intraday event study from the first chapter emerges again. Subsequently a causal identification strategy grounds on the validity of the chosen instrumental variable (i.e., the IV), which requires both relevance (the instrument must be correlated with order flow imbalance) and exclusion (i.e., the instrument must affect returns

only through its effect on contemporaneous order flow imbalance). While the first condition is verified empirically, the exclusion restriction—as with most instrumental variables setups—cannot be tested directly and must be defended on theoretical grounds. The sample of 16 FOMC events, while sufficient for the panel regression approach I adopted, confines the external validity of the findings to the specific monetary policy communication environment of 2023–2024, a period characterized by elevated interest rate uncertainty that may not be representative of other monetary policy regimes.

Notwithstanding the limitations outlined above already constitute a genuine research agenda for future work by themselves, the findings of this dissertation open several promising research avenues. Perhaps the most intuitive research direction that I would pursue relates to whatever is close to the recent advancements in LLMs or the Natural Language Processing (NLP) research field or more in general—though it comes without saying it—the fast-paced integration of Artificial Intelligence into financial ecosystems. The last decade and a half have witnessed the unstoppable surge of these extremely powerful tools, and the amount of information which can nowadays be extracted from raw, unstructured data (i.e., from well-known sentiment, tone, or semantic embeddings of news content via RAG¹⁰, to facial muscular patterns in detecting emotions, etc.,) is unbelievable if not, to some extents, frightening. Indisputably, they have drastically and fundamentally expanded the informational toolkit available to empirical researchers. Under this view, the ESG classification framework developed in the first paper—which, as of today, one would legitimately define as a mere simplistic exercise rather than scientific artifact—represents only an initial step toward exploiting these tools at full potential. A natural extension of my work (if not already existing) could be incorporating a measure of disclosure-novelty (or ‘surprise’) relative to firm-specific news from raw textual data, a dimension that LLMs are uniquely exceptional at measuring. More ambitiously, semantic embeddings of news content via RAG could construct continuous informational distance measures between disclosures, replacing binary classification with a *geometry-aware*¹¹ representations of the information space. Paired with the LOB methodology of the second paper, this would allow a far more precise mapping of how news *content*, and not merely its category, shapes microstructural price formation dynamics. Moreover, the inelasticity of financial markets demands more systematic empirical investigation at the microstructural level. Building on the Inelastic Market Hypothesis [Gabaix and Koijen, 2024] and integrating the methodological tools of the second and third papers, future work could pave the way for a new, revisited version of how the structure of markets acts as mediator in the transmission of information impounded into prices, ultimately advancing the understanding of the speed and efficiency of price discovery in modern financial markets.

A personal note is perhaps appropriate here. The findings presented in this dissertation do not, strictly speaking, constitute formal evidence in favour of the *order-driven view* of markets over the *efficient market* one. The causal decomposition framework developed in the third paper assigns a significant and robust role to the adverse selection component, but it does not rule out order-flow as an important concurrent determinant of price dynamics. In fact, perhaps the most important take-away message from this thesis is that order-flow emerges as the ultimate channel through which information switches from private to public. Nevertheless, the patterns that consistently result from the data (i.e., the heterogeneity of

¹⁰RAG is an architecture for optimizing the performance of language models by connecting them with external knowledge bases [IBM Research, 2024]

¹¹By *geometry-aware* I mean that, instead of assigning news a binary label (E/S/G, positive/negative), one could represent it as a vector in a high-dimensional embedding space, where the geometric relationships between vectors carry meaning. This way, two ESG disclosures that are semantically similar sit close to each other in that space, and dissimilar ones sit far apart.

market responses, the mechanical amplification of order flow, the empirical puzzles together with the stylized facts, and the partial decomposition of price changes from identifiable informational events) are, in my view, more naturally accommodated within an order-driven, *adaptive complex systems* perspective than within the neoclassical paradigm of efficient markets. I offer this not as a conclusion but as a conviction at the end of my PhD experience.

In conclusion, this dissertation started with a simple but consequential question: does public information play a meaningful role in shaping short-term price dynamics in an environment increasingly governed by algorithmic trading, high-frequency data, and order flow dynamics? And, if so, to what extent? Well, after three years of PhD in Economics buried in thorough research, I guess the best answer that would accommodate a reader coming from an economic background would be *it depends*. Surprisingly, the best answer I can propose -as the three papers collectively suggest-, is: 'it depends'. Or, I should say, 'yes, information does have a role', but not in the way the classical efficient markets framework would predict. It is true, information matters, but it only matters through the mechanisms that are deeply intertwined within the strategic behaviour of market participants, the structure of the limit order book, and the inelasticity of modern adaptive complex systems of financial markets. Understanding this intricate interplay of how information arrives, how it is absorbed, and how it translates into prices and order flow, is not merely an academic exercise, rather has direct implications for how financial markets function, for how liquidity is provided, and for how price dynamics are assessed in an increasingly complex and data-rich financial ecosystem. This dissertation has taken several steps in that direction. Nevertheless, it is me the first to say that, still, many more remain to be taken.

Notation and Glossary

Mathematical Notations

Symbol	Description
<i>Time and Indexing</i>	
t	Trade time index, $t = 1, 2, \dots, T$
T	Total number of trades in the sample
$\mathcal{F}_t$	Information set available at time t (sigma-algebra)
h	Impulse response horizon, measured in trade time
H	Maximum horizon for impulse response analysis (set to 20 trades)
<i>Price Variables</i>	
p_t^{bid}	Best bid price immediately after trade t
p_t^{ask}	Best ask price immediately after trade t
q_t	Mid-quote price at trade t , $q_t = (p_t^{\text{bid}} + p_t^{\text{ask}})/2$
r_t	Mid-quote return at trade t , $r_t = (q_t - q_{t-1})/q_{t-1}$
Δq_t	Mid-quote price change, $\Delta q_t = q_t - q_{t-1}$
<i>Trade Variables</i>	
x_t	Trade sign: +1 for buyer-initiated, -1 for seller-initiated
$x_t^{(L_1)}$	Aggregated trade sign over $L_1 = 10$ lags (recent order flow)
$x_t^{(L_2)}$	Aggregated trade sign over lags $L_1 + 1$ to $L_2 = 100$ (medium-term order flow)
v_t	Signed trade volume (positive for buys, negative for sells)
π_t	Probability of a buy order at time t , $\pi_t = P(x_t = +1 \mid \mathcal{F}_{t-1})$
<i>Model Parameters</i>	
<i>Return Equation Parameters</i>	
μ_1	Intercept (drift) in return equation
a_1	Autoregressive coefficient for lagged returns
b_0	Instantaneous market impact parameter (constant in AH/MH models)
$b_{0,t}$	Time-varying instantaneous market impact (SDAMH model)
b_1	Impact of lagged trade sign (x_{t-1}) on returns
b_{10}	Impact of recent aggregated trade signs ($x_t^{(10)}$) on returns
b_{100}	Impact of medium-term aggregated trade signs ($x_t^{(100)}$) on returns
σ	Return volatility (constant in AH/MH models)

Symbol	Description
σ_t	Time-varying return volatility (SDAMH model)
σ^2	Return variance (constant)
σ_t^2	Time-varying return variance (SDAMH model)
<i>Trade Sign Equation Parameters</i>	
μ_2	Intercept in trade sign equation
c_1	Effect of lagged returns on trade probability
d_1	Autocorrelation coefficient for lagged trade sign
d_{10}	Effect of recent aggregated trade signs on trade probability
d_{100}	Effect of medium-term aggregated trade signs on trade probability
<i>Score-Driven Parameters (SDAMH only)</i>	
α	Speed of adaptation for instantaneous impact $b_{0,t}$
γ	Speed of adaptation for volatility σ_t^2
$b_{0,0}$	Initial value of instantaneous impact
σ_0	Initial value of volatility
<i>Conditional Means and States</i>	
$\mu_1^{(t)}$	Conditional mean of return r_t given $\mathcal{F}_{t-1}$ and x_t
$\mu_2^{(t)}$	Linear predictor in logistic model for trade probability
μ_t^*	Market state: conditional mean of r_t given $\mathcal{F}_{t-1}$ excluding $b_{0,t}x_t$
	$\mu_t^* = \mu_1 + a_1 r_{t-1} + b_1 x_{t-1} + b_{10} x_t^{(10)} + b_{100} x_t^{(100)}$
<i>Impulse Responses and Information Content</i>	
$\text{IRF}(h; \delta x_t)$	Impulse response function at horizon h for shock δx_t
$\text{CIRF}(H; \delta x_t)$	Cumulative impulse response function up to horizon H
LRCIRF_t	Long-run cumulative impulse response (permanent impact) at trade t
	$\text{LRCIRF}_t := \text{CIRF}(H; +1 \mid \mathcal{F}_{t-1})$
β_t	Permanent impact coefficient (transaction cost analysis)
$\beta_t^{(s)}$	Permanent impact based on trade signs (not volumes)
<i>Statistical Quantities</i>	
θ	Full parameter vector, $\theta = (\mu_1, a_1, b_0, \dots)$
$\hat{\theta}$	Maximum likelihood estimate of θ
$\mathcal{L}(\theta)$	Log-likelihood function
ε_t	Innovation (shock) in return equation, $\varepsilon_t = r_t - \mu_1^{(t)}$
S	Number of Monte Carlo replications for IRF computation (set to 1,000)
L_1	Short-term aggregation window (10 trades)
L_2	Medium-term aggregation window (100 trades)
M	Number of trades in aggregation window for permanent impact estimation
<i>Functions</i>	
$\text{Logit}^{-1}(z)$	Inverse logistic function, $(1 + e^{-z})^{-1}$

3.6. Conclusions

Symbol	Description
$\mathbb{E}[\cdot]$	Expectation operator
$\mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$	Conditional expectation given information at time $t - 1$
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2

Glossary of Terms

Aggregated Hasbrouck (AH) Model

Linear vector autoregression model for returns and trade signs with aggregated order flow variables. Used as benchmark and for initialization of nonlinear models. Estimated by OLS.

Ask Price The lowest price at which a market participant is willing to sell. The best (lowest) ask price defines one side of the bid-ask spread.

Bid Price The highest price at which a market participant is willing to buy. The best (highest) bid price defines one side of the bid-ask spread.

Bid-Ask Spread

The difference between the best ask and best bid prices. A measure of liquidity and transaction costs. Denoted $s_t = p_t^{\text{ask}} - p_t^{\text{bid}}$.

Buyer-Initiated Trade

A market order that executes against the best available ask price (a buy market order). Coded as $x_t = +1$.

Cumulative Impulse Response Function (CIRF)

The sum of impulse responses from horizon 1 to H , measuring the total price change following a trade shock. In the SDAMH model, this is state-dependent and computed via Monte Carlo simulation.

Dynamic Conditional Score (DCS)

A class of observation-driven time-varying parameter models where parameters evolve according to the score (gradient) of the log-likelihood function. Proposed by [Creal et al. \[2013\]](#).

FOMC Announcement

Federal Open Market Committee announcement regarding monetary policy decisions, typically including interest rate decisions. Known to cause significant market volatility and changes in liquidity.

Hasbrouck Model

Structural vector autoregression (SVAR) model introduced by [Hasbrouck \[1991\]](#) to jointly model price changes and trade signs, used to measure the information content of trades.

Heterogeneous Autoregressive (HAR)

Modeling approach introduced by [Corsi \[2009\]](#) that aggregates variables over multiple time horizons to capture long-memory properties parsimoniously.

High-Frequency Data

Financial market data recorded at tick level (every event) or at very fine time intervals (seconds, milliseconds). Exhibits unique statistical properties including irregular spacing, microstructure noise, and fat tails.

Information Content of Trade

The permanent (long-run) effect of a trade on prices, interpreted as the amount of private information revealed by the trade. Measured by the long-run cumulative impulse response function.

Instantaneous Impact

The immediate effect of a trade on the mid-quote price, measured by parameter b_0 (or $b_{0,t}$ in the time-varying model). Reflects market liquidity at the time of the trade.

Integrated Specification

Score-driven dynamics with unit root (autoregressive coefficient equal to 1), allowing the parameter to track permanent changes. Contrasts with stationary specification.

L-BFGS-B

Limited-memory Broyden-Fletcher-Goldfarb-Shanno algorithm with box constraints. Quasi-Newton optimization method used for maximum likelihood estimation.

Large-Tick Stock

A stock where the bid-ask spread is typically small in units of the tick size (e.g., 1–2 ticks). Examples: AAPL, MSFT. Characterized by high price relative to tick size.

Limit Order Book (LOB)

Electronic record of all outstanding limit orders to buy and sell a security at various price levels. LOBSTER provides reconstructed LOB data.

LOBSTER

Limit Order Book System—The Efficient Reconstructor. Database providing high-frequency order book and message data for NASDAQ-listed stocks with nanosecond timestamps.

Logistic Specification

Modeling the probability of a binary outcome (buy vs. sell) using the inverse logistic function $\pi_t = (1 + e^{-\mu_2^{(t)}})^{-1}$. Ensures probabilities lie in $(0, 1)$.

Long Memory

Statistical property where autocorrelation decays slowly (hyperbolically) rather than exponentially. Characteristic of order flow in financial markets.

Market Impact

The effect of a trade on the price of a security. Distinguished into instantaneous (immediate) impact and permanent (long-run) impact.

Market Order

An order to buy or sell immediately at the best available price. Executes against the limit order book, removing liquidity.

Message Book

LOBSTER file containing all order book events (submissions, cancellations, executions, modifications) with precise timestamps.

Mid-Quote Price

The average of the best bid and ask prices: $q_t = (p_t^{\text{bid}} + p_t^{\text{ask}})/2$. Used as a proxy for the efficient price.

Modified Hasbrouck (MH) Model

Extension of the Hasbrouck model incorporating (i) logistic specification for trade probability and (ii) aggregated order flow variables. Constant parameters, estimated by MLE.

Monte Carlo Simulation

Computational method using repeated random sampling to obtain numerical results. Used to compute impulse response functions in nonlinear models.

Numba JIT Compilation

Just-in-time compilation for Python using the Numba library. Translates Python code to optimized machine code, providing substantial speedups for numerical computations.

Order Flow

The sequence of buy and sell orders arriving at the market. Exhibits strong autocorrelation (persistence) and long memory.

Permanent Impact

The long-run effect of a trade on prices after all price discovery is complete. Related to the information content of the trade. Used in transaction cost analysis.

Quantile Residuals

Residuals for discrete or non-Gaussian models that transform model residuals to be approximately normally distributed. Used for diagnostic testing.

Score (Gradient)

The derivative of the log-likelihood function with respect to parameters. In score-driven models, scaled scores drive parameter dynamics.

Score-Driven Aggregated Modified Hasbrouck (SDAMH) Model

Extension of the MH model where instantaneous impact $b_{0,t}$ and volatility σ_t follow score-driven dynamics. Allows real-time adaptation to changing market conditions.

Seller-Initiated Trade

A market order that executes against the best available bid price (a sell market order). Coded as $x_t = -1$.

Small-Tick Stock

A stock where the bid-ask spread is large in units of the tick size (e.g., 50–100 ticks). Examples: AMZN, GOOG. Characterized by low price relative to tick size.

Structural Vector Autoregression (SVAR)

VAR model with contemporaneous relationships between variables, identified through economic restrictions. Hasbrouck's model is an SVAR.

Temporary Impact

The transient component of market impact that reverts after execution. Related to bid-ask bounce and inventory risk of liquidity providers.

Tick Size The minimum price increment for a security. On NASDAQ, typically \$0.01 for most stocks.

Time-Varying Parameter Model

Econometric model where parameters evolve over time rather than remaining constant. Can be observation-driven (like score-driven) or state-space.

Trade Time

Indexing by trade number rather than calendar/clock time. Accounts for irregular spacing of high-frequency data.

Transaction Cost Analysis (TCA)

Measurement and analysis of costs incurred when trading, including market impact, spread costs, and timing costs. Used for execution quality assessment.

Acronyms and Abbreviations

Acronym	Full Form
AAPL	Apple Inc. (NASDAQ ticker symbol)
AH	Aggregated Hasbrouck (model)
AMH	Aggregated Modified Hasbrouck (model)
AMZN	Amazon.com Inc. (NASDAQ ticker symbol)
BIC	Bayesian Information Criterion
CIRF	Cumulative Impulse Response Function
DCS	Dynamic Conditional Score
DGP	Data Generating Process
ET	Eastern Time (US)
FOMC	Federal Open Market Committee
GOOG	Alphabet Inc. Class C (NASDAQ ticker symbol)
HAR	Heterogeneous Autoregressive
IRF	Impulse Response Function
JIT	Just-In-Time (compilation)
L-BFGS-B	Limited-memory Broyden-Fletcher-Goldfarb-Shanno with Bounds
LOB	Limit Order Book
LOBSTER	Limit Order Book System—The Efficient Reconstructor
LRCIRF	Long-Run Cumulative Impulse Response Function
MAE	Mean Absolute Error
MH	Modified Hasbrouck (model)
MLE	Maximum Likelihood Estimation / Estimator
MSFT	Microsoft Corporation (NASDAQ ticker symbol)
NASDAQ	National Association of Securities Dealers Automated Quotations
OLS	Ordinary Least Squares
SDAMH	Score-Driven Aggregated Modified Hasbrouck (model)
SVAR	Structural Vector Autoregression
TCA	Transaction Cost Analysis
TWAP	Time-Weighted Average Price
USD	United States Dollar
VAR	Vector Autoregression

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Appendix A

Chapter 1

A.1 Summary statistics by stock

In this section, we present the detailed statistics for every company in our sample.

Security	Ticker	Obs	Mean	Std. Dev.	Min	Max
3M Company	MMM.N	24,027	-7.8204	0.4315	-8.2735	-1.8793
American Express Company	AXP.N	23,900	-7.9297	0.3653	-8.2737	-1.0915
Amgen Inc.	AMGN.OQ	24,065	-7.6779	0.5100	-8.2735	-0.6315
Apple Inc.	AAPL.OQ	24,180	-7.8060	0.3678	-8.2105	-2.1372
Caterpillar Inc.	CAT.N	24,030	-7.8017	0.4797	-8.2735	-0.9407
Chevron Corporation	CVX.N	24,118	-7.8576	0.3323	-8.2734	-3.4796
Cisco Systems Inc.	CSCO.OQ	24,173	-7.7739	0.4425	-8.2741	-0.9178
Dow Inc.	DOW.N	24,112	-7.7025	0.4961	-8.2737	-0.2988
Goldman Sachs Group Inc.	GS.N	23,823	-7.7394	0.5352	-8.2730	-0.7647
Honeywell International Inc.	HON.OQ	24,144	-7.6623	0.5000	-8.2736	-1.2453
Intel Corporation	INTC.O	24,180	-6.2092	0.7030	-8.0916	-0.5280
International Business Machines Corp.	IBM.N	24,163	-7.7889	0.4601	-8.2738	0.4226
JPMorgan Chase & Co.	JPM.N	24,178	-7.9141	0.4028	-8.2739	-1.0227
Johnson & Johnson	JNJ.N	24,179	-7.9246	0.3751	-8.2740	-0.9750
McDonald's Corporation	MCD.N	24,055	-7.8988	0.4160	-8.2737	-1.7759
Merck & Co. Inc.	MRK.N	24,074	-7.9906	0.2909	-8.2740	-3.2461
Microsoft Corporation	MSFT.OQ	24,180	-7.7387	0.4036	-8.2178	-1.8591
NIKE Inc.	NKE.N	24,171	-7.8674	0.4056	-8.2739	-1.3541
Procter & Gamble Company	PG.N	24,078	-7.9583	0.2894	-8.2740	-2.3797
Salesforce.com Inc.	CRM.N	24,179	-7.5972	0.5200	-8.2738	-1.2222
The Boeing Company	BA.N	24,125	-7.7019	0.5239	-8.2735	-0.4923
The Coca-Cola Company	KO.N	24,090	-7.9666	0.3629	-8.2741	-1.7210
The Home Depot Inc.	HD.N	24,093	-7.9525	0.3823	-8.2738	-1.6484
The Travelers Companies Inc.	TRV.N	22,518	-7.7484	0.5517	-8.2726	-0.5130
The Walt Disney Company	DIS.N	24,114	-7.8458	0.3677	-8.2740	-2.5542
UnitedHealth Group Incorporated	UNH.N	24,003	-7.9643	0.3832	-8.2738	-1.8502
Verizon Communications Inc.	VZ.N	23,998	-7.9077	0.3960	-8.2741	-0.7728
Visa Inc.	V.N	24,171	-7.9553	0.3685	-8.2740	-0.9948
Walgreens Boots Alliance Inc.	WBA.OQ	24,055	-7.4847	0.5698	-8.2737	-0.1796
Walmart Inc.	WMT.N	24,168	-8.1678	0.2225	-8.2741	-3.0809

Table A.1: Summary statistics of Trading Volume Turnover by DJIA security.

Security	Ticker	Obs	Mean	Std. Dev.	Min	Max
3M Company	MMM.N	18,729	-0.001194	0.0975	-5.6362	2.7015
American Express Company	AXP.N	17,596	-0.000047	0.1238	-3.7123	7.4511
Amgen Inc.	AMGN.OQ	17,919	0.000561	0.0680	-1.6201	1.4940
Apple Inc.	AAPL.OQ	24,118	0.000920	0.0747	-0.5240	0.5509
Caterpillar Inc.	CAT.N	17,909	-0.001232	0.0973	-2.9134	1.9348
Chevron Corporation	CVX.N	23,833	-0.000323	0.0905	-3.8439	2.8789
Cisco Systems Inc.	CSCO.OQ	24,001	0.000533	0.0624	-1.1774	1.7130
Dow Inc.	DOW.N	23,134	0.000281	0.0891	-3.0090	2.1614
Goldman Sachs Group Inc.	GS.N	15,699	-0.000566	0.1121	-4.2323	2.6676
Honeywell International Inc.	HON.OQ	20,658	-0.000884	0.0775	-4.1836	1.2965
Intel Corporation	INTC.O	24,179	0.000816	0.1346	-11.2292	2.7500
International Business Machines Corp.	IBM.N	22,765	-0.000392	0.0604	-2.1978	1.0243
JPMorgan Chase & Co.	JPM.N	23,945	-0.000135	0.0889	-3.2079	2.6544
Johnson & Johnson	JNJ.N	23,886	-0.000529	0.0522	-0.9235	0.8603
McDonald's Corporation	MCD.N	18,568	0.000083	0.0971	-8.1392	7.2890
Merck & Co. Inc.	MRK.N	23,705	-0.000153	0.0650	-2.2209	1.7677
Microsoft Corporation	MSFT.OQ	24,118	0.000480	0.0766	-0.5424	0.5533
NIKE Inc.	NKE.N	23,588	0.000202	0.1134	-8.3778	7.9481
Procter & Gamble Company	PG.N	23,797	-0.000056	0.0538	-1.3343	1.1955
Salesforce.com Inc.	CRM.N	23,833	0.001571	0.1445	-4.0423	13.5374
The Boeing Company	BA.N	20,668	0.000105	0.1104	-3.9495	1.8271
The Coca-Cola Company	KO.N	23,830	-0.000104	0.0495	-1.0490	1.2065
The Home Depot Inc.	HD.N	19,142	-0.000728	0.0894	-3.3419	1.5905
The Travelers Companies Inc.	TRV.N	11,516	-0.000744	0.0960	-3.1042	1.6776
The Walt Disney Company	DIS.N	23,773	0.000511	0.0995	-2.7301	5.0429
UnitedHealth Group Incorporated	UNH.N	17,410	-0.000475	0.0806	-1.8274	2.3356
Verizon Communications Inc.	VZ.N	23,706	-0.000099	0.1018	-9.6793	6.9682
Visa Inc.	V.N	23,368	0.000251	0.0944	-7.3413	6.0031
Walgreens Boots Alliance Inc.	WBA.OQ	23,180	0.000192	0.0799	-1.4523	1.9838
Walmart Inc.	WMT.N	23,758	0.000083	0.0848	-7.1846	6.2832

Table A.2: Summary statistics of log-returns for DJIA constituent securities. Tickers followed by .N denote stocks listed on the New York Stock Exchange (NYSE) and those followed by .OQ denote stocks listed on the NASDAQ Stock Market.

A.2 ESG news example

In this section, we present some examples of the ESG-related news classified by ESG class and Sentiment using GPT-4 Turbo, which also computes the associated ESG Score and Sentiment Score.

A.2. ESG news example

Security	Date	GPT-4o		ESG News	
		Class	Sentiment	Class	Sentiment
		Score	Score		Score
Goldman S.	24/01/2023 03:00	G	0.90	Negative	0.80
GOLDMAN SACHS ALERT: Bragar Eigel & Squire, P.C. is Investigating The Goldman Sachs Group, Inc. on Behalf of Goldman Sachs Stockholders and Encourages Investors to Contact the Firm					
Goldman S.	18/01/2023 10:48	E	0.90	Neutral	0.50
Fed sets out climate risk analysis for US banks					
IBM	02/03/2023 19:15	S	0.70	Positive	0.80
IBM's Lydia Logan at SXSW EDU: How to Build Meaningful Relationships with HBCUs					
Microsoft	18/01/2023 17:45	S	0.70	Negative	0.90
Microsoft axing 10,000 roles in latest US tech jobs cull					
JPMorgan	23/02/2023 20:44	G	0.70	Neutral	0.50
JP Morgan restricts staff from using AI bot ChatGPT					
Amgen	31/03/2023 15:21	G	0.90	Negative	0.80
Securities Fraud Class Action Lawsuit Pending: Contact Levi & Korsinsky Before May 12, 2023 - AMGN					
Walt Disney	01/01/2023 16:29	O	0.00	Neutral	0.50
OUTGOING REP. KEVIN BRADY (R-TEXAS) IS INTERVIEWED ON FOX NEWS SUN-DAY					
Goldman S.	18/02/2023 00:05	G	0.90	Neutral	0.50
ROSEN, TOP RANKED INVESTOR COUNSEL, Encourages The Goldman Sachs Group, Inc. Investors to Inquire About Securities Class Action Investigation - GS					
Amgen	22/03/2023 06:45	G	0.90	Negative	0.80
AMGEN INC. (NASDAQ: AMGN) SHAREHOLDER CLASS ACTION ALERT: Bernstein Liebhart LLP Reminds Investors of the Deadline to File a Lead Plaintiff Motion					
Chevron	27/03/2023 14:55	E	0.85	Neutral	0.50
Star Petroleum Refining PCL - Sustainability Report 2022					

Table A.3: Examples of ESG-related news classified by ESG pillar and sentiment by GPT-4o Turbo, along with the associated classification and sentiment scores.

Appendix B

Chapter 2

B.1 Distribution of order arrivals

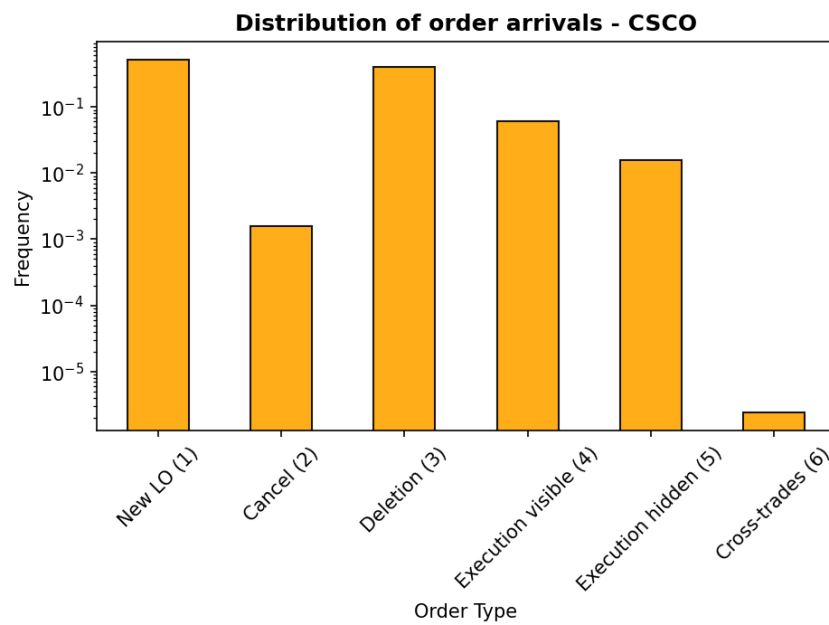


Figure B.1: Distribution of order arrivals for AMZN on March 7,2023 on a log scale.

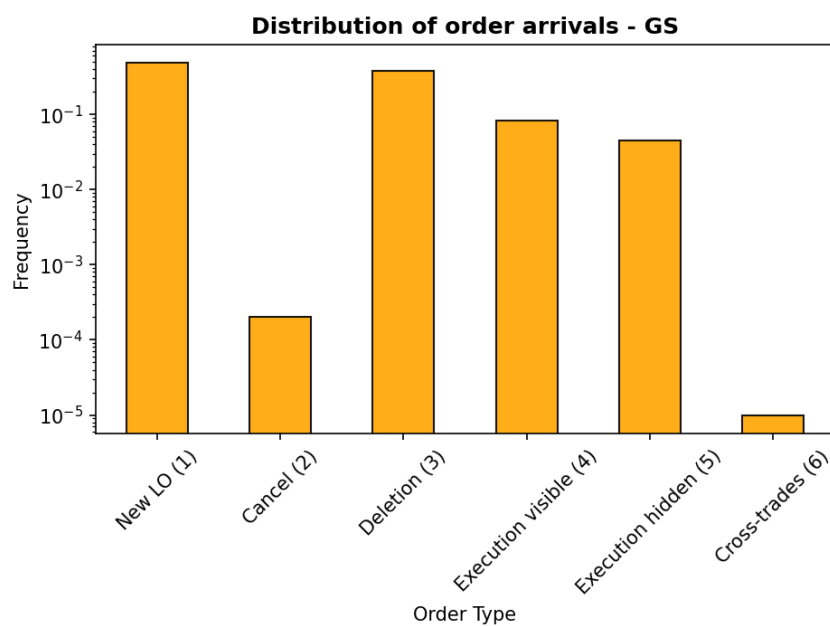


Figure B.2: Distribution of order arrivals for AMZN on March 7,2023 on a log scale.

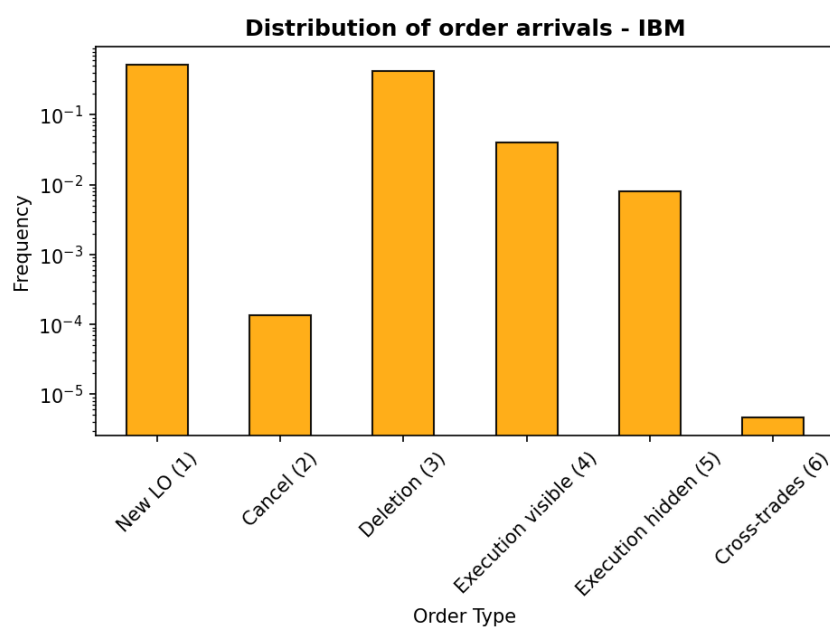


Figure B.3: Distribution of order arrivals for AMZN on March 7,2023 on a log scale.

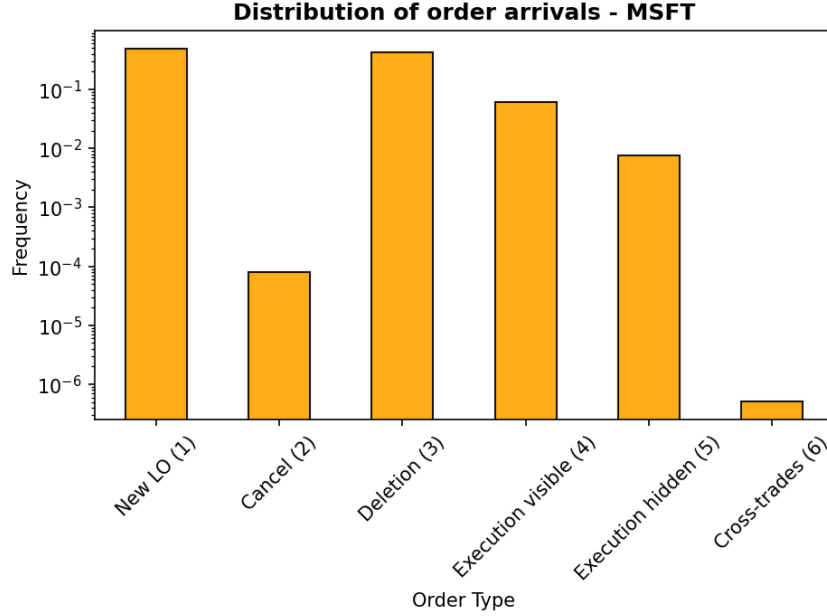


Figure B.4: Distribution of order arrivals for AMZN on March 7, 2023 on a log scale.

B.2 Stylized Facts

Financial markets are well known to exhibit a set of universal empirical regularities, commonly referred to as stylized facts [Cont, 2001].

Fat Tails Figures B.6 to B.10 plot the empirical frequency distribution of intraday mid-price log returns against a Gaussian benchmark for each stock. The empirical distribution consistently lies above the Gaussian in the tails, confirming that large price movements occur far more frequently than a normal distribution would predict. This excess is particularly pronounced for GS, whose wider tick-size and lower trading frequency allow larger individual price increments. These findings are corroborated by Figure B.5, which overlays the complementary cumulative distribution functions (CCDFs) across all stocks and shows that all series share heavier-than-Gaussian tails, a well-documented feature of high-frequency returns [Cont, 2001].

Volatility Clustering and Return Uncorrelation Figures B.11 to B.15 display the ACF of raw returns $r(t)$ (left panels) and absolute returns $|r(t)|$ (right panels). The ACF of $r(t)$ drops to near zero after lag 1 for all stocks, confirming that price changes are essentially uncorrelated at the intraday level (consistently with weak-form market efficiency). In contrast, the ACF of $|r(t)|$ decays slowly and remains significantly positive across all lags shown, providing clear evidence of volatility clustering: large price movements tend to be followed by further large movements, regardless of sign. This effect is most persistent for GS and IBM, where the ACF of $|r(t)|$ remains above the confidence band beyond lag 50.

Trade Sign Autocorrelation Figures B.16 to B.20 report the ACF and PACF of trade signs $x_t \in \{+1, -1\}$. The ACF decays slowly and remains significantly positive across all

lags, indicating strong persistence in the direction of order flow. Interestingly, the PACF cuts-off sharply after lag 1 for all stocks. This might suggesting that the long-range dependence in trade signs is not driven by direct multi-lag interactions but rather by a latent process, which would be consistent with the *order-splitting* mechanism of metaorders, whereby institutional traders fragment large metaorders into sequences of smaller orders (i.e., *child* orders) in the same direction [Lillo et al., 2021].

Figures B.21 to B.25 examine this persistence more formally. The log-log plots (left panels) reveal a power-law decay $C(\ell) \sim \ell^{-\gamma}$ over several lags, with the empirical auto-correlation (blue) broadly following the power-law fit (red). The linear-scale ACF (right panels) confirms that trade sign correlation remains positive and well above zero even at lag 100 for actively traded stocks such as AMZN and MSFT, while it decays more rapidly for lower-volume names such as IBM and GS.

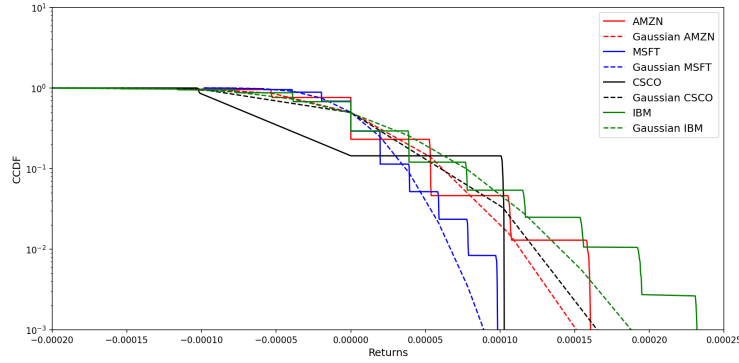


Figure B.5: Complementary cumulative distribution functions (CCDFs) of mid-price log returns for AMZN, MSFT, CSCO, and IBM, compared against Gaussian benchmarks (dashed)

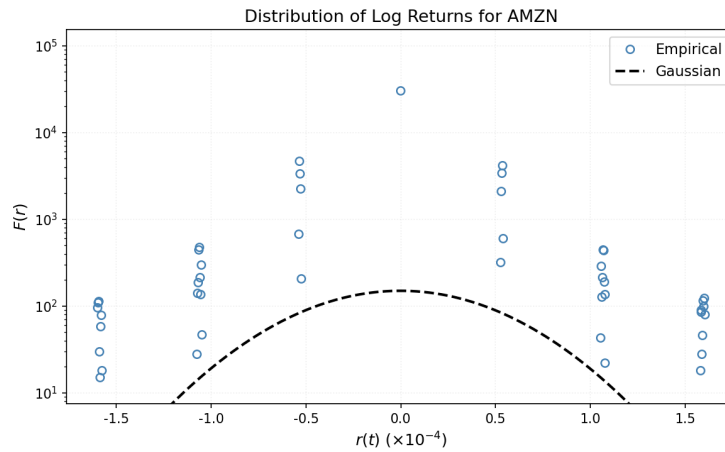


Figure B.6: Distribution of log returns for AMZN.

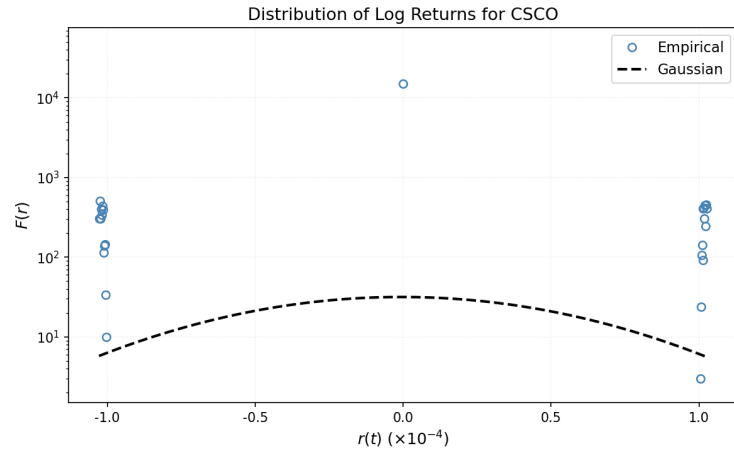


Figure B.7: Distribution of log returns for CSCO.

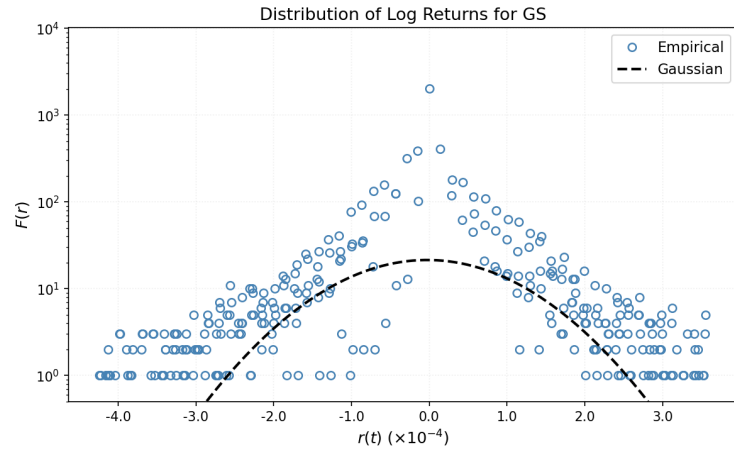


Figure B.8: Distribution of log returns for GS.

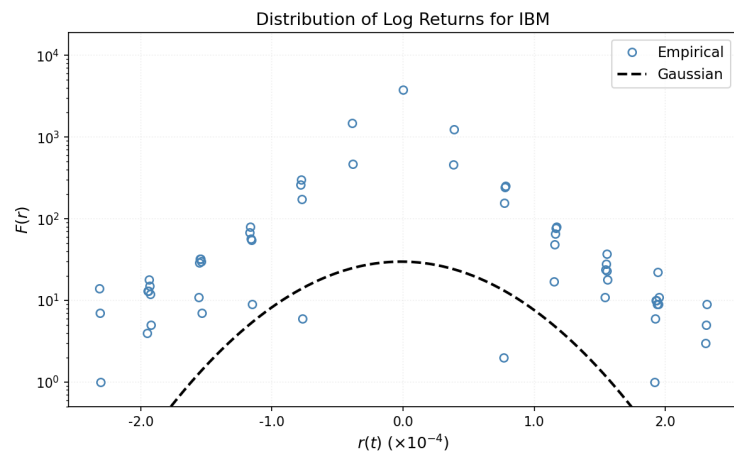


Figure B.9: Distribution of log returns for IBM.

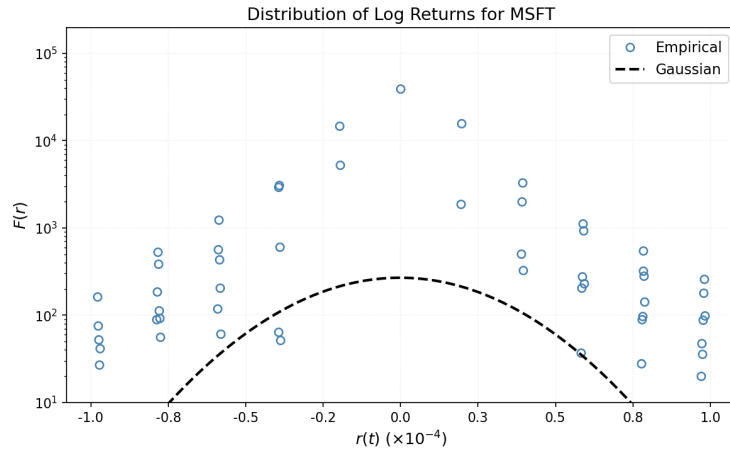


Figure B.10: Distribution of log returns for MSFT.

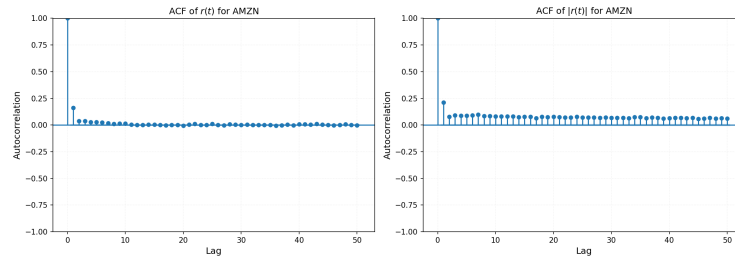


Figure B.11: ACF of raw returns $r(t)$ (left) and absolute returns $|r(t)|$ (right) for AMZN.

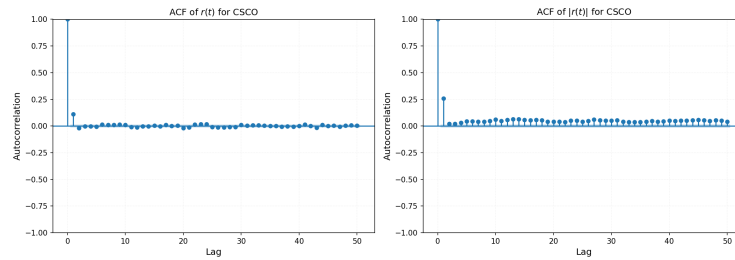


Figure B.12: ACF of raw returns $r(t)$ (left) and absolute returns $|r(t)|$ (right) for CSCO.

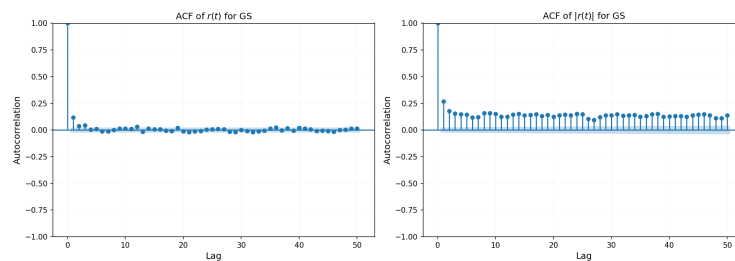


Figure B.13: ACF of raw returns $r(t)$ (left) and absolute returns $|r(t)|$ (right) for GS.

B.2. Stylized Facts

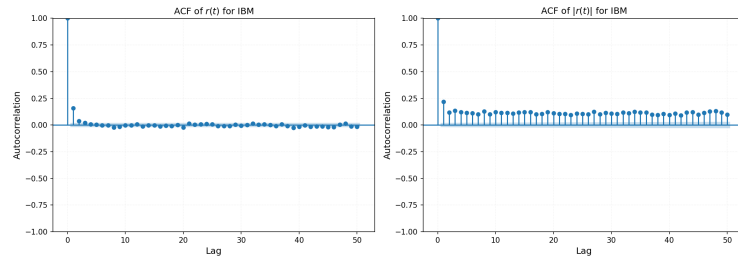


Figure B.14: ACF of raw returns $r(t)$ (left) and absolute returns $|r(t)|$ (right) for IBM.

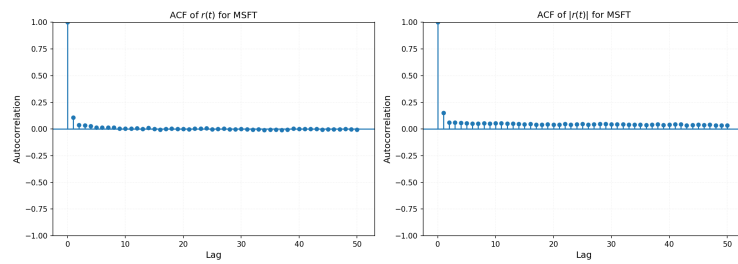


Figure B.15: ACF of raw returns $r(t)$ (left) and absolute returns $|r(t)|$ (right) for MSFT.

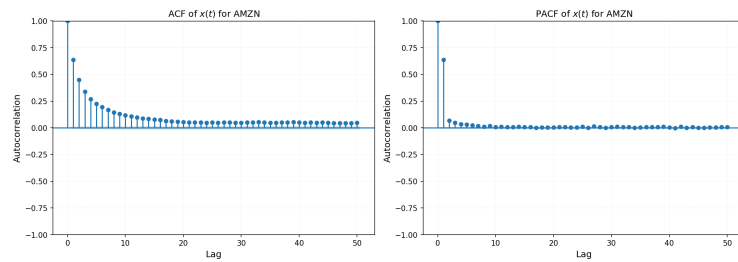


Figure B.16: ACF and PACF of trade signs (x_t , $+1,-1$, respectively left and right panel) for AMZN.

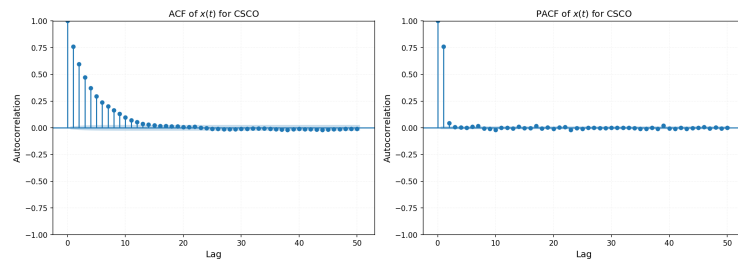


Figure B.17: ACF and PACF of trade signs (x_t , $+1,-1$, respectively left and right panel) for CSCO.

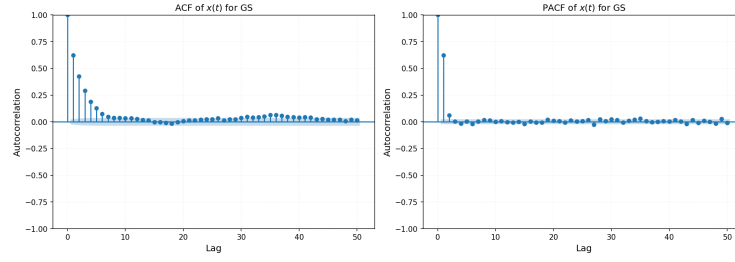


Figure B.18: ACF and PACF of trade signs (x_t , $+1,-1$, respectively left and right panel) for GS.

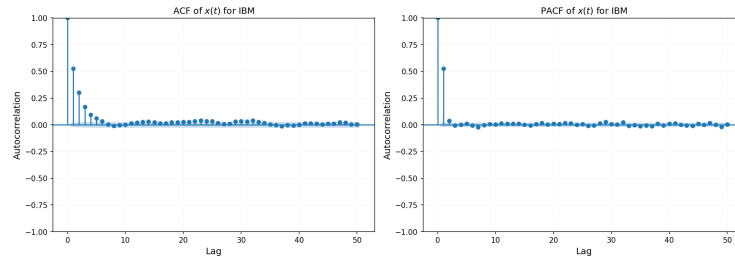


Figure B.19: ACF and PACF of trade signs (x_t , $+1,-1$, respectively left and right panel) for IBM.

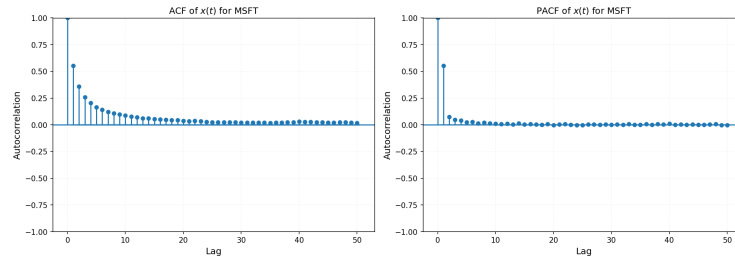


Figure B.20: ACF and PACF of trade signs (x_t , $+1,-1$, respectively left and right panel) for MSFT.

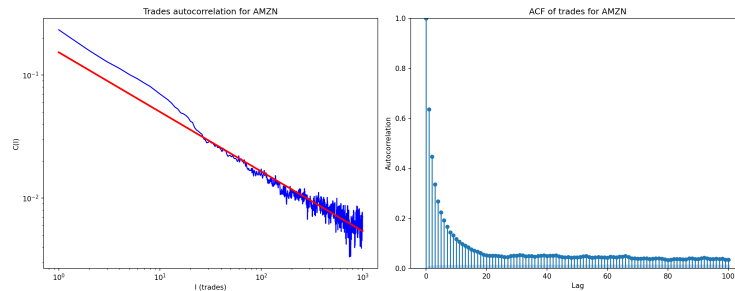


Figure B.21: Long-memory autocorrelation of trade signs for AMZN. Left: log-log plot of the trade sign autocorrelation $C(\ell)$ as a function of lag ℓ (blue), with a power-law fit (red) indicating slow hyperbolic decay $C(\ell) \sim \ell^{-\gamma}$. Right: linear-scale ACF of trade signs up to lag 100, confirming strong positive serial correlation consistent with order-splitting by institutional traders.

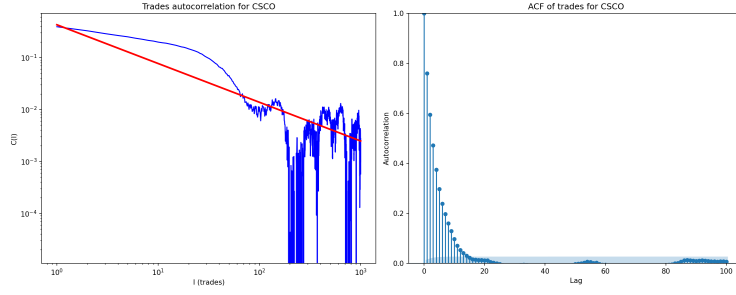


Figure B.22: Long-memory autocorrelation of trade signs for CSCO. Left: log-log plot of the trade sign autocorrelation $C(\ell)$ as a function of lag ℓ (blue), with a power-law fit (red) indicating slow hyperbolic decay $C(\ell) \sim \ell^{-\gamma}$. Right: linear-scale ACF of trade signs up to lag 100, confirming strong positive serial correlation consistent with order-splitting by institutional traders.

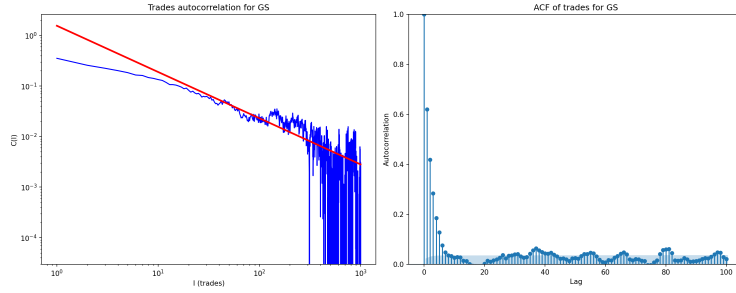


Figure B.23: Long-memory autocorrelation of trade signs for GS. Left: log-log plot of the trade sign autocorrelation $C(\ell)$ as a function of lag ℓ (blue), with a power-law fit (red) indicating slow hyperbolic decay $C(\ell) \sim \ell^{-\gamma}$. Right: linear-scale ACF of trade signs up to lag 100, confirming strong positive serial correlation consistent with order-splitting by institutional traders.

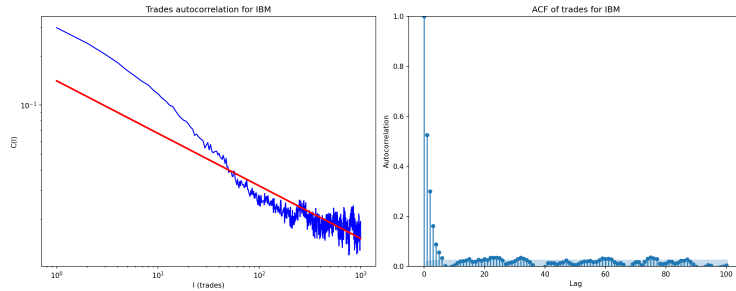


Figure B.24: Long-memory autocorrelation of trade signs for IBM. Left: log-log plot of the trade sign autocorrelation $C(\ell)$ as a function of lag ℓ (blue), with a power-law fit (red) indicating slow hyperbolic decay $C(\ell) \sim \ell^{-\gamma}$. Right: linear-scale ACF of trade signs up to lag 100, confirming strong positive serial correlation consistent with order-splitting by institutional traders.

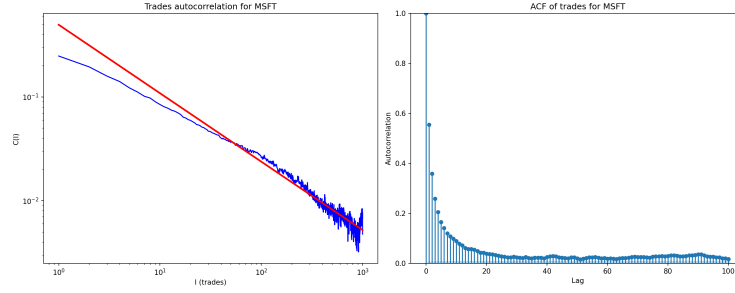


Figure B.25: Long-memory autocorrelation of trade signs for MSFT. Left: log-log plot of the trade sign autocorrelation $C(\ell)$ as a function of lag ℓ (blue), with a power-law fit (red) indicating slow hyperbolic decay $C(\ell) \sim \ell^{-\gamma}$. Right: linear-scale ACF of trade signs up to lag 100, confirming strong positive serial correlation consistent with order-splitting by institutional traders.

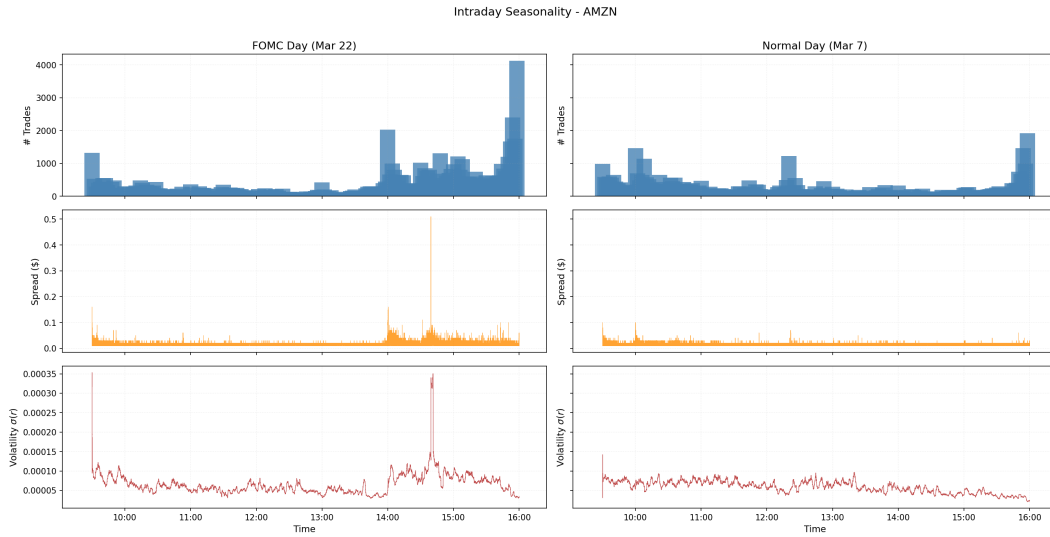


Figure B.26: Intraday seasonality of AMZN on the FOMC day (March 22, left) and a normal trading day (March 7, right). Each panel shows the number of trades per interval (top, blue), bid-ask spread in dollars (middle, orange), and realized volatility (bottom, red).

B.2. Stylized Facts

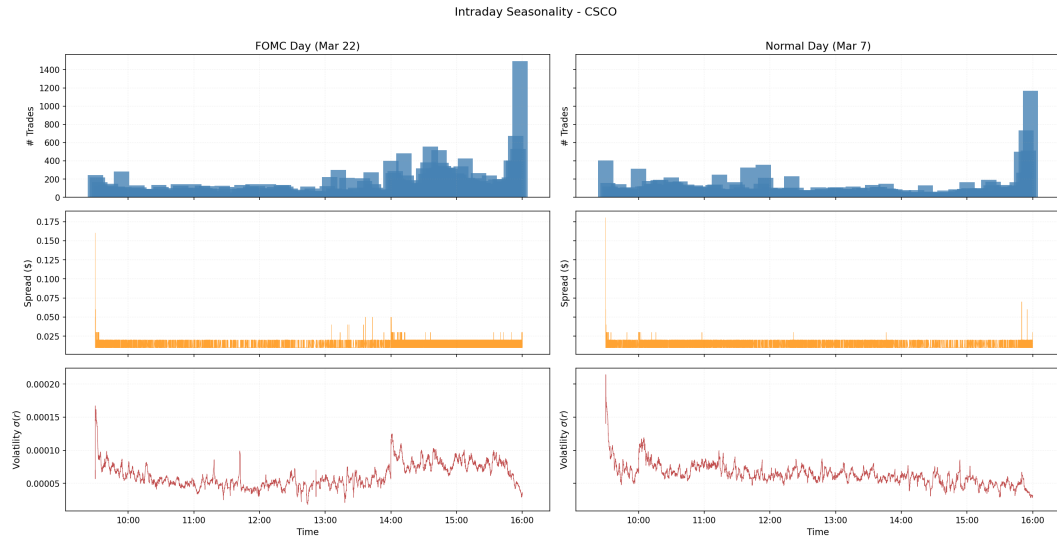


Figure B.27: Intraday seasonality of CSCO on the FOMC day (March 22, left) and a normal trading day (March 7, right). Each panel shows the number of trades per interval (top, blue), bid-ask spread in dollars (middle, orange), and realized volatility (bottom, red).

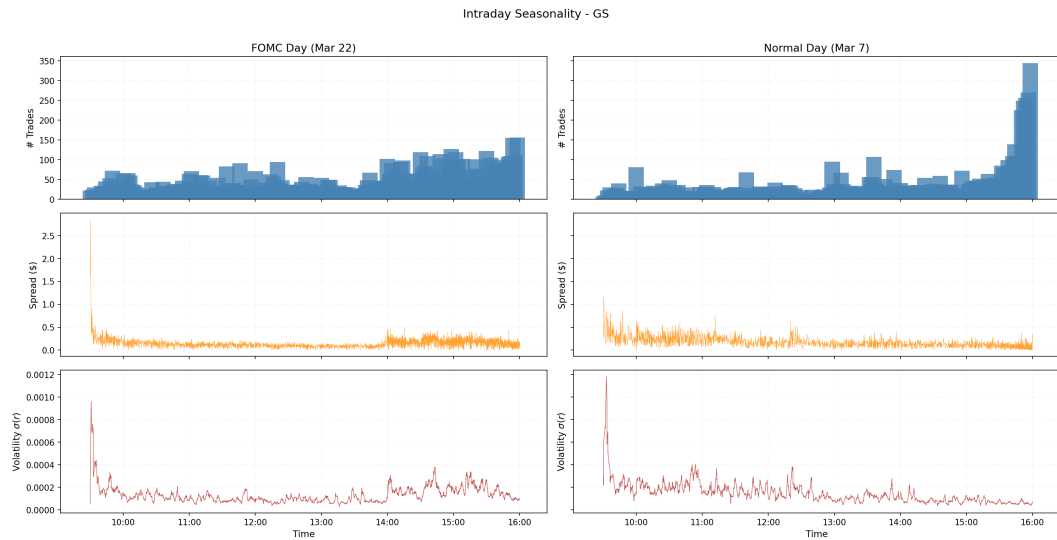


Figure B.28: Intraday seasonality of GS on the FOMC day (March 22, left) and a normal trading day (March 7, right). Each panel shows the number of trades per interval (top, blue), bid-ask spread in dollars (middle, orange), and realized volatility (bottom, red).

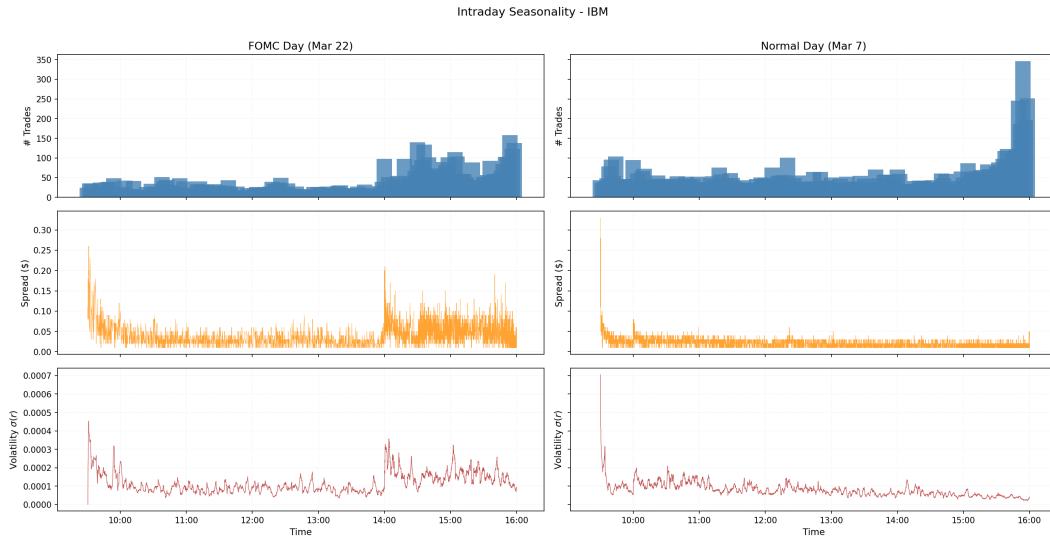


Figure B.29: Intraday seasonality of IBM on the FOMC day (March 22, left) and a normal trading day (March 7, right). Each panel shows the number of trades per interval (top, blue), bid-ask spread in dollars (middle, orange), and realized volatility (bottom, red).

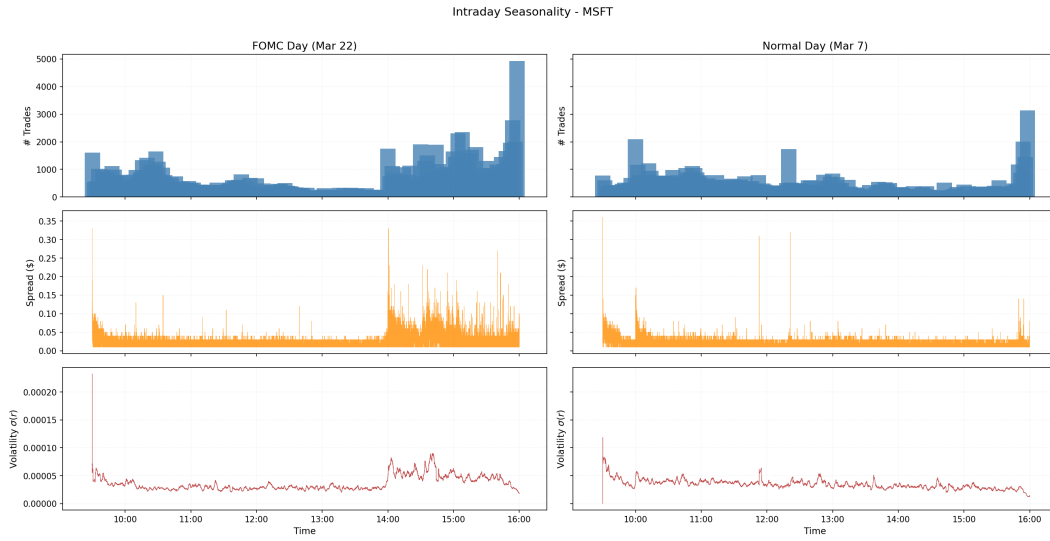
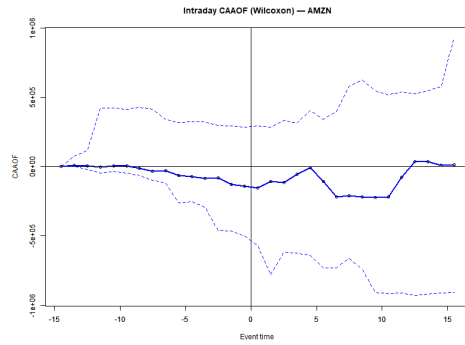


Figure B.30: Intraday seasonality of MSFT on the FOMC day (March 22, left) and a normal trading day (March 7, right). Each panel shows the number of trades per interval (top, blue), bid-ask spread in dollars (middle, orange), and realized volatility (bottom, red).

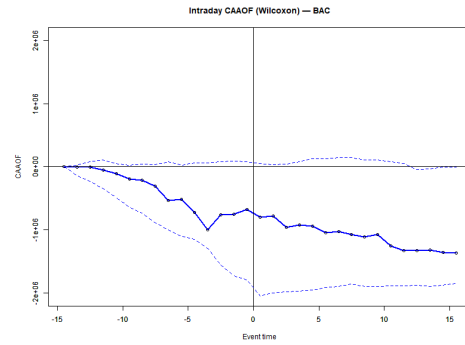
Appendix C

Chapter 3

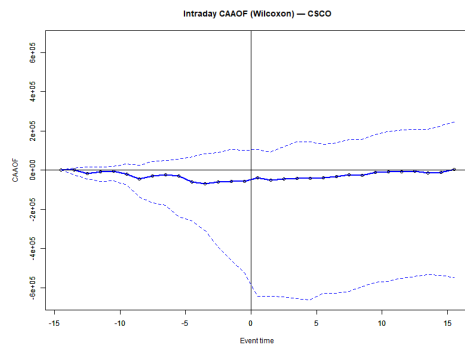
C.1 CAAOFI Event Study Robustness



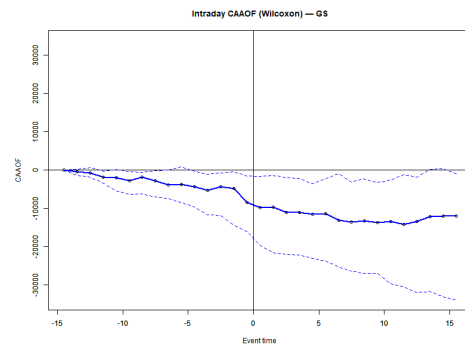
(a) AMZN



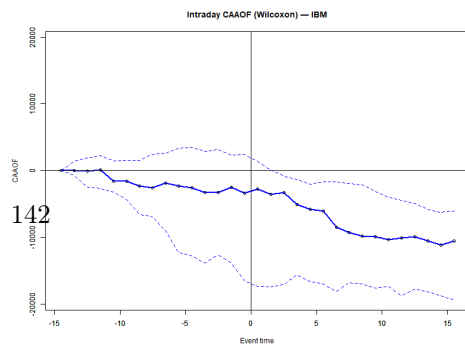
(b) BAC



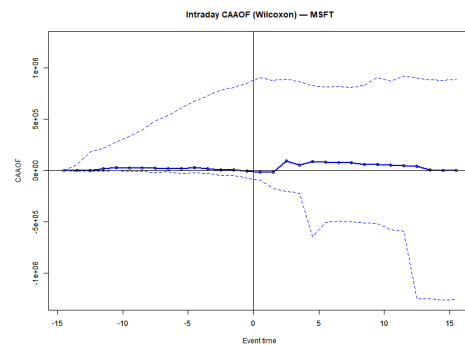
(c) CSCO



(d) GS



(e) IBM



(f) MSFT

Figure C.1: Wilcoxon test results for CAAOFI across six stocks: Amazon (AMZN), Bank of America (BAC), Cisco (CSCO), Goldman Sachs (GS), IBM, and Microsoft (MSFT). Each panel shows the Wilcoxon distributions of CAAOFI coefficient estimates.